



CAPSTONE PROJECT REPORT

Report 4 – Software Design Document

Table of Contents

I. Record of Changes.....	1
II. Software Design Document.....	2
1. System Design.....	2
1.1 System Architecture.....	2
1.1.1 Monolithic architecture.....	2
1.1.2 Component architecture.....	2
1.2 Package Diagram.....	4
1.2.1 Front-end packages.....	4
1.2.2 Back-end packages.....	6
2. Database Design.....	10
3. Detailed Design.....	13
3.1 User and Access Management.....	13
3.1.1 View list employee profile.....	13
3.1.2 Create Receptionist Account.....	14
3.1.3 Update Receptionist Account.....	16
3.1.5 Update Doctor Account.....	17
3.1.6 Create patient account.....	18
3.1.7 Login.....	19
3.1.8 Register.....	20
3.1.9 Forgot Password.....	22
3.1.10 Update profile.....	23
3.2 Schedule Management.....	24
3.2.1 Create Appointment.....	24
3.2.2 View appointment.....	25
3.2.3 Cancel Appointment.....	26
3.2.4 Approve the Appointment.....	28
3.2.5 Create Doctor Schedule.....	29
3.2.6 Update Doctor Schedule.....	30
3.2.7 Delete Doctor Schedule.....	31
3.2.8 View Doctor Schedule.....	32
3.2.9 View Appointment(Admin).....	33
3.2.10 Schedules follow-up appointment.....	34
3.3 Marketing.....	36
3.3.1 Create Blog Medical.....	36
3.3.2 Update Blog Medical.....	37

3.3.3 Delete Blog Medical.....	38
3.3.4 View Blog Medical.....	39
3.3.5 Search blog.....	40
3.3.6 View list of services.....	41
3.3.7 View service detail.....	42
3.3.8 View Doctor's Information.....	43
3.3.9 View List of Doctors.....	44
3.3.10 Update answer.....	45
3.3.11 Update question.....	46
3.3.12 Create Question for doctor.....	47
3.4 Medical Notebooks.....	49
3.4.1 Doctor View Medical Notebooks.....	49
3.4.2 Receptionist View Medical Notebooks.....	51
3.4.3 Doctor create medical Notebooks.....	53
3.4.4 Patient View Medical Notebooks.....	54
3.4.5 Active patient status.....	56
3.4.6 Import result image.....	58
3.4.7 Search Medical Notebook.....	59
3.5 Online Consulting.....	60
3.5.1 Reminder follow-up appointment(patient).....	60
3.5.2 Reminder follow-up appointment(admin).....	61
3.5.3 Post treatment follow-up(Doctor).....	63
3.5.4 Online advising(Patient).....	64
3.5.5 Online advising(Receptionist).....	65

I. Record of Changes

Date	A* M, D	In charge	Change Description
05/07/2024	A	Nguyen Duc Phuong	Create report item(1,2,3)

*A - Added M - Modified D - Deleted

II. Software Design Document

1. System Design

1.1 System Architecture

1.1.1 Monolithic architecture

Explain about Monolithic architecture: Monolithic architecture is a traditional model of a software program, which is built as a unified unit that is self-contained and independent from other applications.

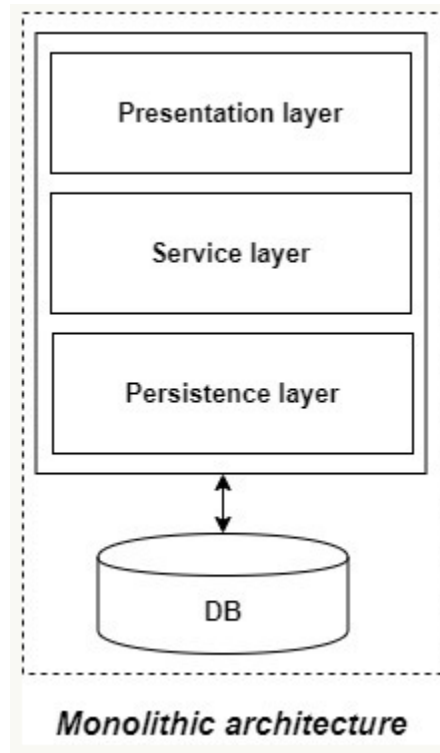


Figure 1.1.1.1: Monolithic architecture

This system is self-contained; all the components - data storage, business logic, client-side user interface, and server-side application - are packaged together and managed as one unit.

1.1.2 Component architecture

1.1.2.1 Front-end

ReactJS is known for its component-based architecture, which allows developers to build encapsulated components that manage their own state, and then compose them to make complex UIs. React's primary focus is on rendering data to the DOM, and it provides a set of tools and patterns for managing component states and lifecycle.

- **Component:** A React component is a self-contained module that renders some output. You can think of a React component as a JavaScript function or class that optionally

accepts inputs (props) and returns a React element that describes how a section of the UI should appear. Components can be either class-based or functional.

- + **Class Component:** Defined using ES6 classes, they can manage their own state and lifecycle methods.
- + **Functional Component:** A simpler way to write components using functions, enhanced with hooks to manage state and side effects.
- **JSX :** stands for JavaScript XML. It is a syntax extension for JavaScript that looks similar to HTML or XML. JSX is used to describe what the UI should look like. React elements are written using JSX.
- **Props:** Props, short for properties, are read-only attributes used to pass data from a parent component to a child component. Props allow you to pass dynamic data and event handlers to components.
- **State:** State is a built-in object that stores property values that belong to a component. When the state object changes, the component re-renders. State is managed within the component and can be updated using the `setState` method in class components or the `useState` hook in functional components.
- **Hooks:** Hooks are functions that let you "hook into" React state and lifecycle features from function components. Common hooks include `useState` for state management, `useEffect` for side effects, and `useContext` for context API.
- **Router:** is a library for managing routing in React applications, allowing you to navigate between different components based on the URL.

1.1.2.2 Back-end

Back-end is built using C# and the ASP.NET Web API framework. API controllers handle HTTP requests and responses, ensuring data interaction between the front-end and the database. The backend architecture includes

- + **RESTful API Design:** The API follows REST principles, ensuring a consistent and predictable interface for client-server communication.
- + **Authentication and Authorization:** Security measures, such as token-based authentication (JWT) or OAuth, are implemented to safeguard data and resources, ensuring access control based on user roles and permissions.
- + **Data Validation:** Input data is validated to prevent unauthorised access and ensure data integrity using validation libraries or custom validation mechanisms.
- + **Service:** The gateway (API controller) receives requests from clients and redirects to the suitable service in order to handle the request and the service returns a response. Grpc is used as a communication way between the gateway and services.
- + **Command and Query :** The Command and Query Responsibility Segregation (CQRS) pattern separates the responsibilities of commands (write operations that change the state of the system) from queries (read operations that retrieve data without changing the state). This segregation helps to simplify the design of complex systems and provides several benefits, including improved scalability, performance, and maintainability.

- + Repository : This class acts as a mediator between the domain model entities and the data source, providing a straightforward way to perform CRUD (Create, Read, Update, Delete) operations without exposing the underlying data access logic to the rest of the application.

1.1.2.3 Database

SQL Server is a powerful and scalable relational database management system (RDBMS) developed by Microsoft. It is designed to manage and store large amounts of data with high performance, reliability, and security.

- + Data Modeling: involves organizing the database structure with appropriate tables, relationships, and constraints to ensure efficient data storage and retrieval.
- + Normalization is the process of organizing the database to reduce redundancy and ensure data consistency. This involves dividing large tables into smaller, related tables and defining relationships between them to minimize duplicate data and ensure data integrity.
- + Data Security in SQL Server involves implementing measures to protect sensitive information such as authentication, authorization, encryption.
- + Backup and Recovery: SQL Server provides robust backup and recovery solutions to protect data from loss and ensure business continuity. This includes regular backups, transaction log backups, and point-in-time recovery capabilities.

1.2 Package Diagram

1.2.1 Front-end packages

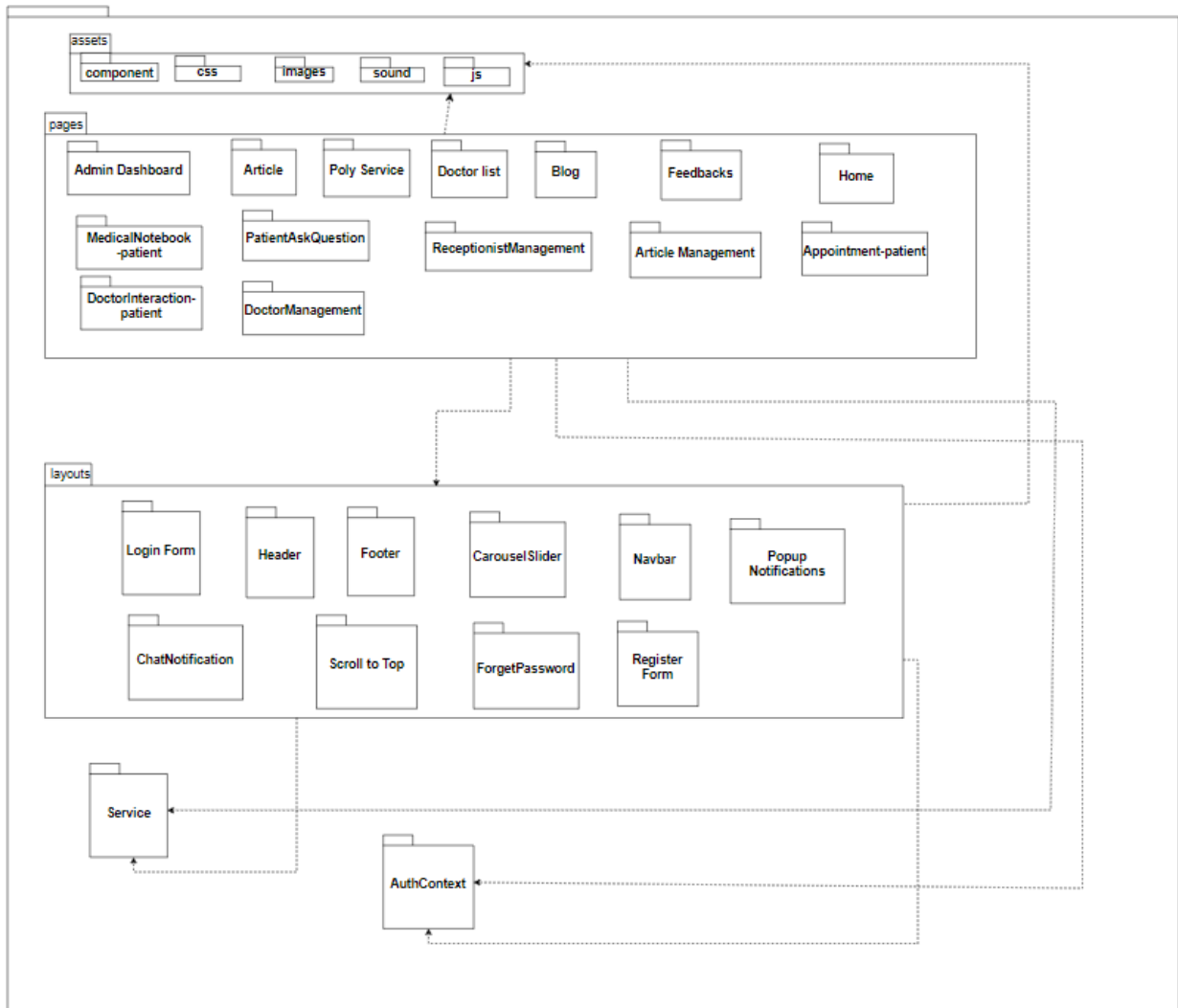


Figure 1.2.1.1 Front-end package diagram

1.2.2 Package descriptions

Table 1.2.2.1 Package descriptions

No	Package	Description
1	<i>asset</i>	<i>Stores frameworks, image files, sound files, CSS</i>
1.1	Component	Common components for pages
1.2	CSS	CSS files
1.3	sound	Sound files for buttons
1.4	js	Bootstrap files
2	<i>pages</i>	<i>Function screens</i>
2.1	Admin Dashboard	Admin screen
2.2	Article	Clinic information screen
2.3	Poly Service	Services screen
2.4	Doctor list	Doctor list screen
2.5	Blog	Medical blog post screen
2.6	Feedbacks	Customer feedback screen
2.7	Home	Main screen
2.8	Medical Notebook patient	Screen of the patient's medical notebook
2.9	Receptionists Management	Functions in the reception management screen
2.10	Article Management	Functions in the article manager screen
2.11	Appointment patient	Patient appointment scheduling screen
2.12	Doctor Interaction Patient	Post treatment service for patients
2.13	Doctor Management	Functions in the doctor management screen
2.14	Patient Ask Question	Patients consult medical professionals

3	<i>layouts</i>	<i>Components for different screens</i>
3.1	Login form	Display the login form
3.2	Header	Displays the page header
3.3	Footer	Displays the footer of the page
3.4	Forget Password	Display the password reset screen
3.5	CarouselSlider	Image for home page
3.6	NavBar	Used to navigate the page
3.7	Register Form	Display the registration form
3.8	Popup Notifications	Function result messages
3.9	Chat Notification	Notification for chat
3.10	Scroll to Top	When turning pages, it helps to always stay on top of the page
4	<i>Service</i>	<i>The files contain API calls</i>
5	<i>AuthContext</i>	<i>File contains user authentication functions</i>

1.2.3 Back-end packages

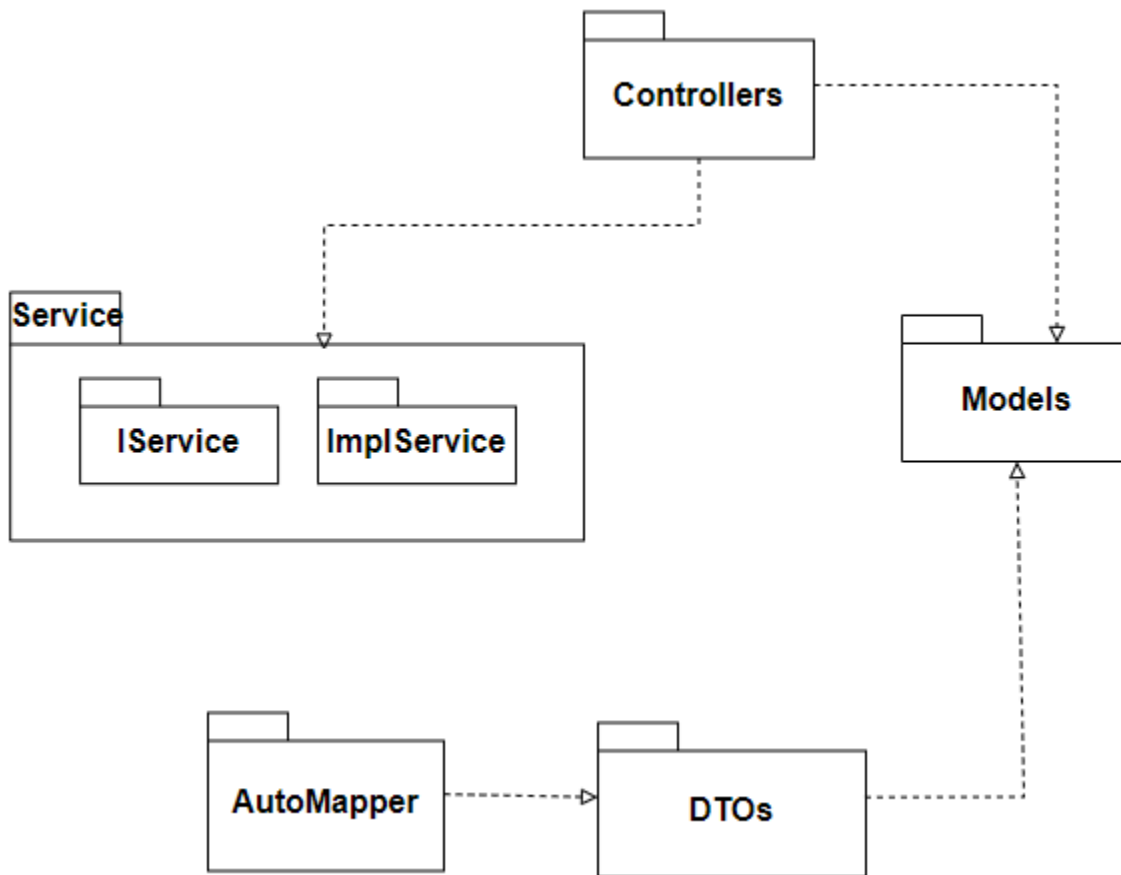


Figure 1.2.3.1: Back-End Package Diagram

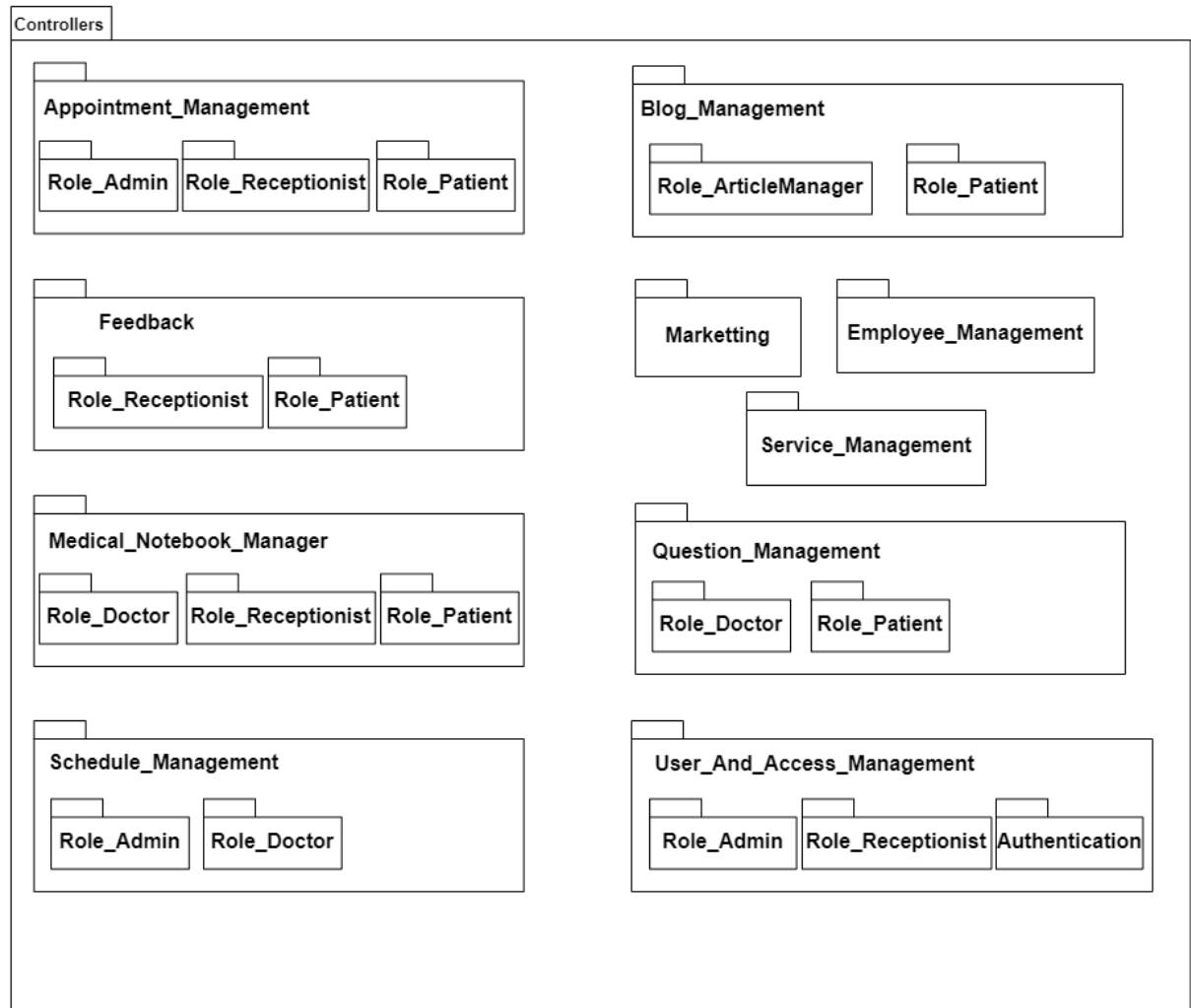


Figure 1.2.3.2: Controller Package

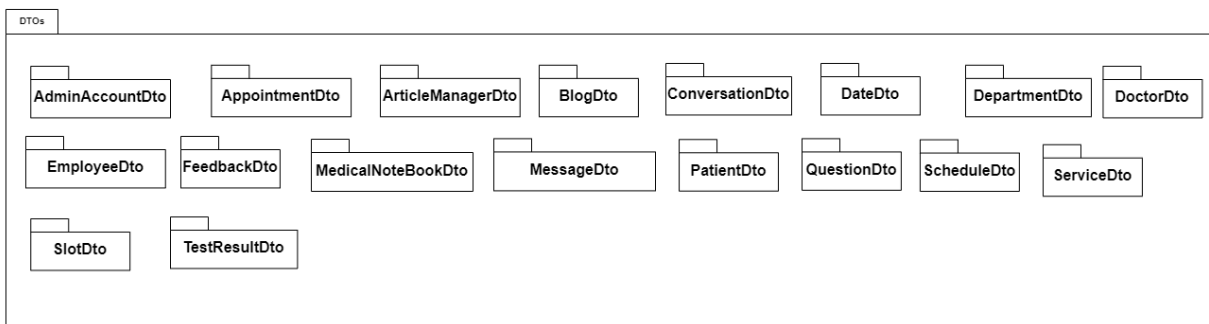


Figure 1.2.3.3: DTO Package

1.2.4 Package descriptions

Table 1.2.4.1 Package descriptions

No	Package	Description
1	Service	<i>Saving all service provider(SMS, Send Email, OTP,...)</i>
1.1	IService	Saving interface component for Service
1.2	ImplService	Saving implement component for Service
2	Controller	<i>API controller(CRUD, transition,.....)</i>
2.1	Appointment_Management	Saving API for appointment management assigns permissions according to folder roles(Ex: folder Role_Admin saving API for what admin can do to the appointment,...)
2.2	Blog_Management	Saving API for blog management assign permissions according to folder roles(Ex: folder Role_Article saving API for what article can do to the blog,...)
2.3	Marketing	Saving API for marketing like: content header, introduction
2.4	Feedback	Saving API for feedback assigns permissions according to folder roles(Ex: folder Role_Patient saving API for what patient can do to the feedback,...)
2.5	Question_Manager	Saving API for question management assigns permissions according to folder roles(Ex: folder Role_Doctor saving API for what doctor can do to the question,...)
2.6	Medical_Notebook_Manager	Saving API for medical notebook management assigns permissions according to folder roles(Ex: folder Role_Doctor saving API for what doctor can do to the medical notebook,...)
2.7	Schedule_Management	Saving API for schedule management assigns permissions according to folder roles(Ex: folder Role_Doctor saving API for what doctor can do to the blog,...)

2.8	User_And_Access_Management	Saving API for blog management assign permissions according to folder roles(Ex: folder Role_Admin saving API for what admin can do to the account employee,...) and also saving API for authentication & authorization
2.9	Employee_Management	Saving API for employee management
2.10	Service_Management	Saving API for service management
3	Models	Encompasses the fundamental entities, relationships, and behaviors that define the application's domain.
4	DTOs	Visually represent how the data is structured and transferred in the application. Help to maintain a clear separation between data representation and business logic, promoting a more modular and maintainable architecture.
4.1	AdminAccountDto	Saving class DTO for admin account
4.2	AppointmentDto	Saving class DTO for appointment
4.3	ArticleManagerDto	Saving class DTO for article manager
4.4	BlogDto	Saving class DTO for blog
4.5	ConversationDto	Saving class DTO for conversation
4.6	DateDto	Saving class DTO for date
4.7	DepartmentDto	Saving class DTO for department
4.8	DoctorDto	Saving class DTO for doctor
4.9	EmployeeDto	Saving class DTO for employee
4.10	FeedbackDto	Saving class DTO for feedback
4.11	MedicalNoteBookDto	Saving class DTO for medical notebook
4.12	MessageDto	Saving class DTO for message
4.13	PatientDto	Saving class DTO for patient
4.14	QuestionDto	Saving class DTO for question
4.15	ScheduleDto	Saving class DTO for schedule
4.16	ServiceDto	Saving class DTO for service
4.17	Slotdto	Saving class DTO for slot
4.18	TestResultDto	Saving class DTO for test result
5	AutoMapper	Configure the transfer between Models and DTOs

2. Database Design

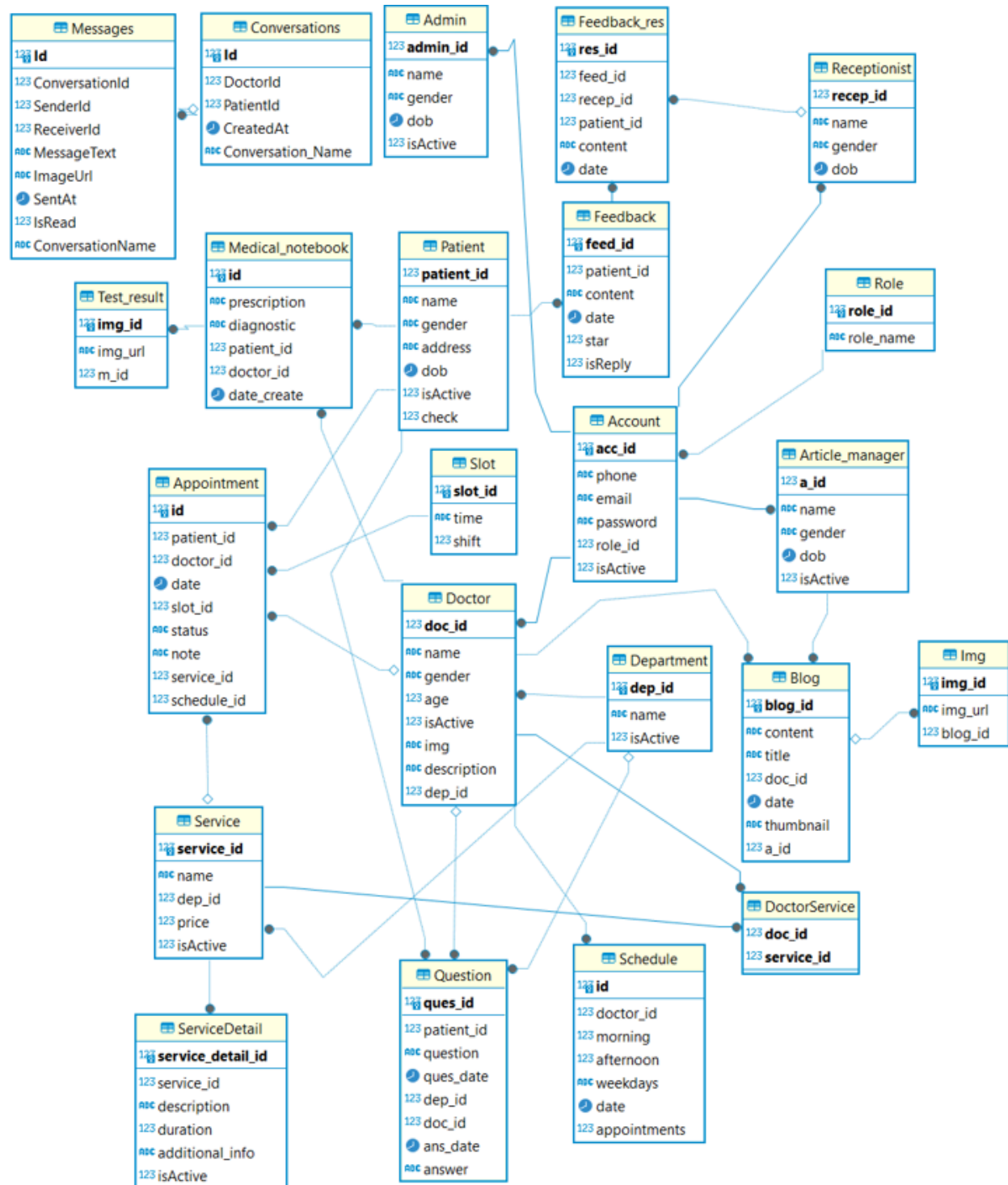


Figure 2.1: Database diagram

2.1 Table Descriptions

Table 2.1.1 Table Descriptions

No	Table	Description
01	Account	Saving all account in system - Primary keys: <i>acc_id</i> - Foreign keys: <i>role_id</i> (Role)
02	Doctor	Saving information about doctor - Primary keys: <i>doc_id</i> - Foreign keys: <i>dep_id</i> (Department), <i>doc_id</i> (Account)
03	Article_Manager	Saving information for article manager - Primary keys: <i>a_id</i> - Foreign keys: <i>a_id</i> (Account)
04	Admin	Saving information for admin - Primary keys: <i>admin_id</i> - Foreign keys: <i>admin_id</i> (Account)
05	Patient	Saving information for patient - Primary keys: <i>patient_id</i> - Foreign keys: <i>patient_id</i> (Account)
06	Slot	Saving slot by time(ex: 7h30) - Primary keys: <i>slot_id</i> - Foreign keys: None
07	Medical_Notebook	Saving medical result for patients - Primary keys: <i>id</i> - Foreign keys: <i>patient_id</i> (Patient)
08	Feedback	Saving all feedback that patient has sent - Primary keys: <i>feed_id</i> - Foreign keys: <i>patient_id</i> (Patient)
09	Question	Saving all question that patient has sent - Primary keys: <i>ques_id</i> - Foreign keys: <i>doc_id</i> (Doctor), <i>patient_id</i> (Patient), <i>dep_id</i> (Department)
10	Feedback_res	Saving response from receptionist - Primary keys: <i>res_id</i> - Foreign keys: <i>feed_id</i> (Feedback), <i>recep_id</i> (Receptionist)
11	Service	Saving service provided - Primary keys: <i>service_id</i> - Foreign keys: <i>dep_id</i> (Department)
12	Department	Saving doctor department - Primary keys: <i>dep_id</i> - Foreign keys: None
13	Schedule	Saving doctor schedule - Primary keys: <i>id</i>

		- Foreign keys: <i>doctor_id</i> (Doctor)
14	Messages	Saving information of messages - Primary keys: <i>Id</i> - Foreign keys: <i>ConversationId</i> (Conversations)
15	Role	Saving all role in system - Primary keys: <i>role_id</i> - Foreign keys: None
16	Blog	Saving blog that article manager created - Primary keys: <i>blog_id</i> - Foreign keys: <i>a_id</i> (Article_Manager), <i>doc_id</i> (Doctor)
17	Conversations	Saving information about conversations - Primary keys: <i>Id</i> - Foreign keys: None
18	Img	Saving image of blog - Primary keys: <i>Img_id</i> - Foreign keys: <i>blog_id</i> (Blog)
19	Appointment	Saving appointment that patient has order - Primary keys: <i>id</i> - Foreign keys: <i>slot_id</i> (Slot), <i>patient_id</i> (Patient), <i>doctor_id</i> (Doctor), <i>service_id</i> (Service)
20	Receptionist	Saving information of receptionist - Primary keys: <i>recep_id</i> - Foreign keys: <i>recep_id</i> (Account)
21	ServiceDetail	Saving information about service detail - Primary keys: <i>service_detail_id</i> - Foreign keys: <i>service_id</i> (Service)
22	DoctorService	A table to represent the many-to-many relationship between Doctor and Service - Primary keys: <i>doc_id</i> , <i>service_id</i> - Foreign keys: <i>doc_id</i> (Doctor), <i>service_id</i> (Service)
23	Test_Result	Saving information about conversations - Primary keys: <i>Img_id</i> - Foreign keys: <i>m_id</i> (Medical_Notbook)

3. Detailed Design

3.1 User and Access Management

3.1.1 View list employee profile

3.1.1.1 Class Diagram

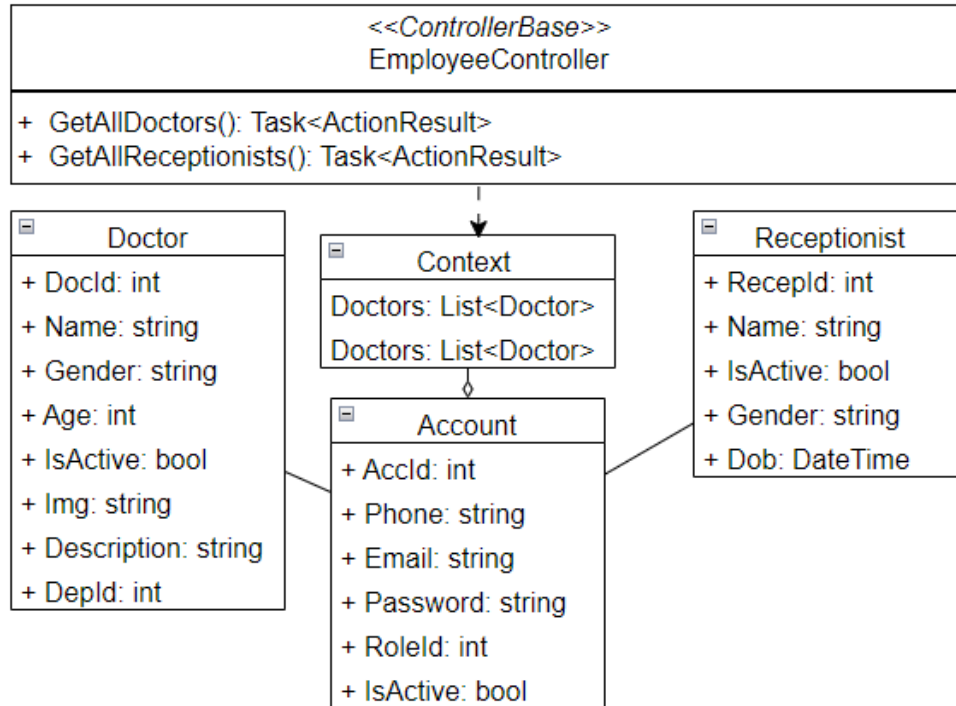


Figure 3.1.1.1.1 View list employee Class Diagram

3.1.1.2 Sequence Diagram

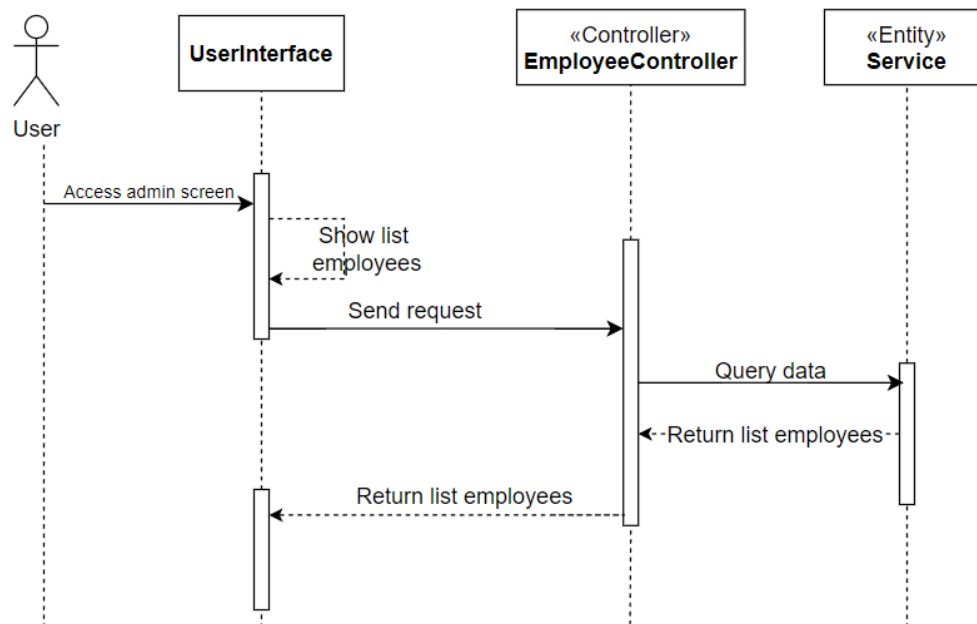


Figure 3.1.1.2.1 View list employee Sequence diagram

3.1.2 Create Receptionist Account

3.1.2.1 Class Diagram

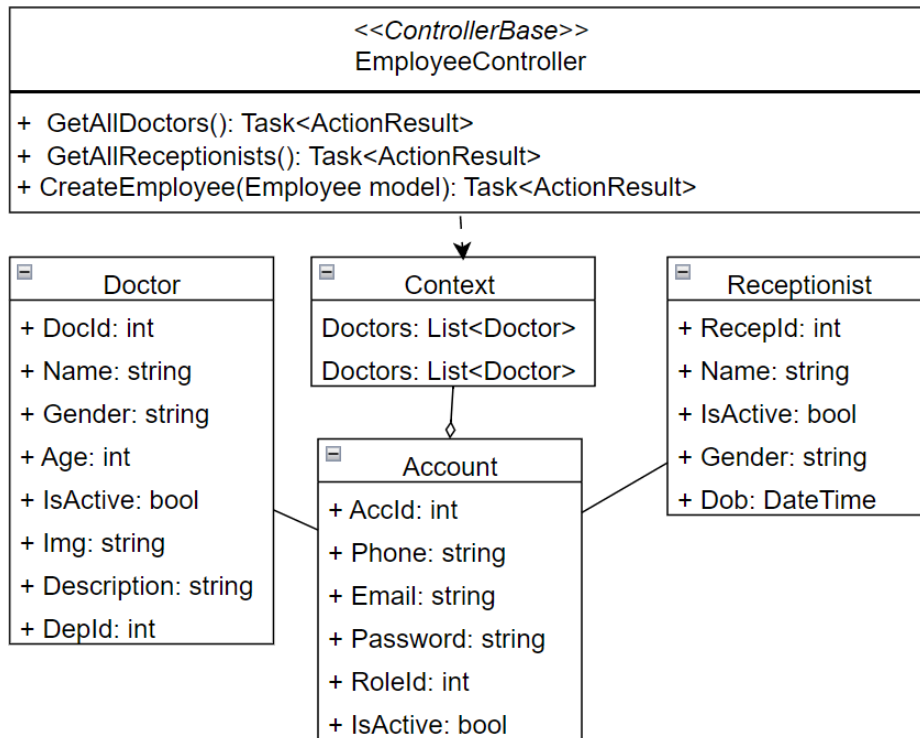


Figure 3.1.2.1.1 Create Receptionist account Class diagram

3.1.2.2 Sequence Diagram

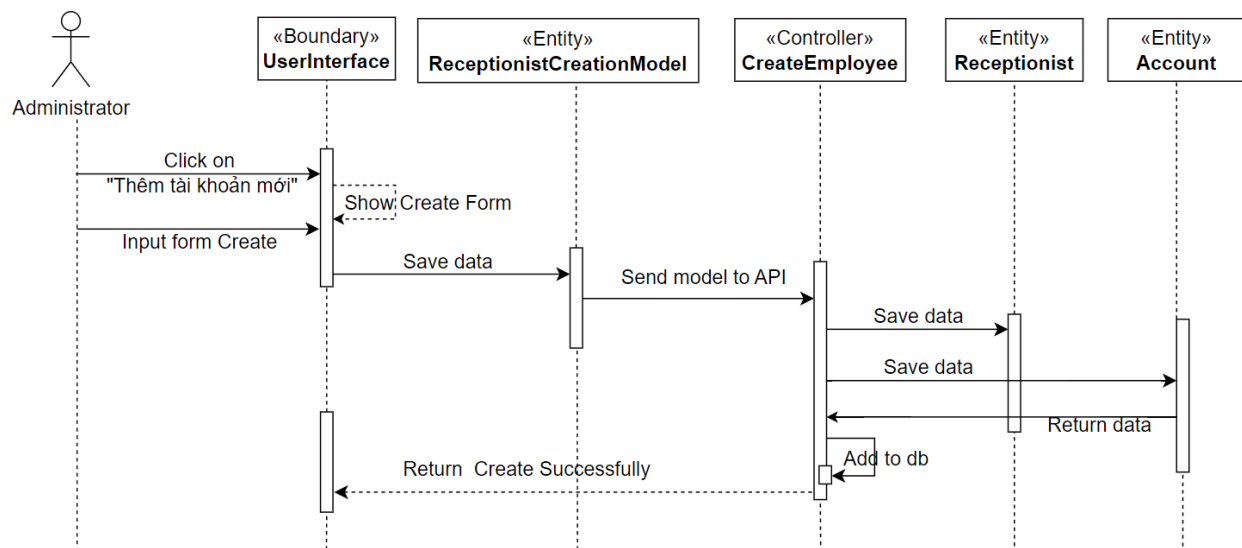


Figure 3.1.2.2.1 Create Receptionist account sequence diagram

3.1.3 Update Receptionist Account

3.1.3.1 Class Diagram

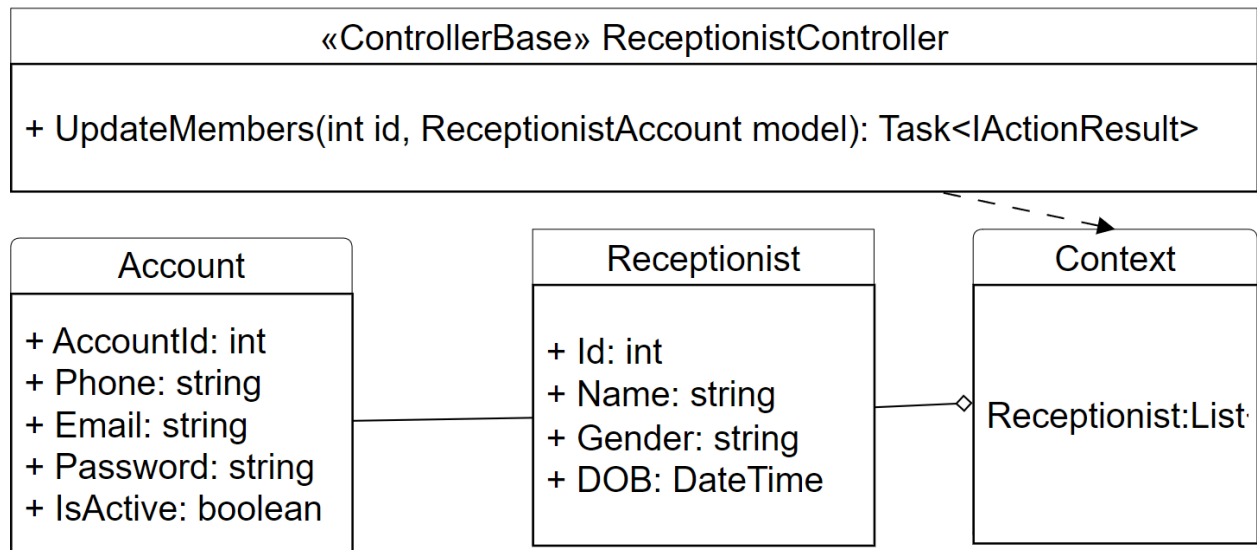


Figure 3.1.3.1.1 Update Receptionist class diagram

3.1.3.2 Sequence Diagram

3.1.5 Update Doctor Account

3.1.5.1 Class Diagram

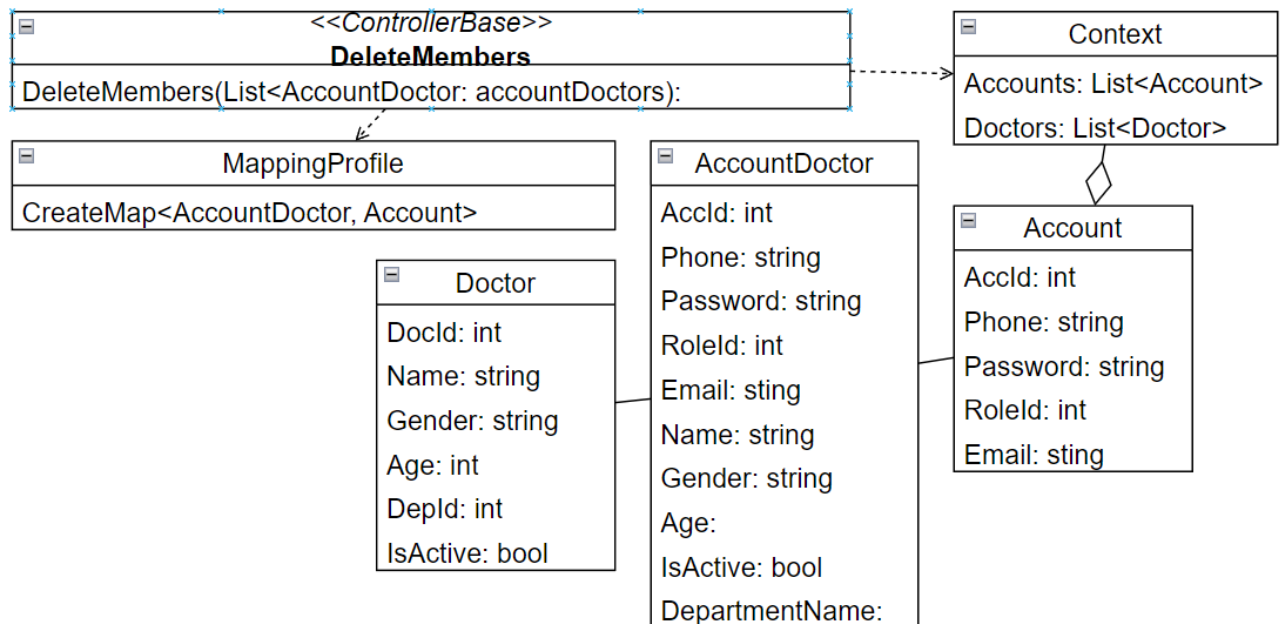


Figure 3.1.5.1.1 Update doctor account Class diagram

3.1.5.2 Sequence Diagram

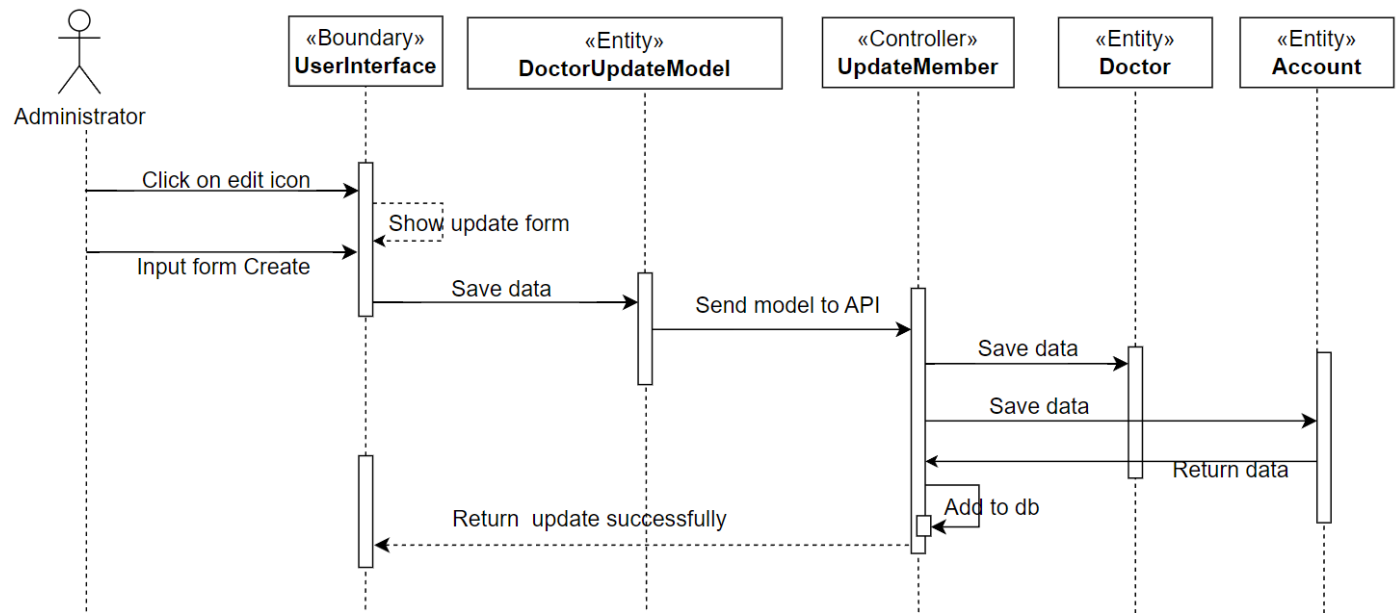


Figure 3.1.5.2.1 Update doctor account Sequence diagram

3.1.6 Create patient account

3.1.6.1 Class diagram

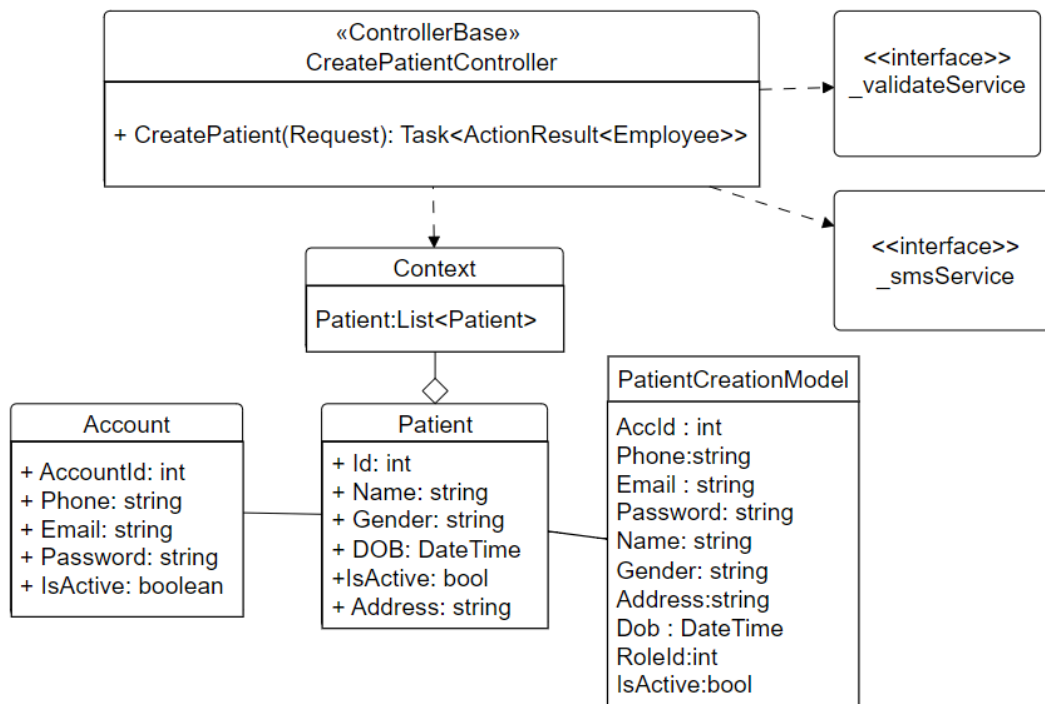


Figure 3.1.6.1.1 Create patient account class diagram

3.1.6.2 Sequence diagram

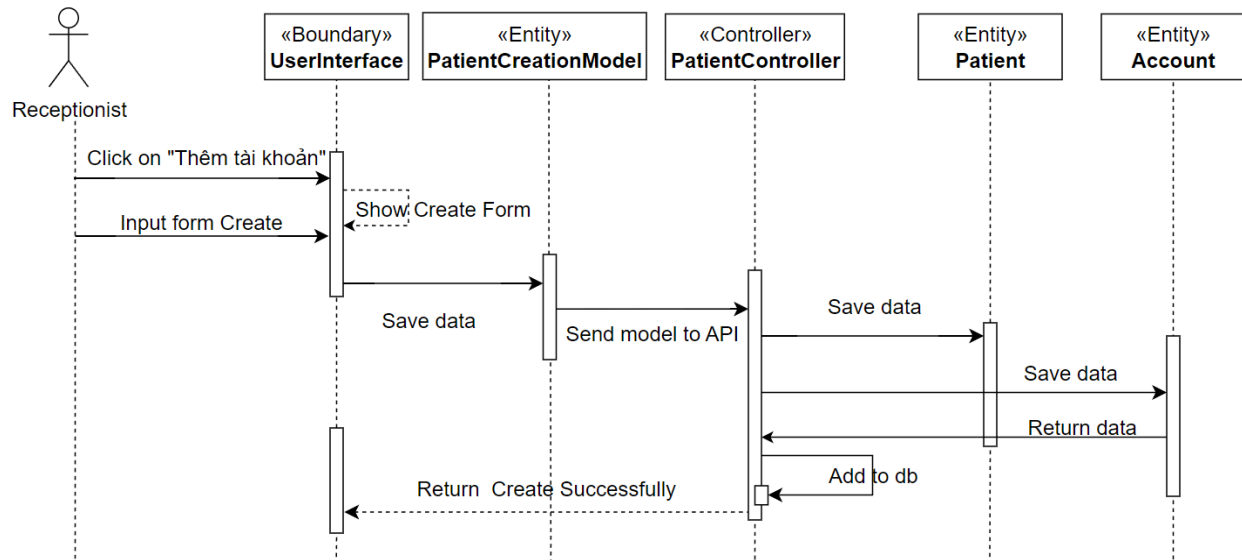


Figure 3.1.6.2.1 Create patient account sequence diagram

3.1.7 Login

3.1.7.1 Class Diagram

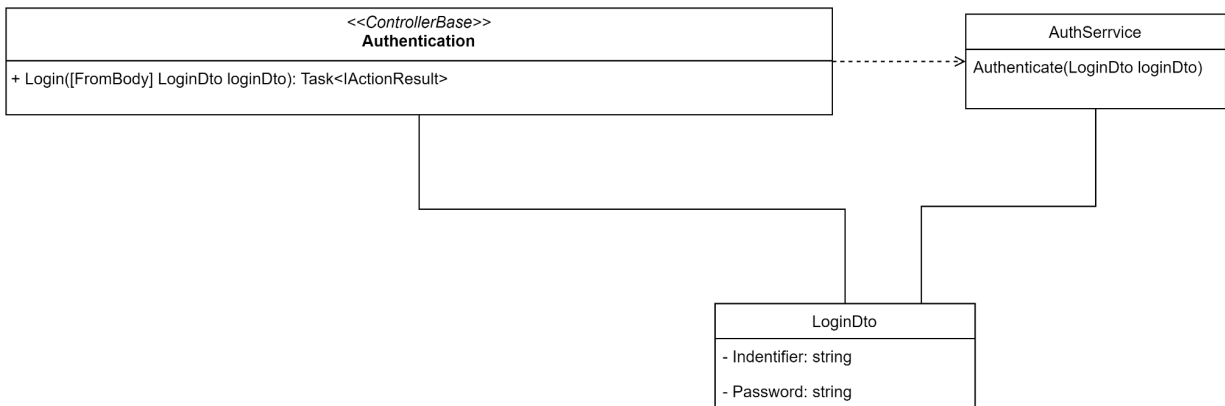


Figure 3.1.7.1.1 Login class diagram

3.1.7.2 Sequence Diagram

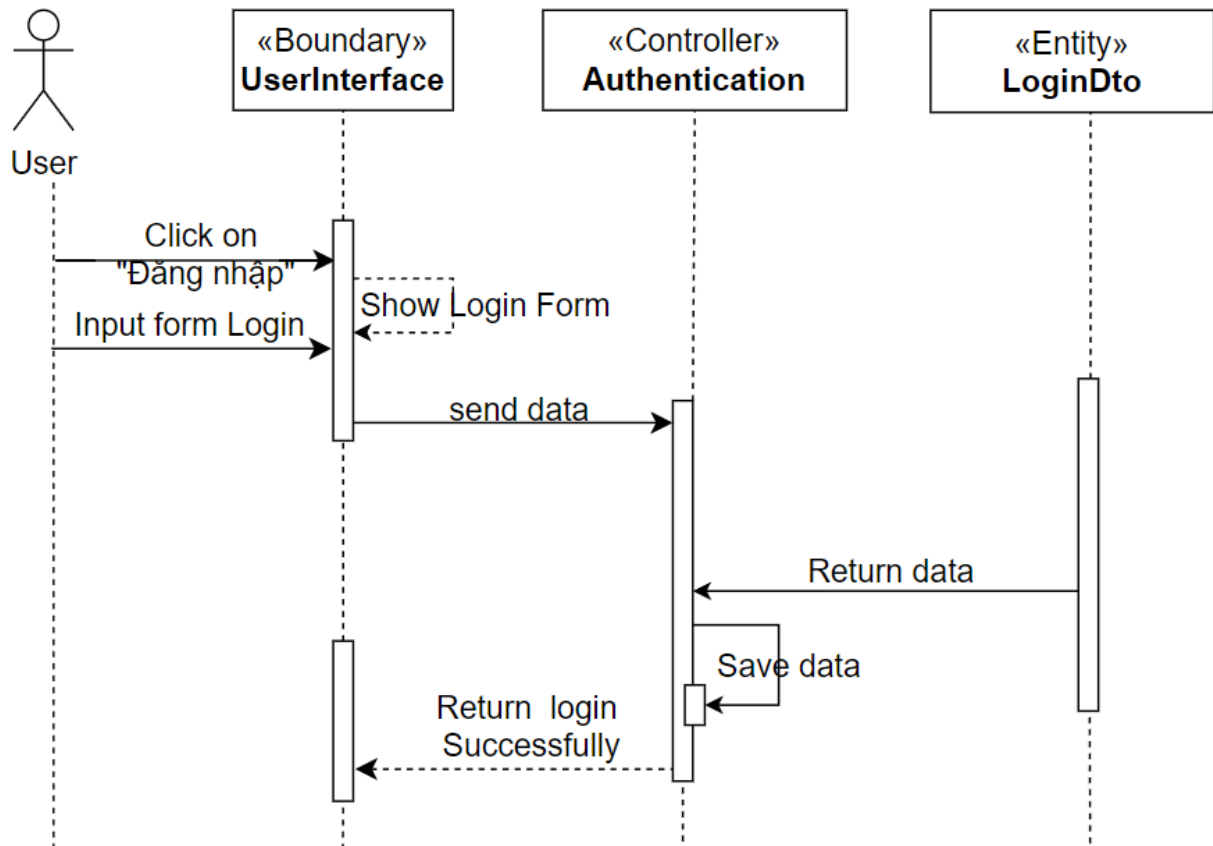


Figure 3.1.7.2.1 Login sequence diagram

3.1.8 Register

3.1.8.1 Class Diagram

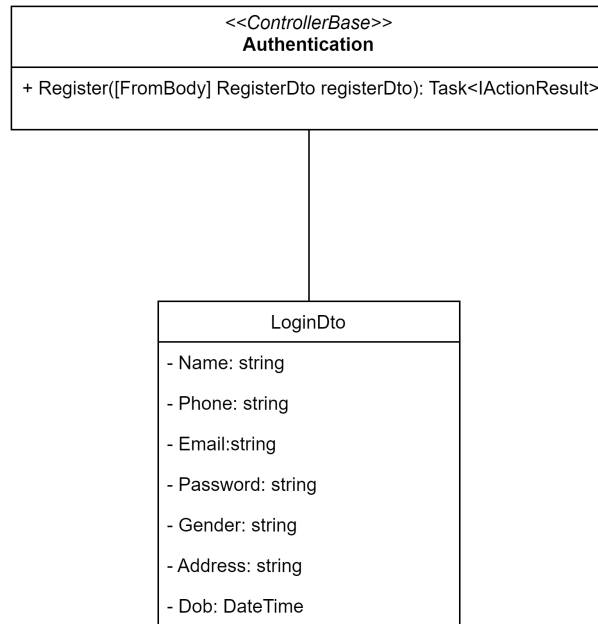


Figure 3.1.8.1.1 Register class diagram

3.1.8.2 Sequence Diagram

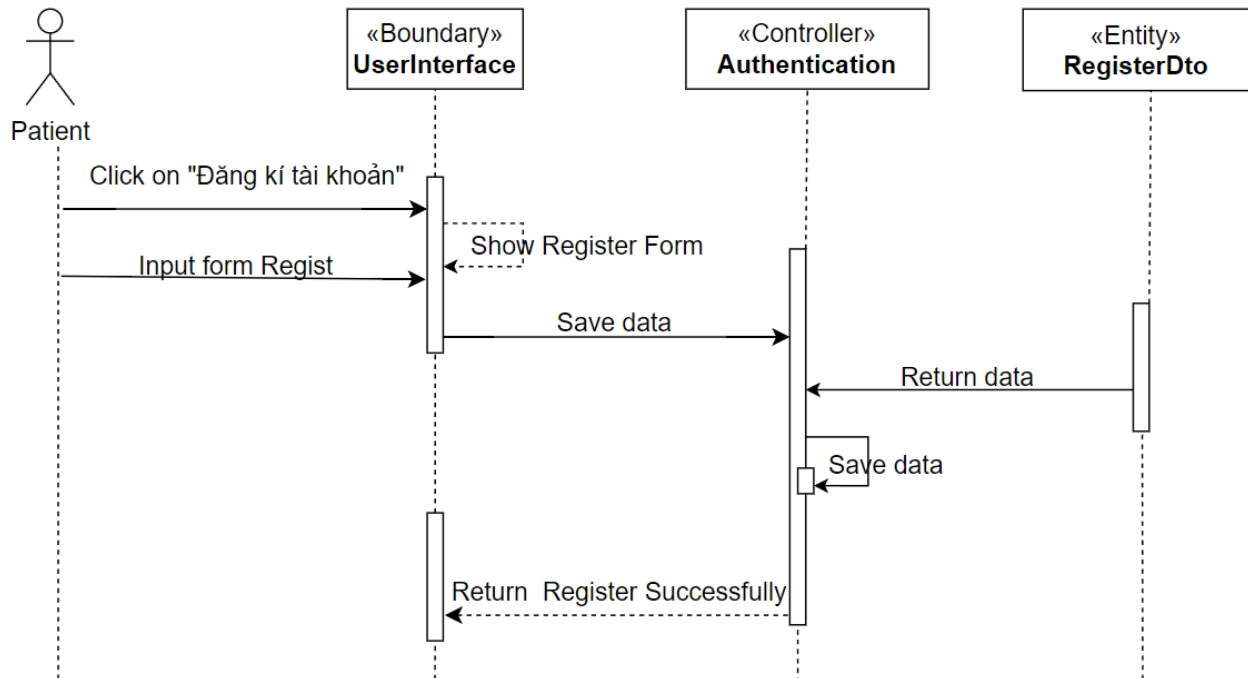


Figure 3.1.8.2.1 Register sequence diagram

3.1.9 Forgot Password

3.1.9.1 Class Diagram

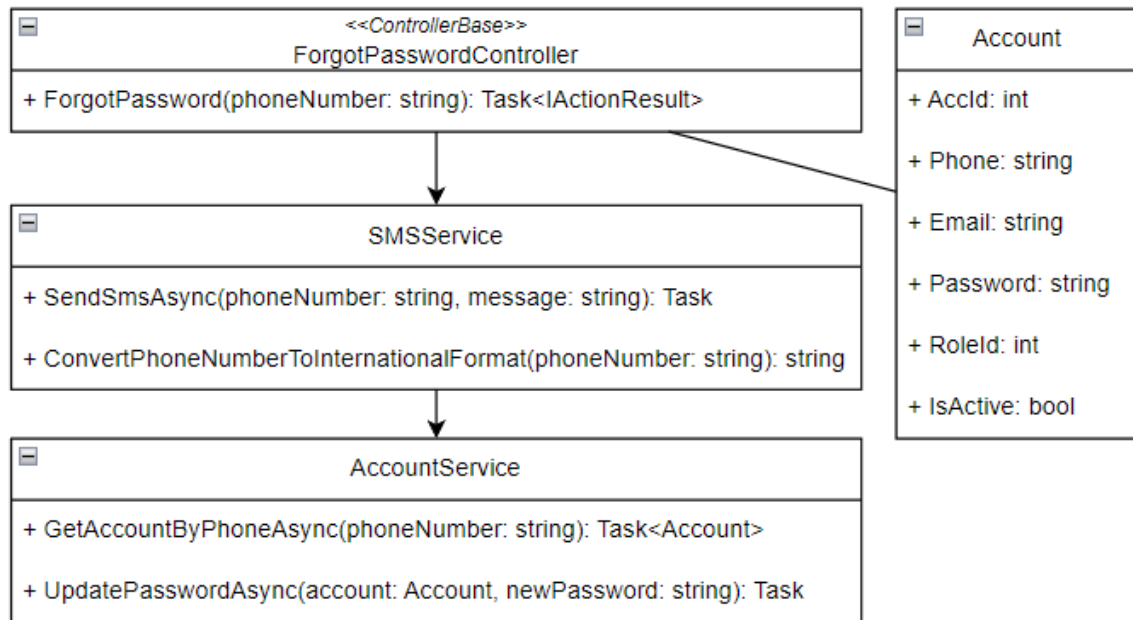


Figure 3.1.9.1.1 Forgot password class diagram

3.1.9.2 Sequence Diagram

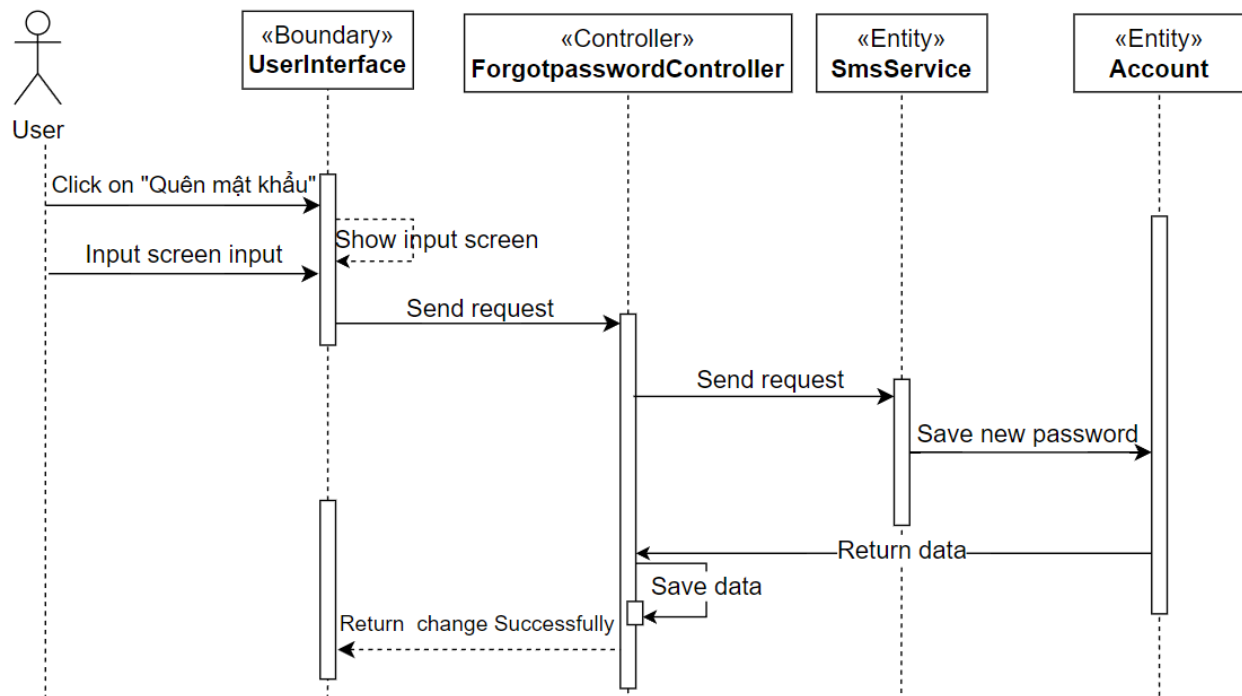


Figure 3.1.9.2.1 Forgot password sequence diagram

3.1.10 Update profile

3.1.10.1 Class Diagram

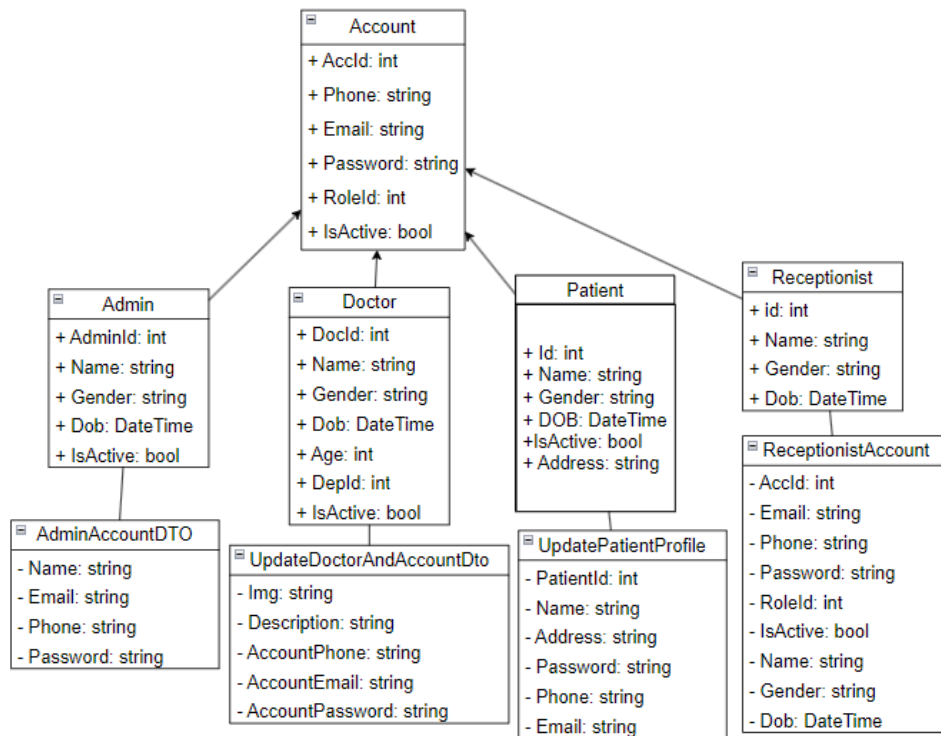


Figure 3.1.10.1.1 Update profile class diagram

3.1.10.2 Sequence Diagram

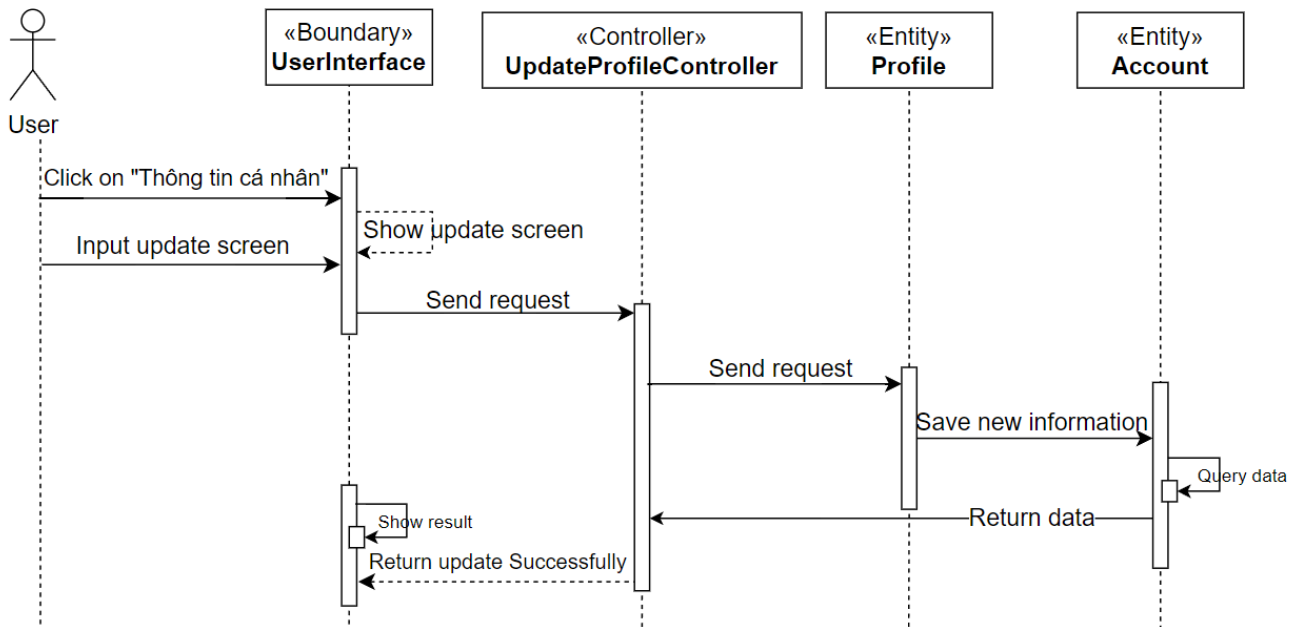
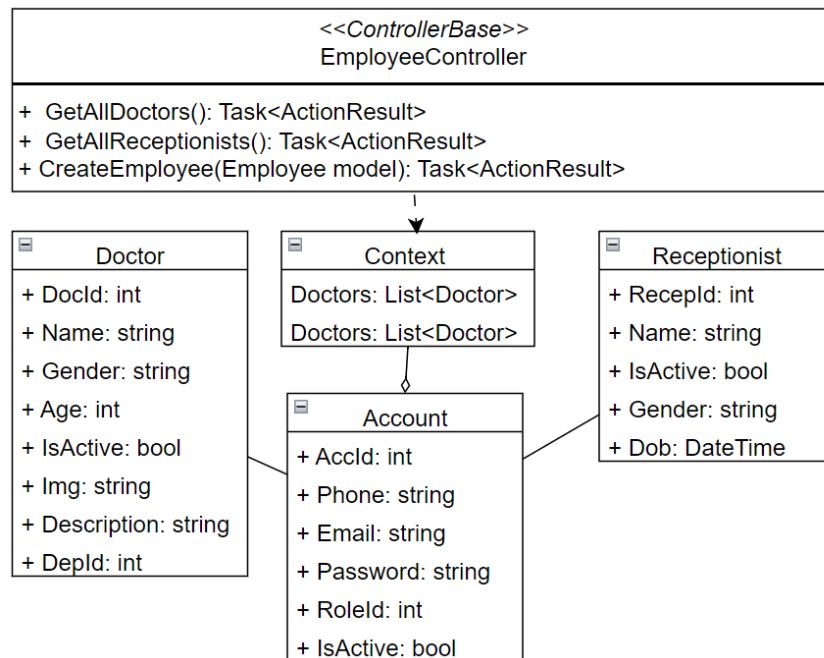


Figure 3.1.10.2.1 Update profile sequence diagram

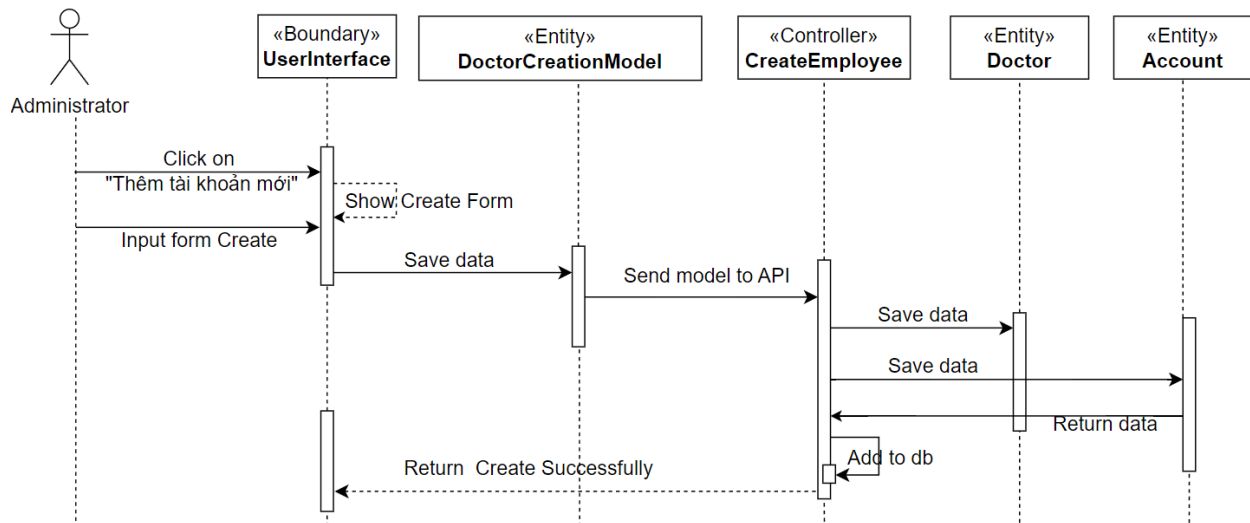
3.1.11 Create doctor account

3.1.11.1 Class Diagram



3.1.11.1.1 Create doctor class diagram

3.1.11.2 Sequence Diagram



3.1.11.2 Create doctor sequence diagram

3.2 Schedule Management

3.2.1 Create Appointment

3.2.1.1 Class Diagram

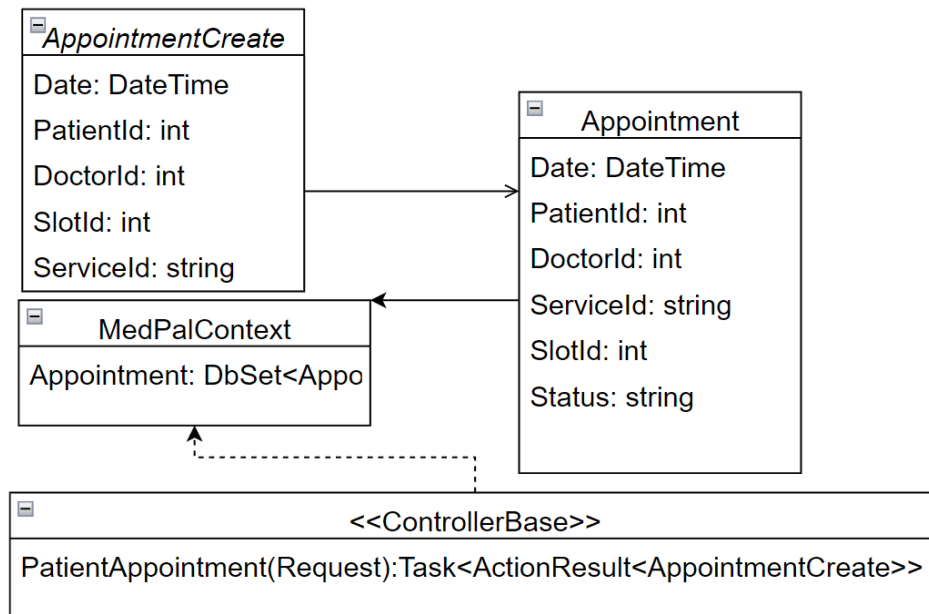


Figure 3.2.1.1.1 Create appointment class diagram

3.2.1.2 Sequence Diagram

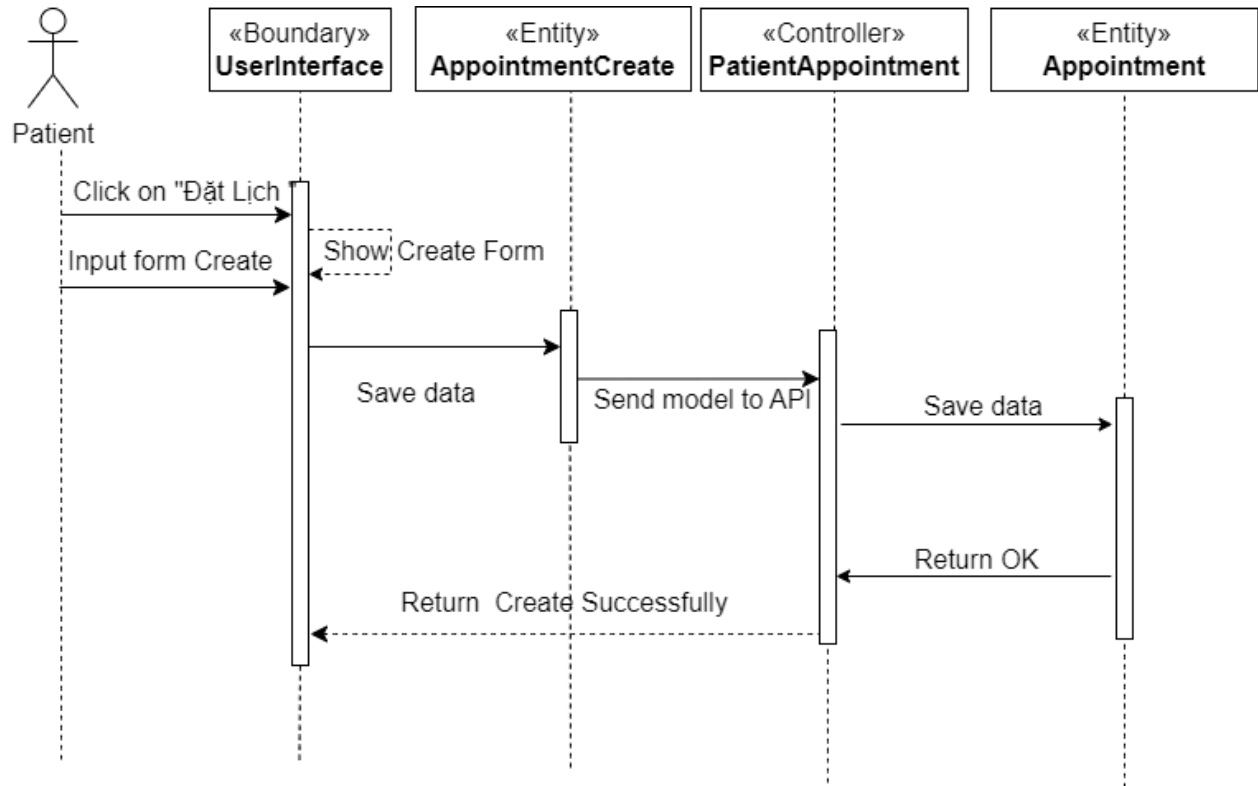


Figure 3.2.1.2.1 Create appointment sequence diagram

3.2.2 View appointment

3.2.2.1 Class Diagram

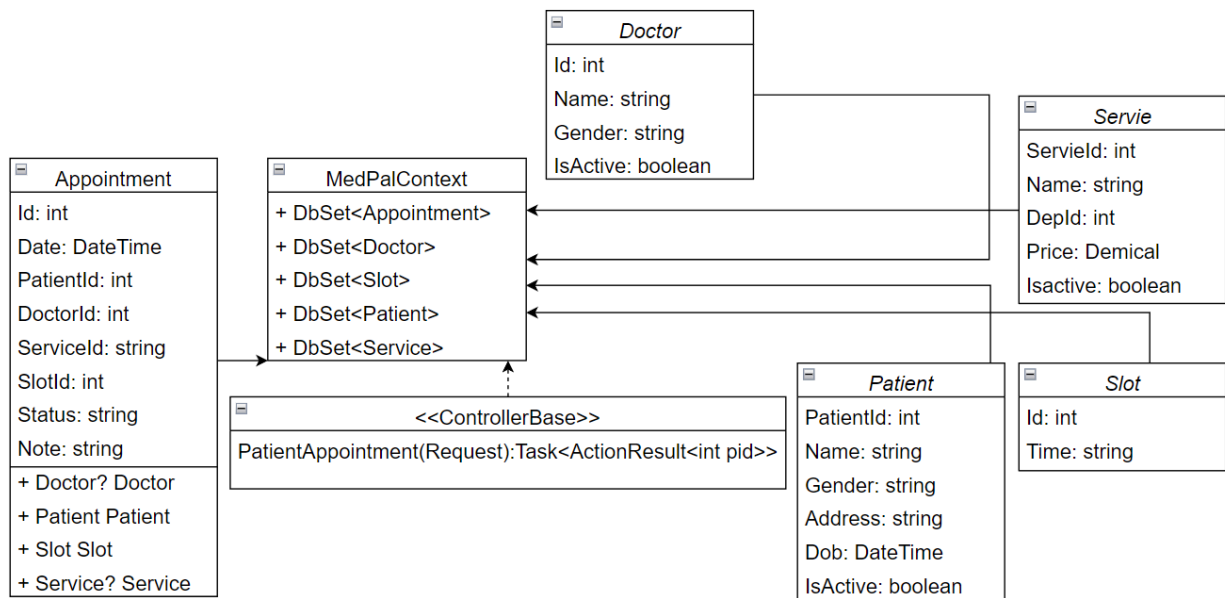


Figure 3.2.2.1.1 View appointment class diagram

3.2.2.2 Sequence Diagram

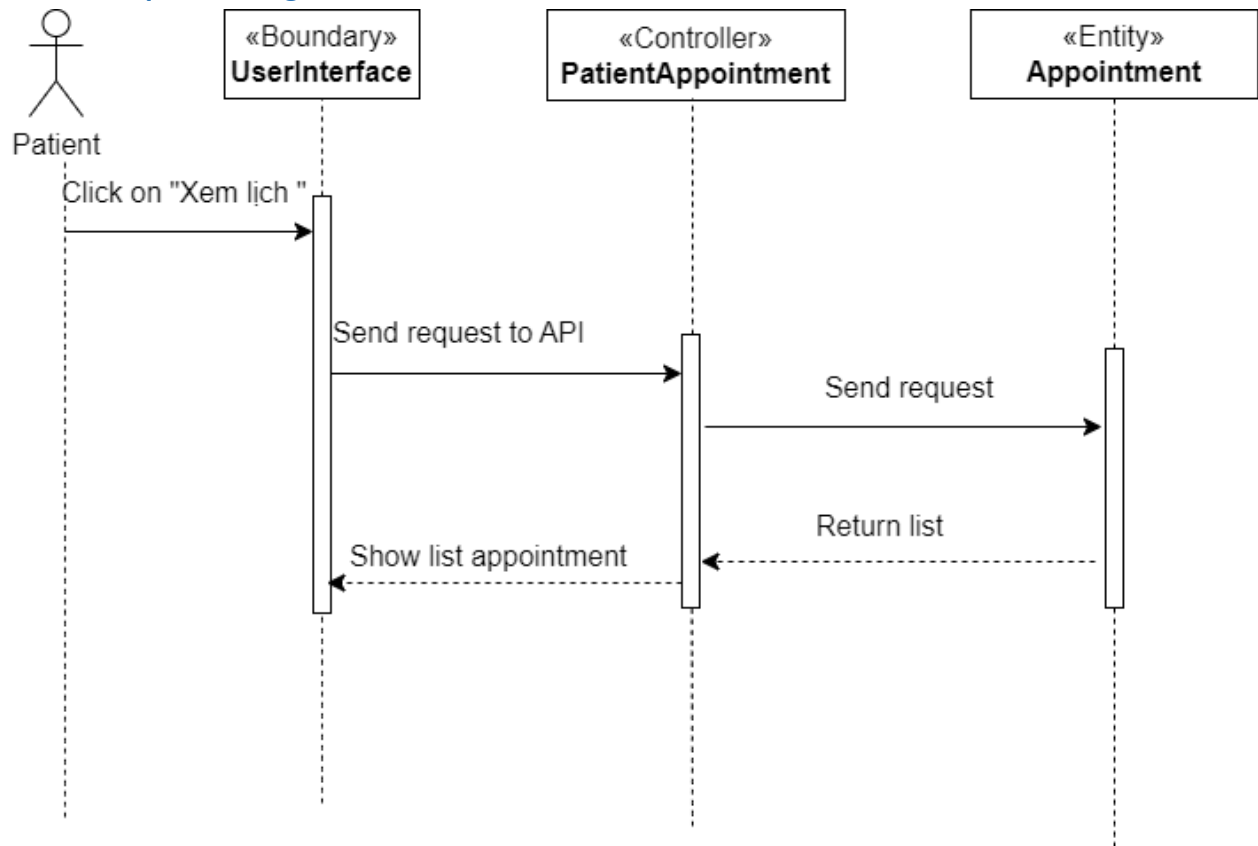


Figure 3.2.2.2.1 View appointment sequence diagram

3.2.3 Cancel Appointment

3.2.3.1 Class Diagram

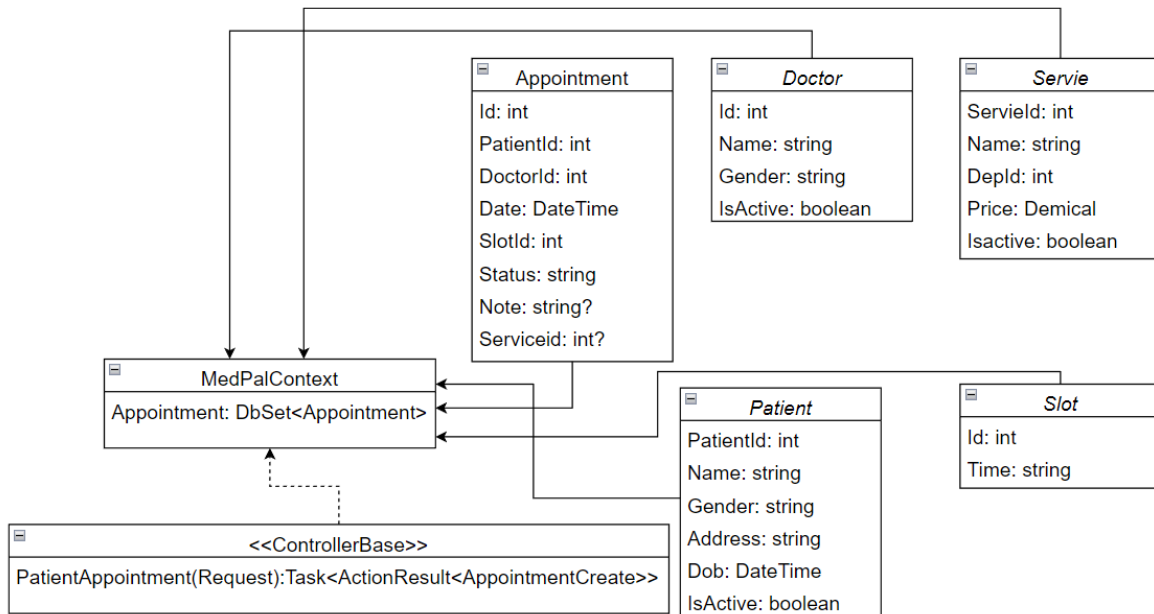


Figure 3.2.3.1.1 Cancel appointment class diagram

3.2.3.2 Sequence Diagram

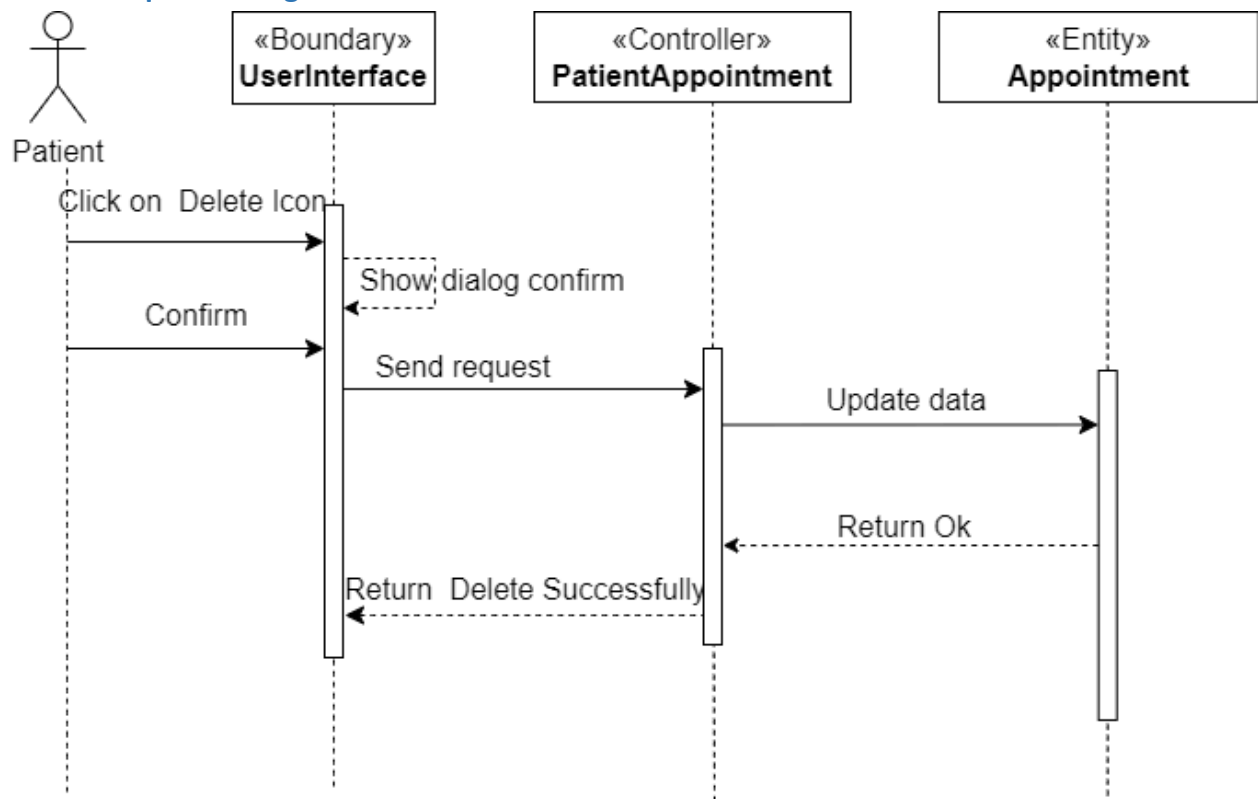


Figure 3.2.3.2.1 Cancel appointment sequence diagram

3.2.4 Approve the Appointment

3.2.4.1 Class Diagram

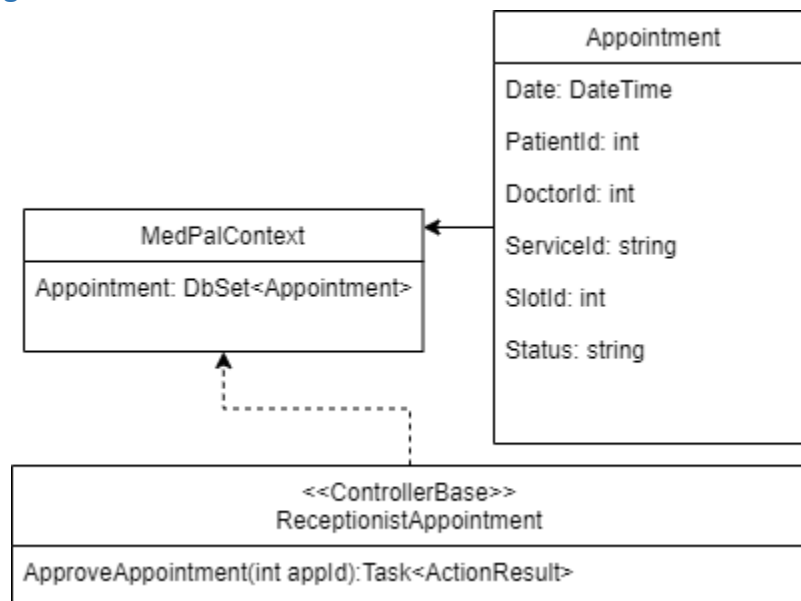


Figure 3.2.4.1.1 Approve the appointment class diagram

3.2.4.2 Sequence Diagram

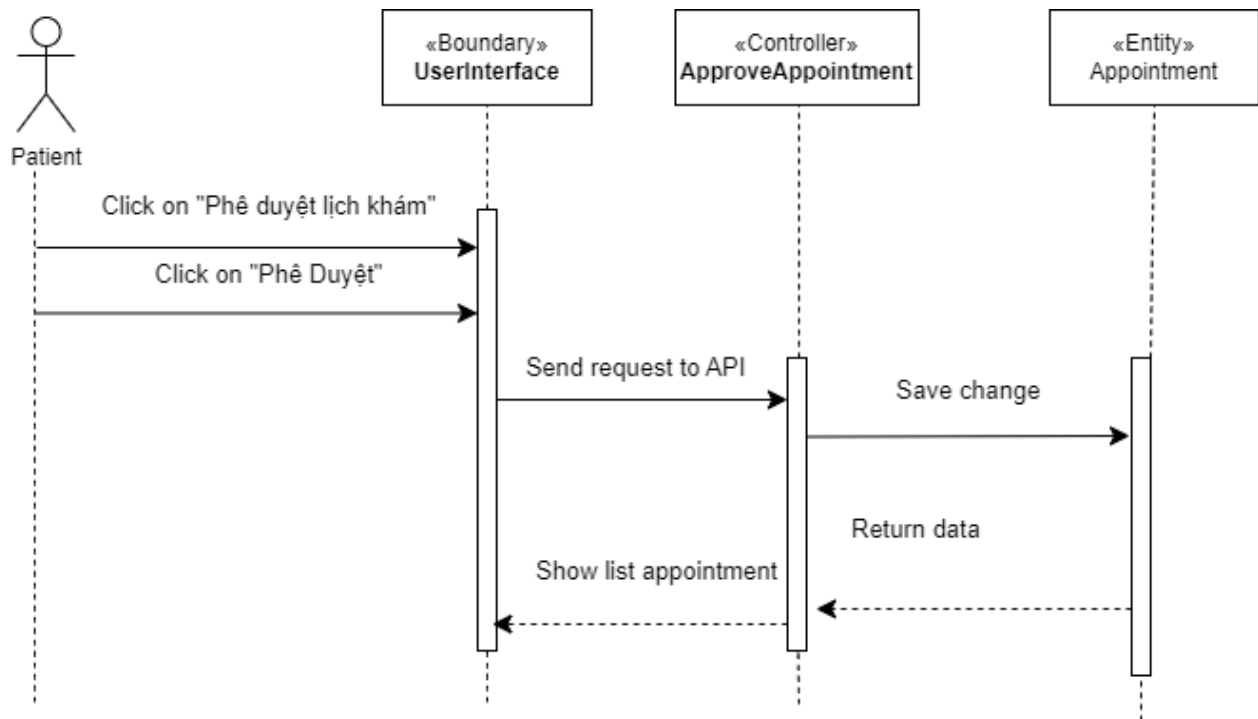


Figure 3.2.4.2.1 Approve the appointment sequence diagram

3.2.5 Create Doctor Schedule

3.2.5.1 Class Diagram

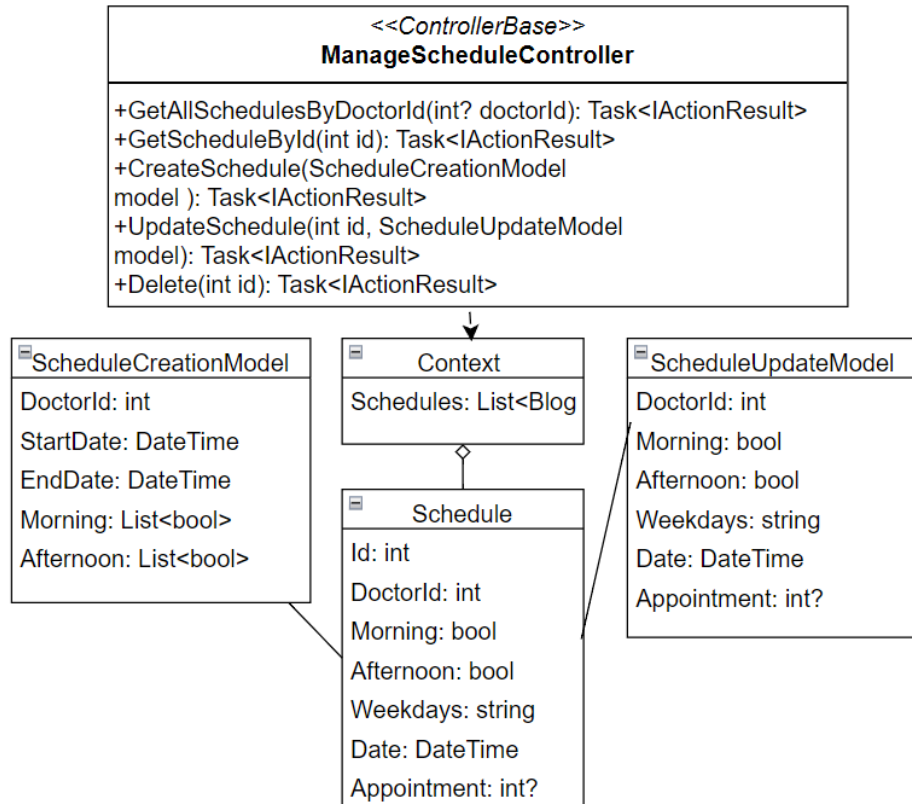


Figure 3.2.5.1.1 Create doctor schedule class diagram

3.2.5.2 Sequence Diagram

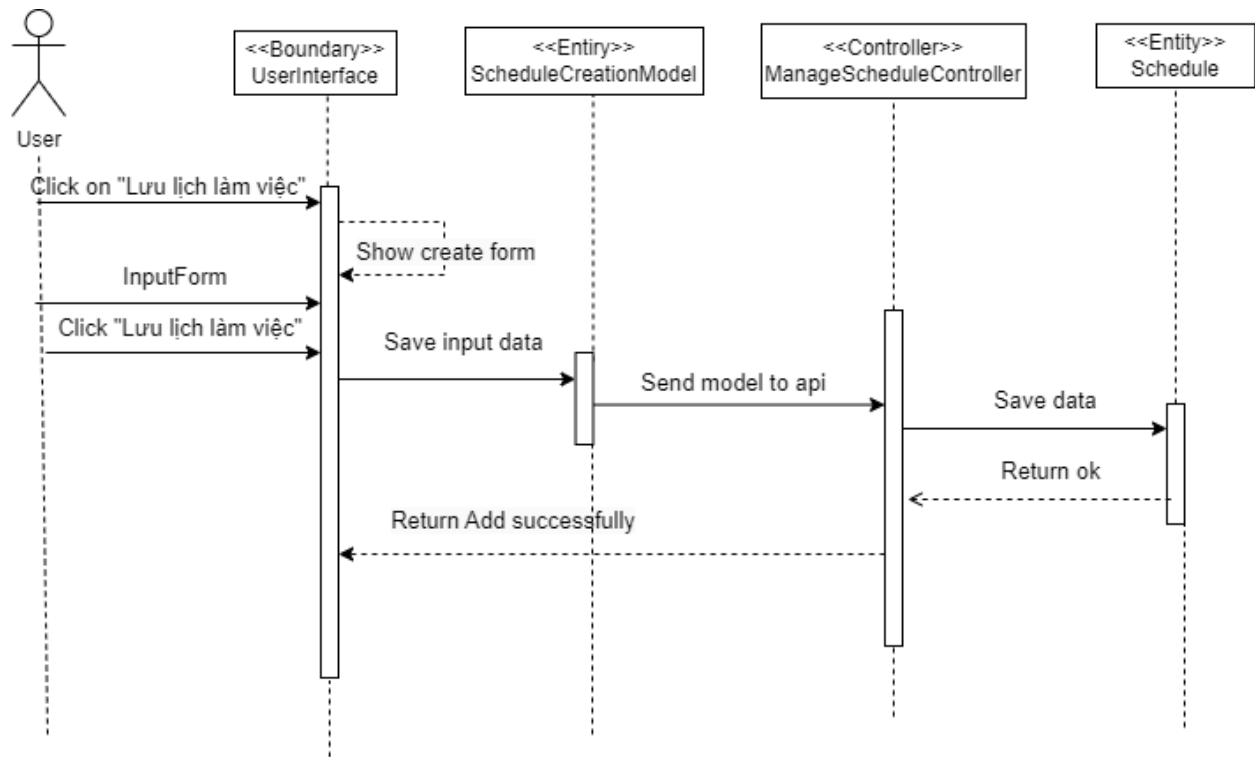


Figure 3.2.5.2.1 Create doctor schedule class diagram

3.2.6 Update Doctor Schedule

3.2.6.1 Class Diagram

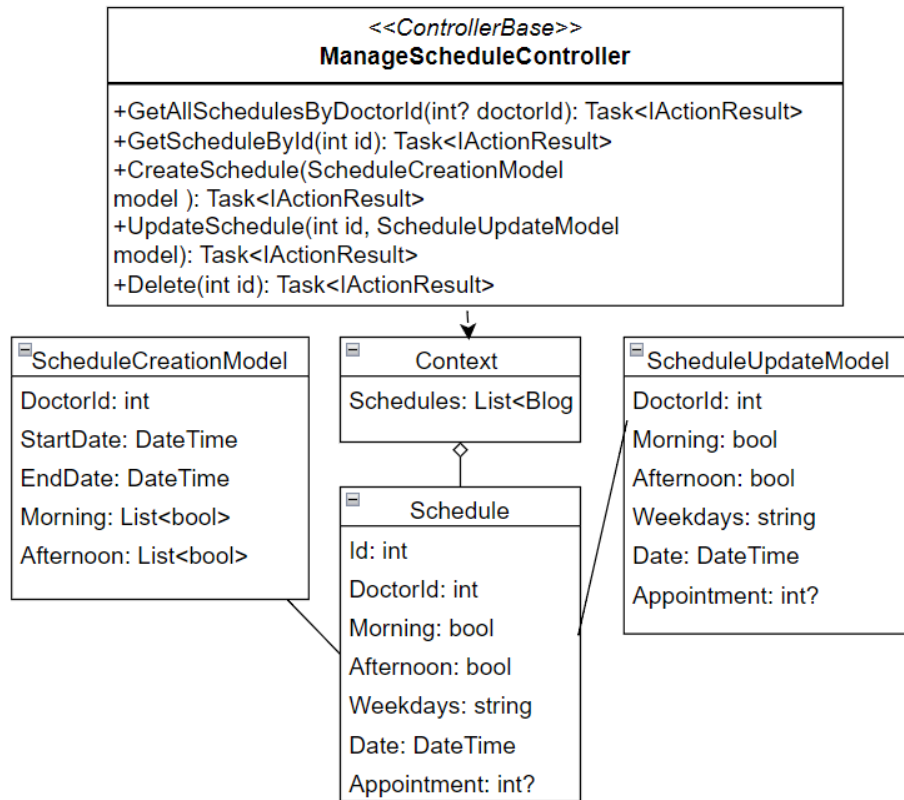


Figure 3.2.6.1.1 Update doctor schedule class diagram

3.2.6.2 Sequence Diagram

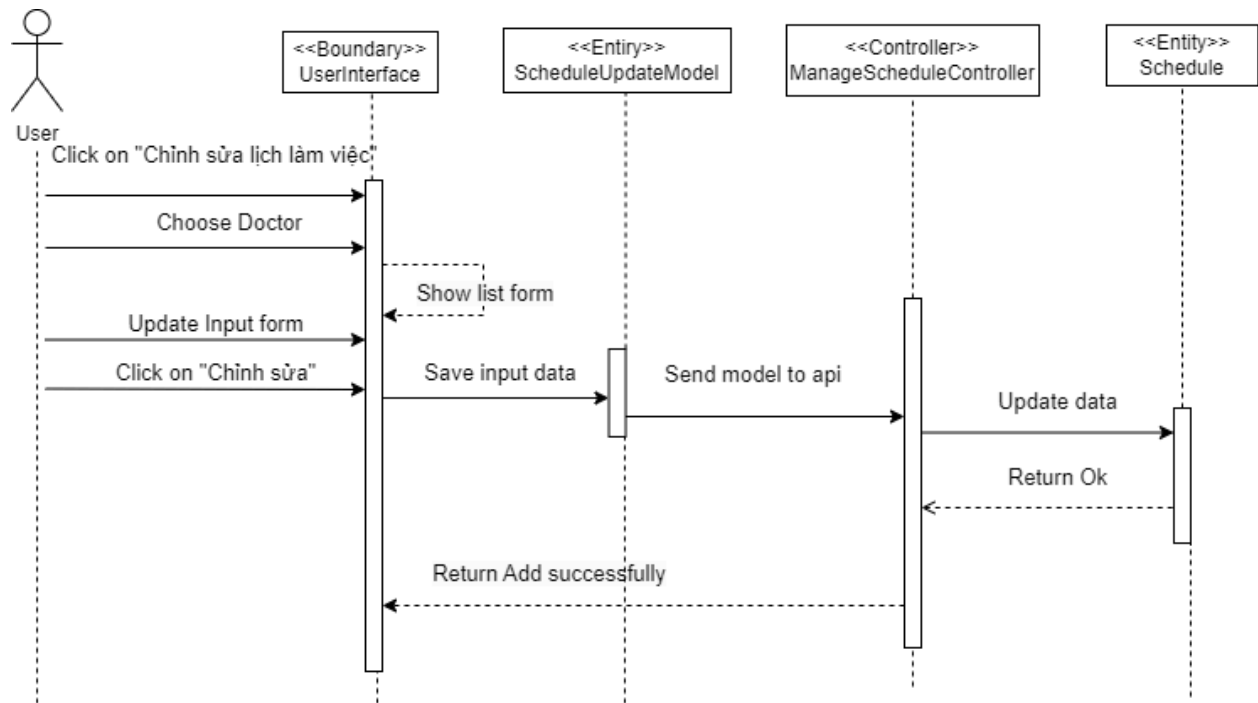


Figure 3.2.6.2.1 Update doctor schedule sequence diagram

3.2.7 Delete Doctor Schedule

3.2.7.1 Class Diagram

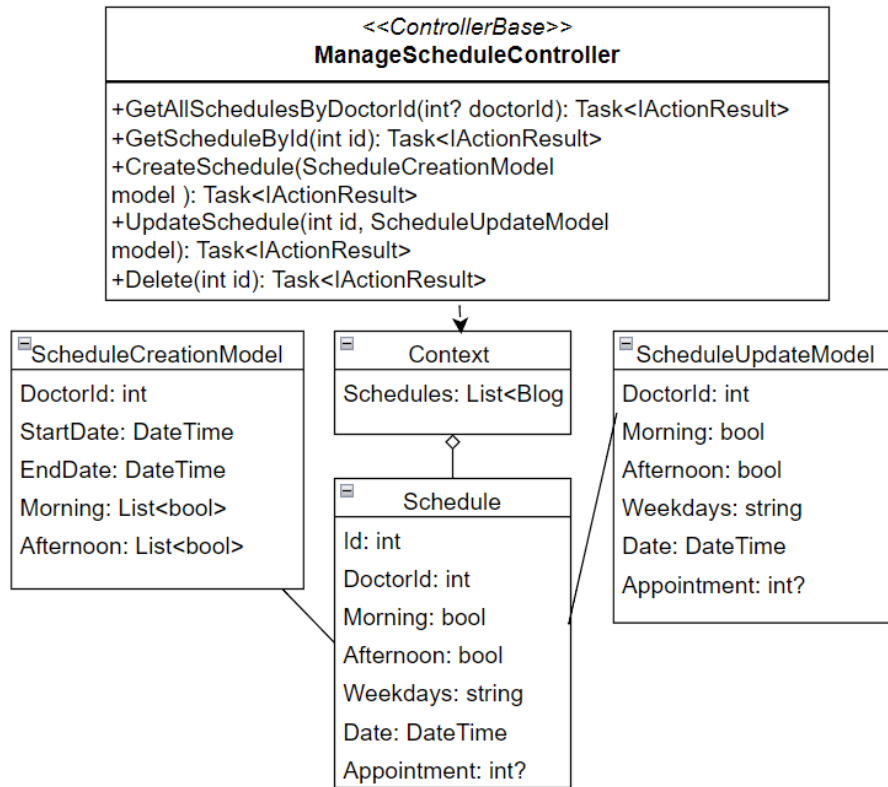


Figure 3.2.7.1.1 Delete doctor schedule class diagram

3.2.7.2 Sequence Diagram

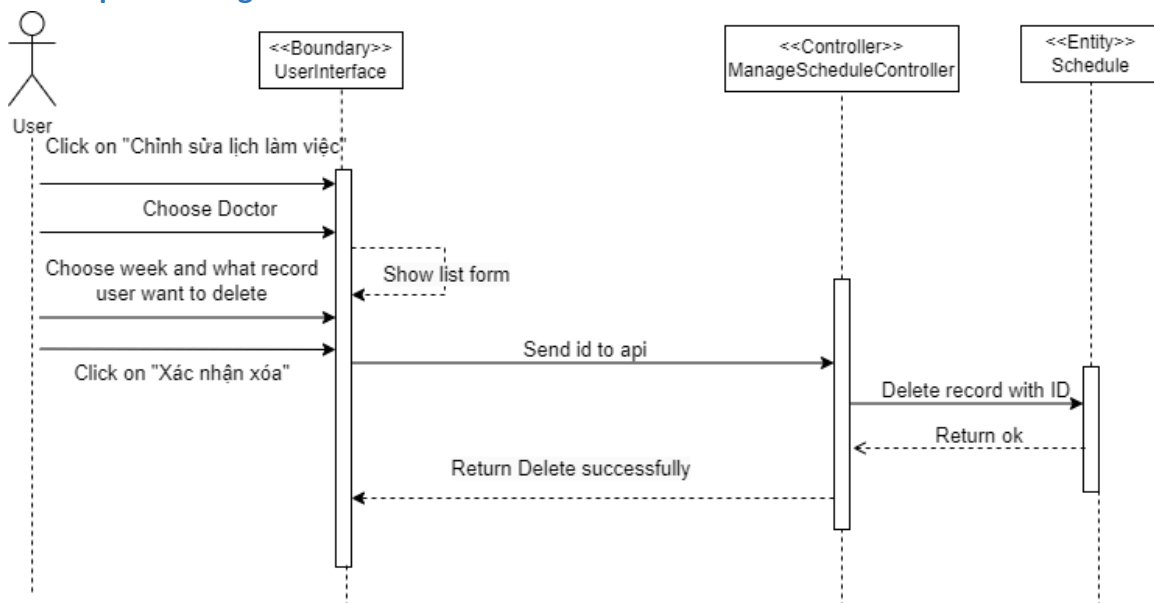


Figure 3.2.7.2.1 Delete doctor schedule sequence diagram

3.2.8 View Doctor Schedule

3.2.8.1 Class Diagram

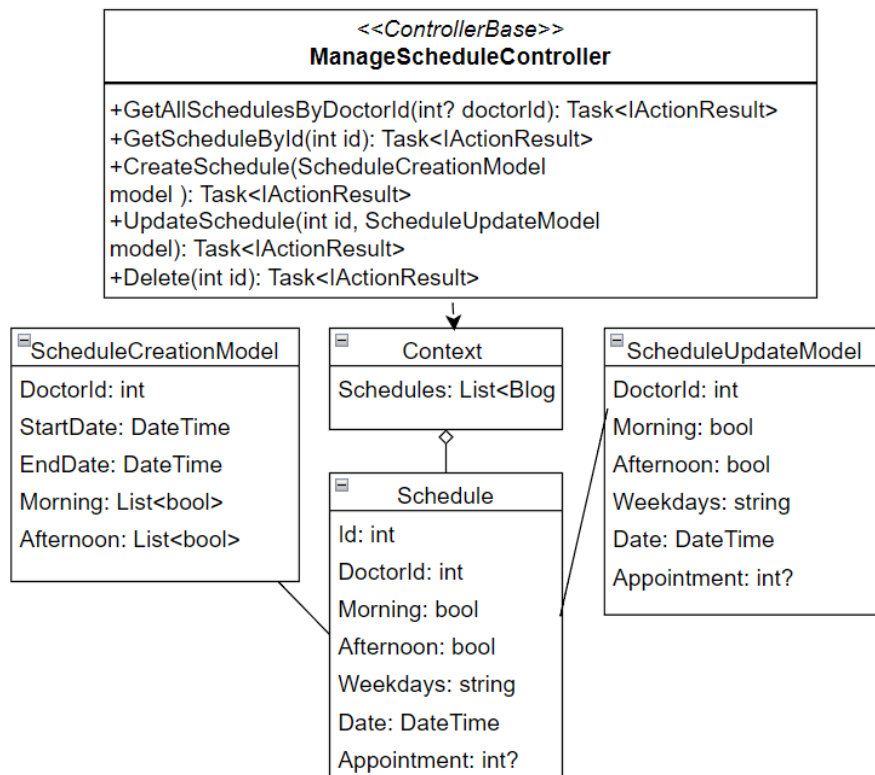


Figure 3.2.8.1.1 View doctor schedule class diagram

3.2.8.2 Sequence Diagram

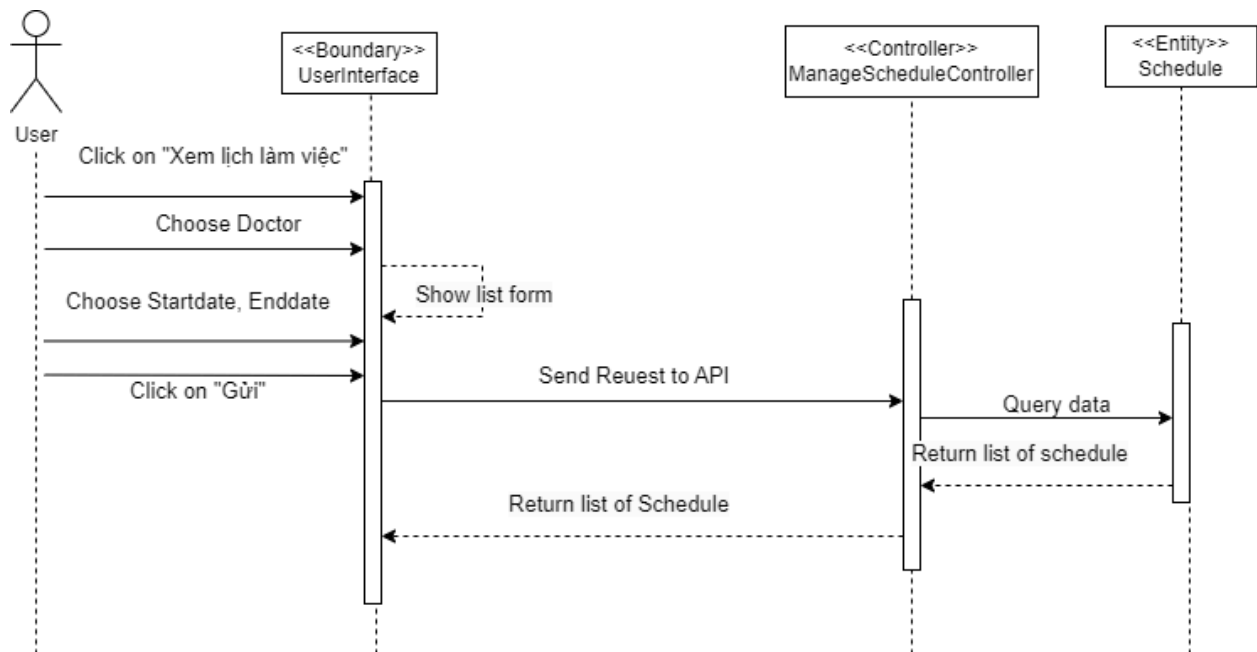


Figure 3.2.8.2.1 View doctor schedule Sequence diagram

3.2.9 View Appointment(Admin)

3.2.9.1 Class Diagram

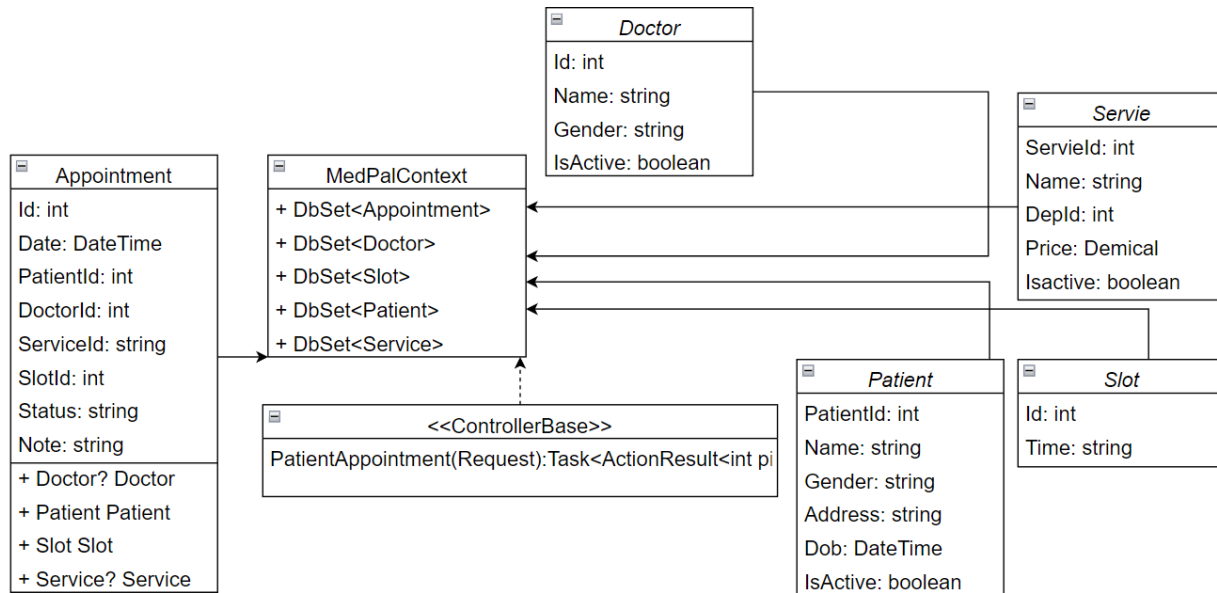


Figure 3.2.9.1.1 View appointment class diagram

3.2.9.2 Sequence Diagram

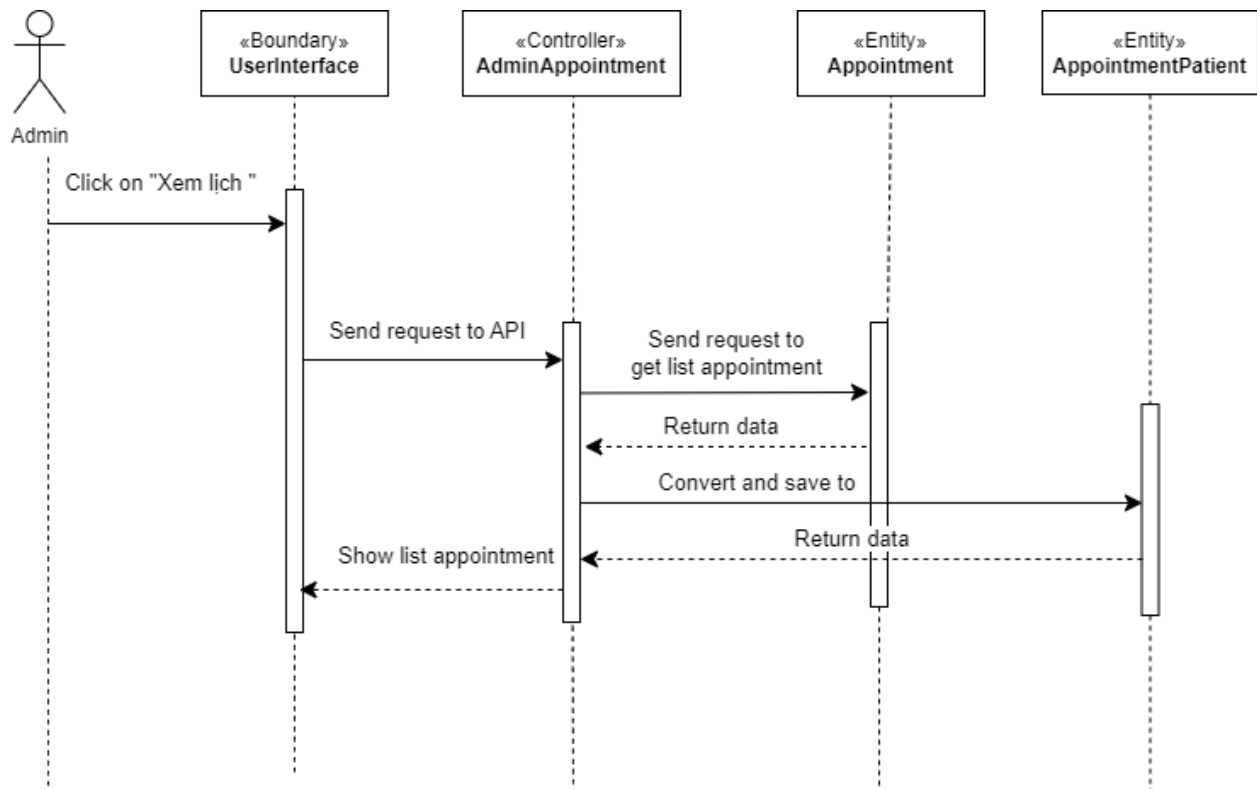


Figure 3.2.9.2.1 View appointment class diagram

3.2.10 Schedules follow-up appointment

3.2.10.1 Class Diagram

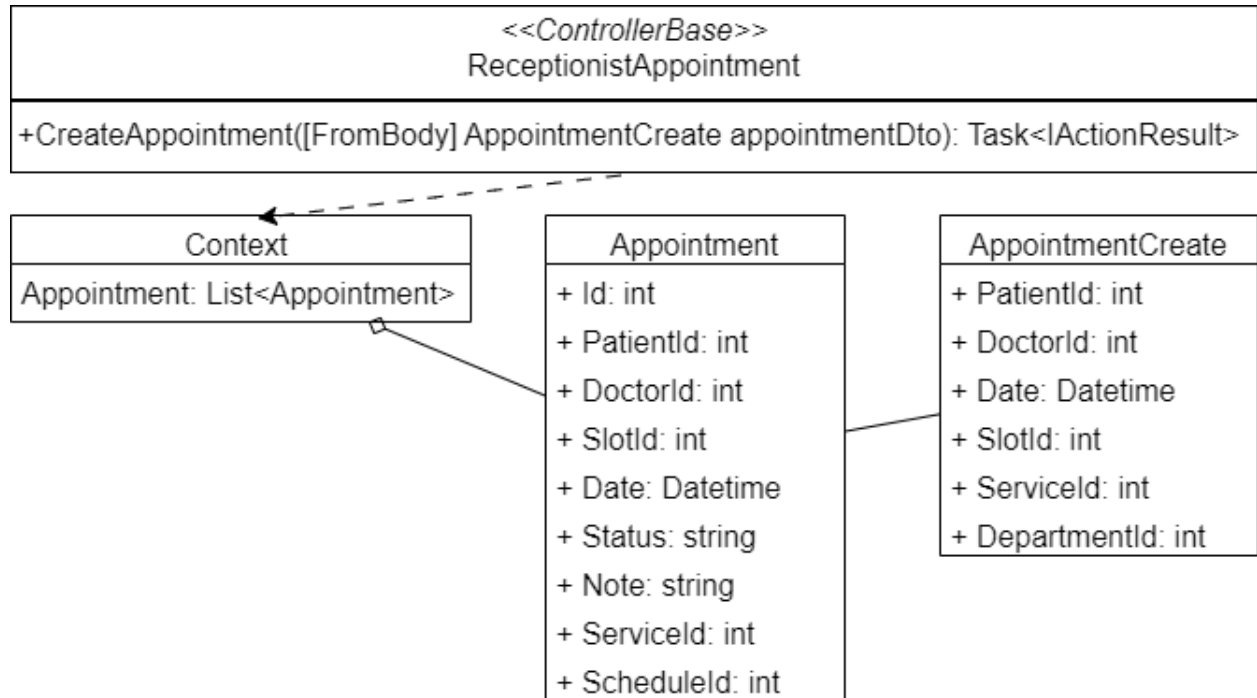


Figure 3.2.10.1.1 Schedule follow-up appointment class diagram

3.2.10.2 Sequence Diagram

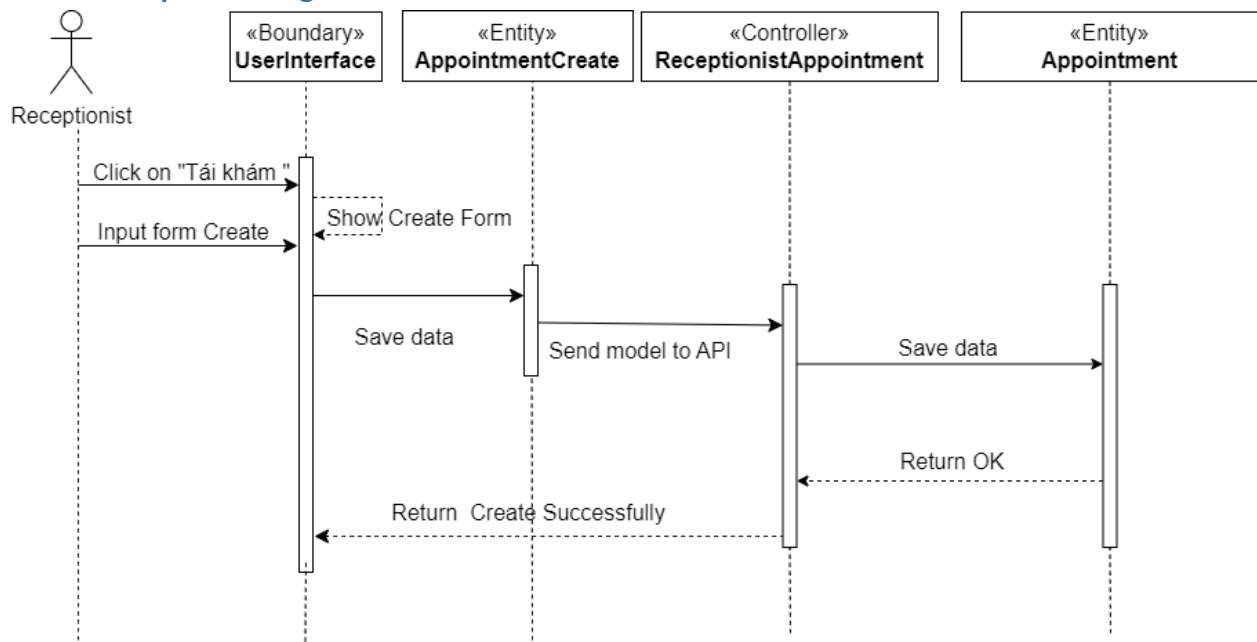


Figure 3.2.10.2.1 Schedule follow-up appointment sequence diagram

3.3 Marketing

3.3.1 Create Blog Medical

3.3.1.1 Class Diagram

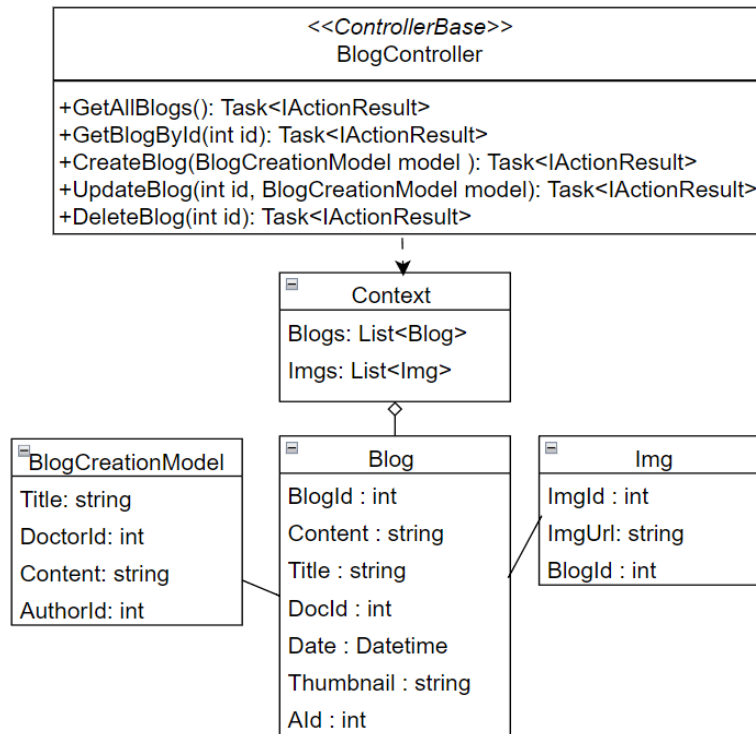


Figure 3.3.1.1.1 Create blog medical class diagram

3.3.1.2 Sequence Diagram

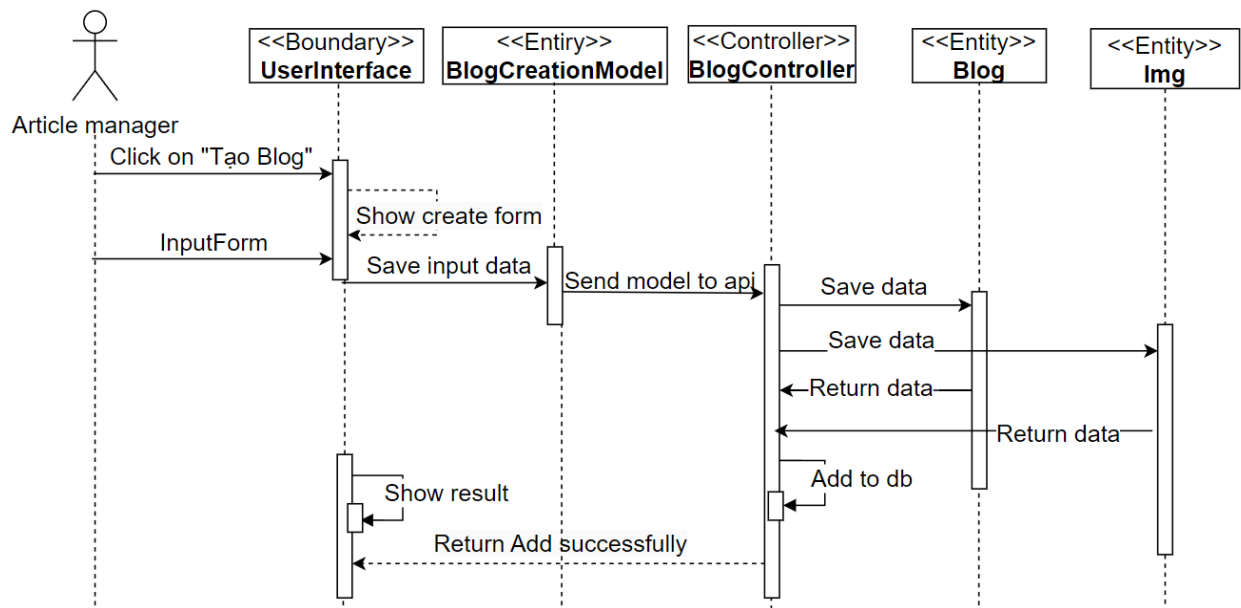


Figure 3.3.1.2.1 Create blog medical sequence diagram

3.3.2 Update Blog Medical

3.3.2.1 Class Diagram

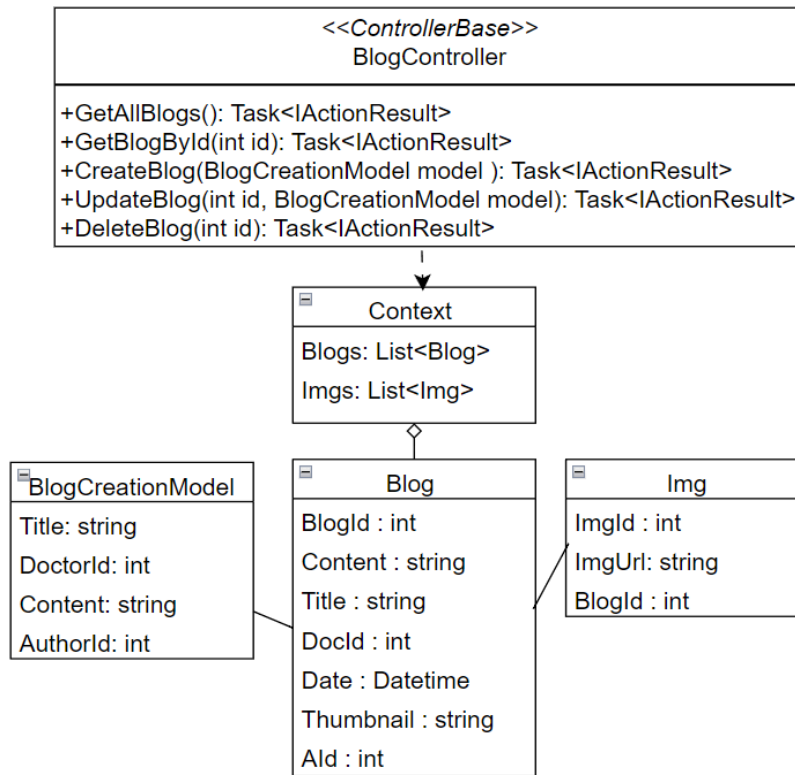


Figure 3.3.2.1.1 Update blog medical class diagram

3.3.2.2 Sequence Diagram

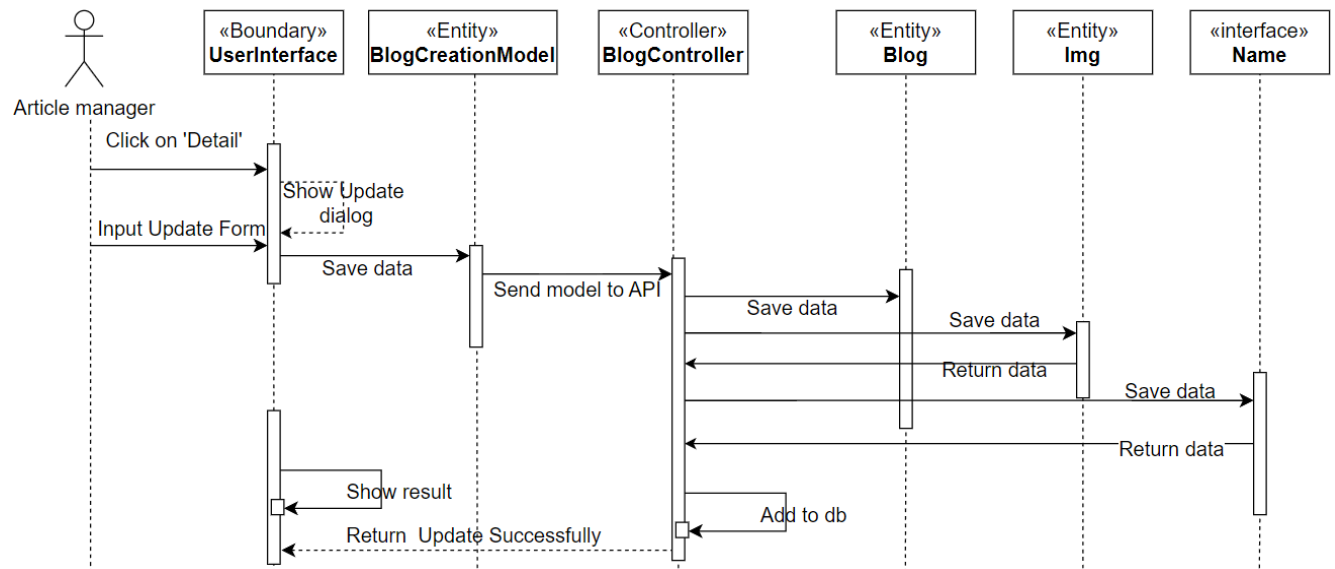


Figure 3.3.2.2.1 Update blog medical sequence diagram

3.3.3 Delete Blog Medical

3.3.3.1 Class Diagram

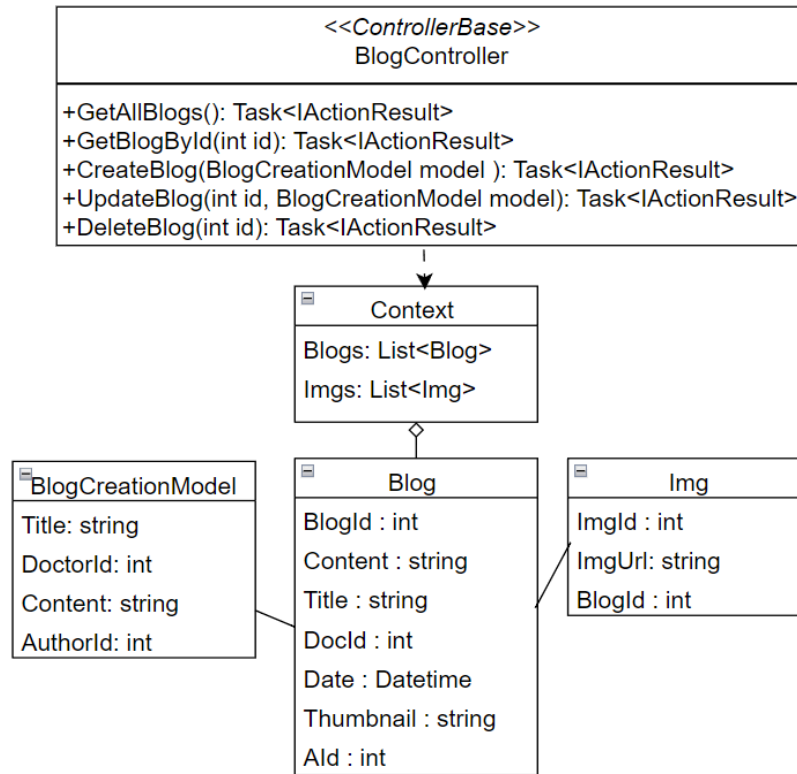


Figure 3.3.3.1.1 Delete blog medical class diagram

3.3.3.2 Sequence Diagram

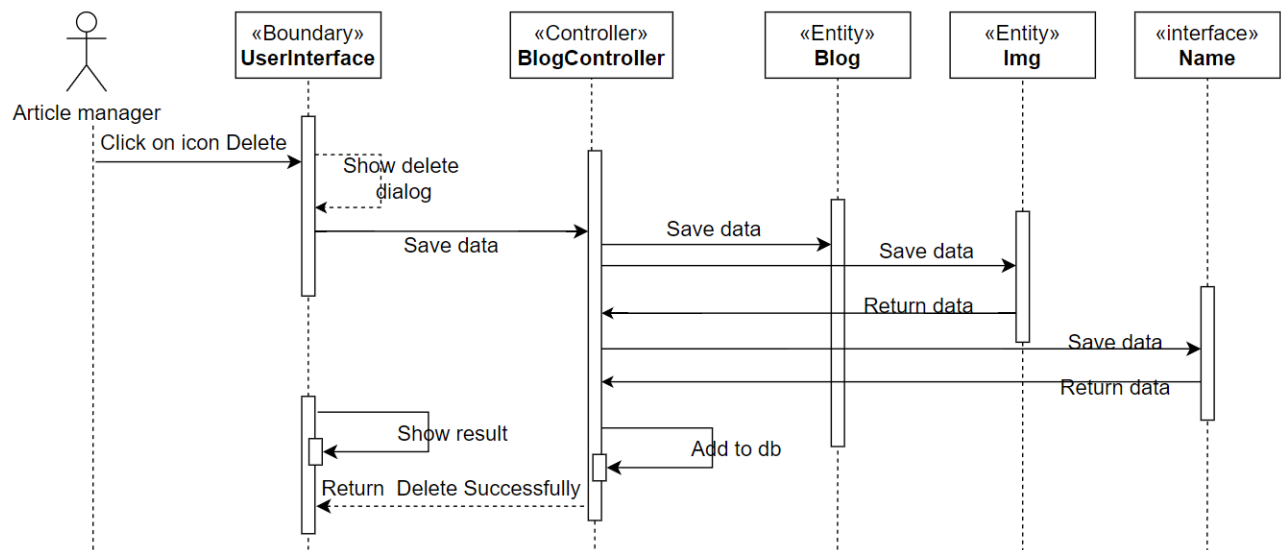


Figure 3.3.3.2.1 Delete blog medical sequence diagram

3.3.4 View Blog Medical

3.3.4.1 Class Diagram

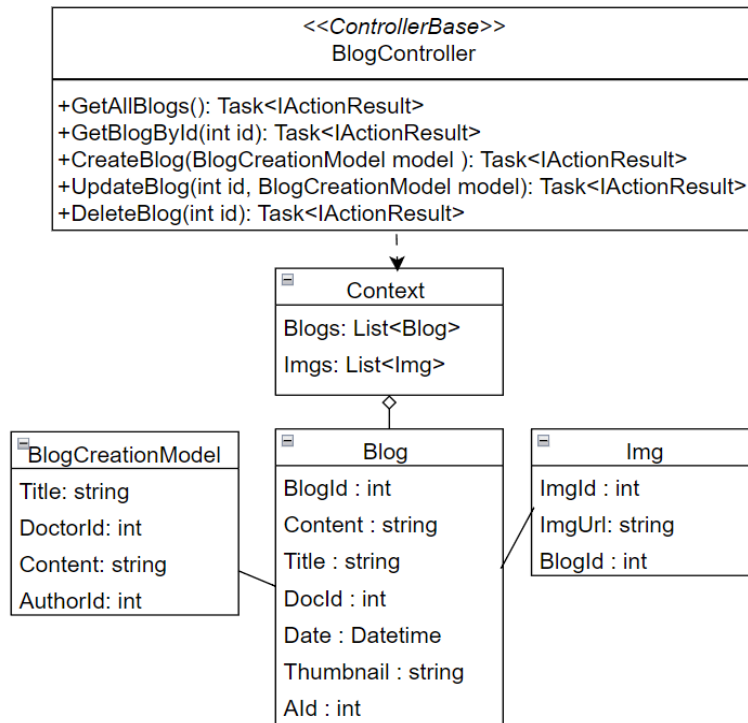


Figure 3.3.4.1.1 View blog medical class diagram

3.3.4.2 Sequence Diagram

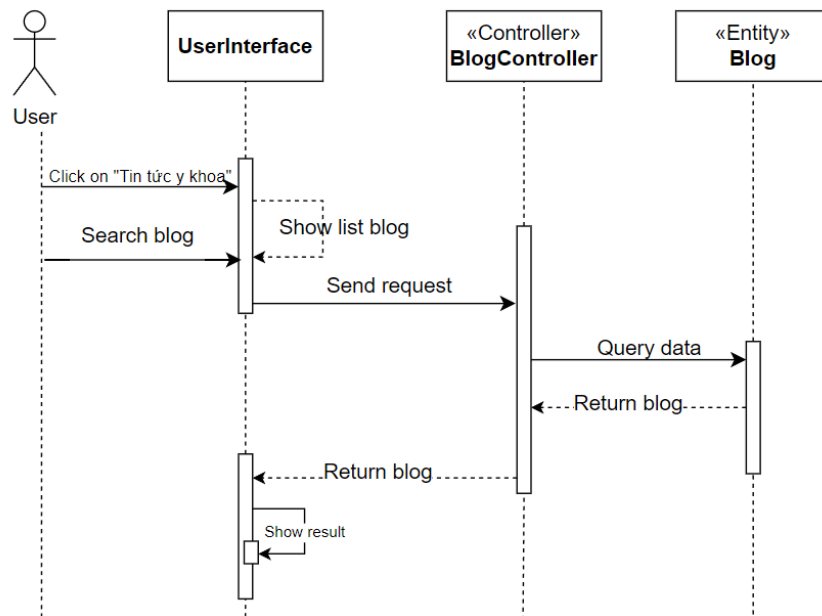


Figure 3.3.4.2.1 View blog medical sequence diagram

3.3.5 Search blog

3.3.5.1 Class Diagram

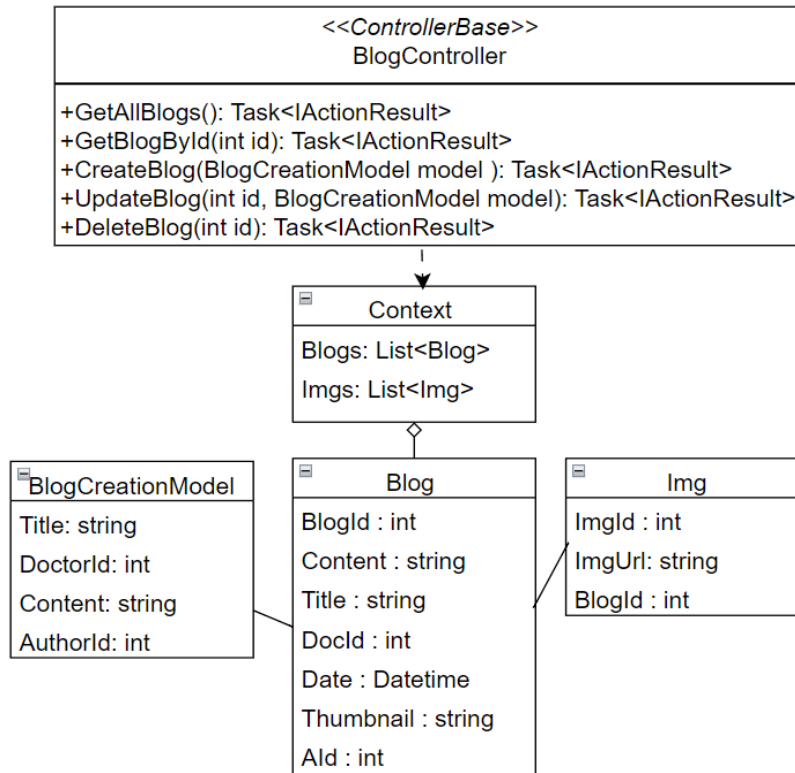


Figure 3.3.5.1.1 Search blog medical class diagram

3.3.5.2 Sequence Diagram

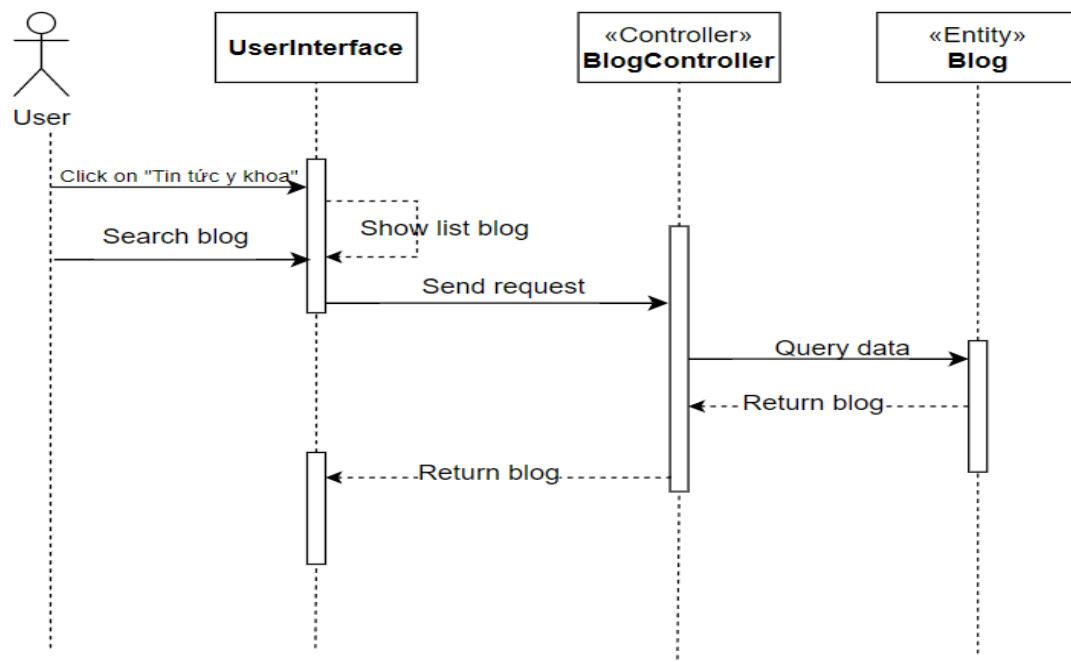


Figure 3.3.5.2.1 Search blog medical sequence diagram

3.3.6 View list of services

3.3.6.1 Class Diagram

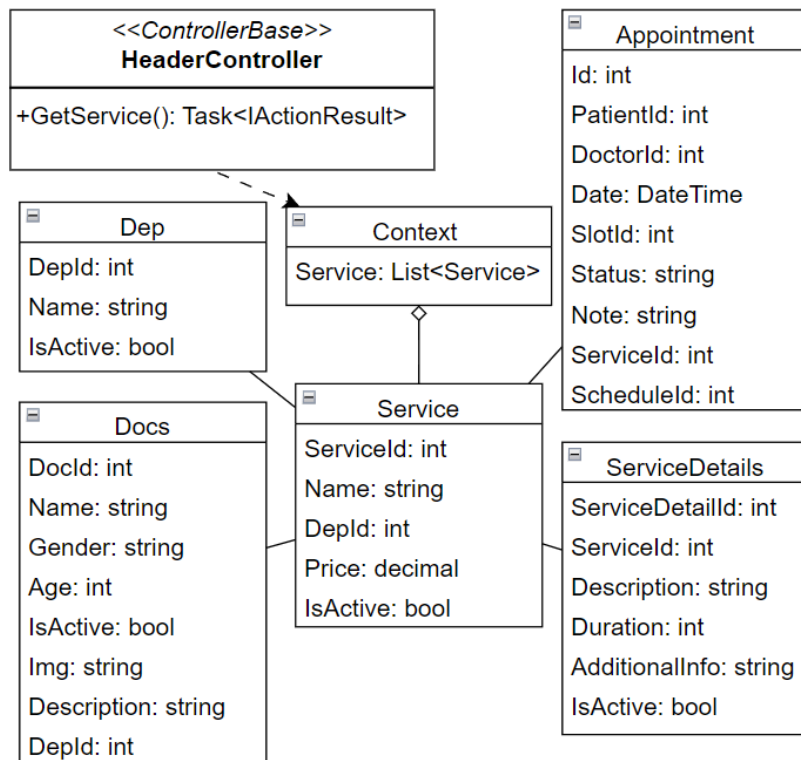


Figure 3.3.6.1.1 View list of services class diagram

3.3.6.2 Sequence Diagram

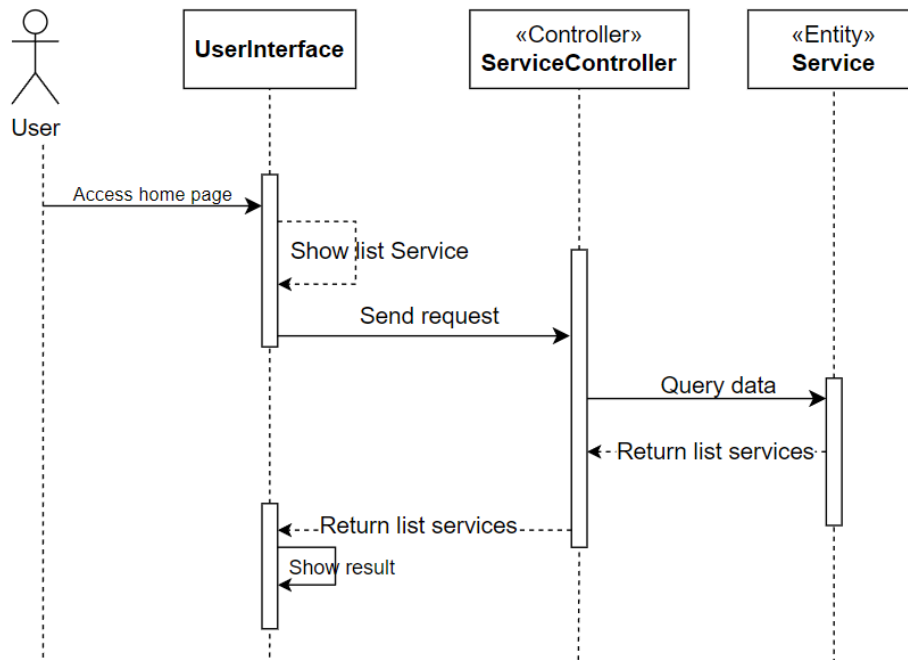


Figure 3.3.6.2.1 View list of services sequence diagram

3.3.7 View service detail

3.3.7.1 Class Diagram

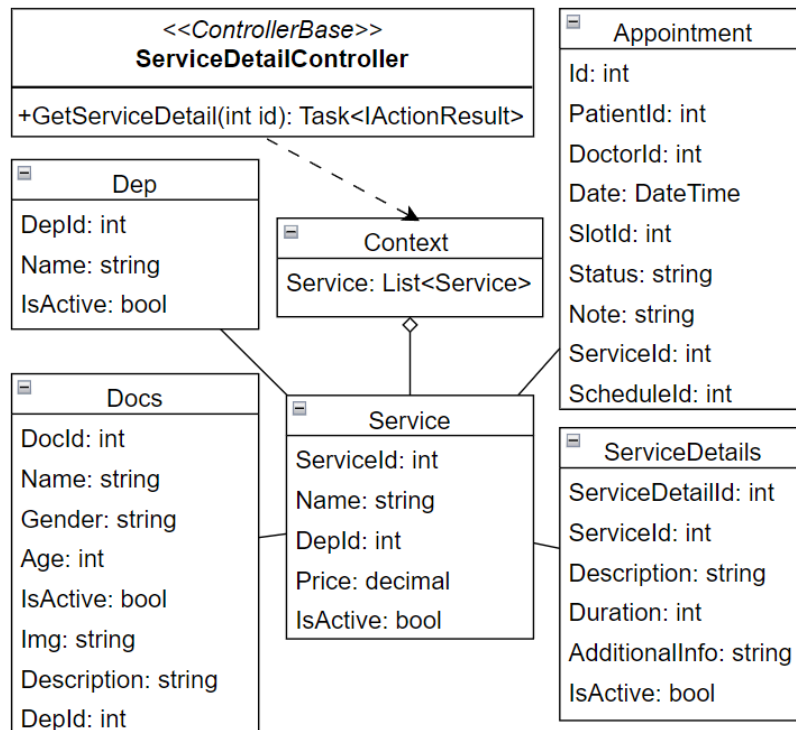


Figure 3.3.7.1.1 View service detail class diagram

3.3.7.2 Sequence Diagram

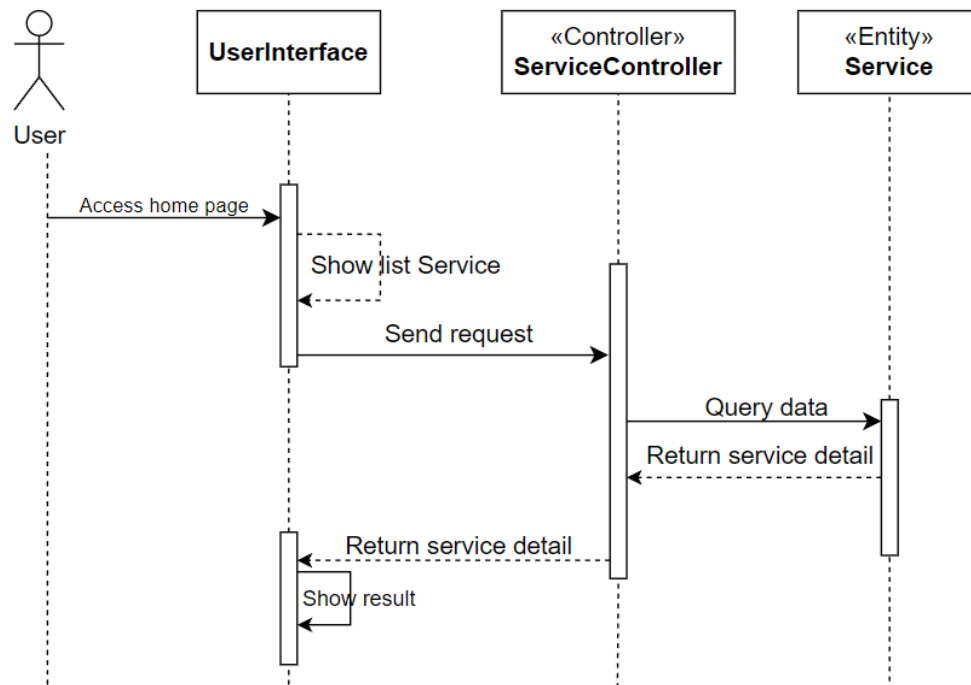


Figure 3.3.7.2.1 View service detail sequence diagram

3.3.8 View Doctor's Information

3.3.8.1 Class Diagram

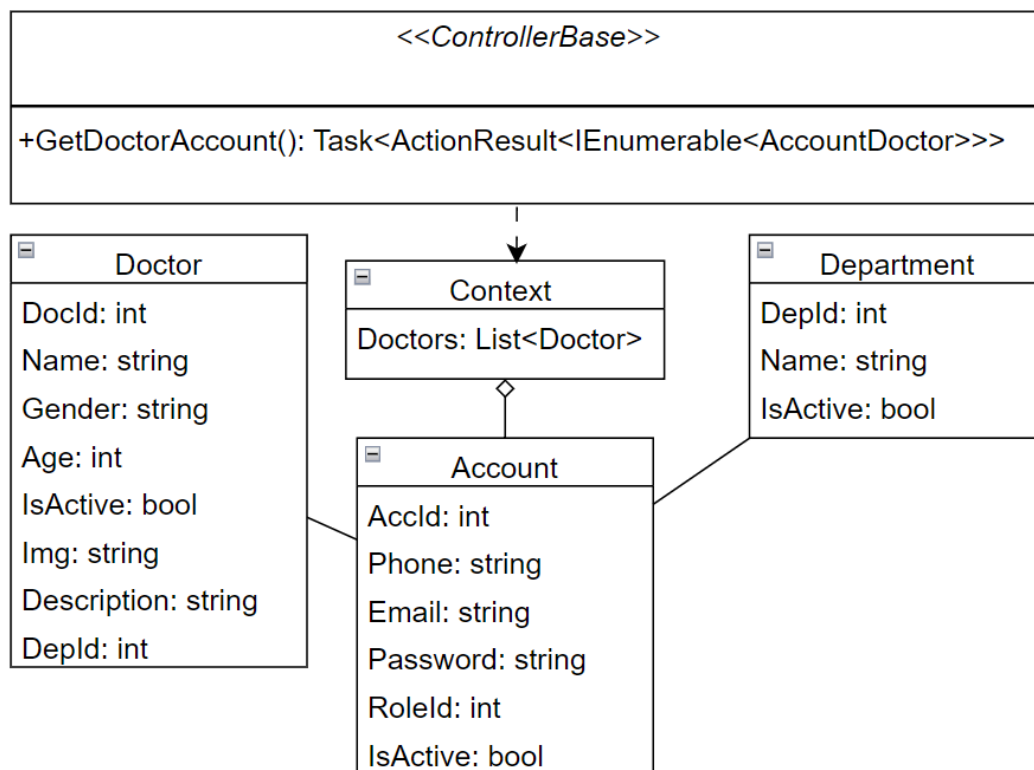


Figure 3.3.8.1.1 View doctor's information class diagram

3.3.8.2 Sequence Diagram

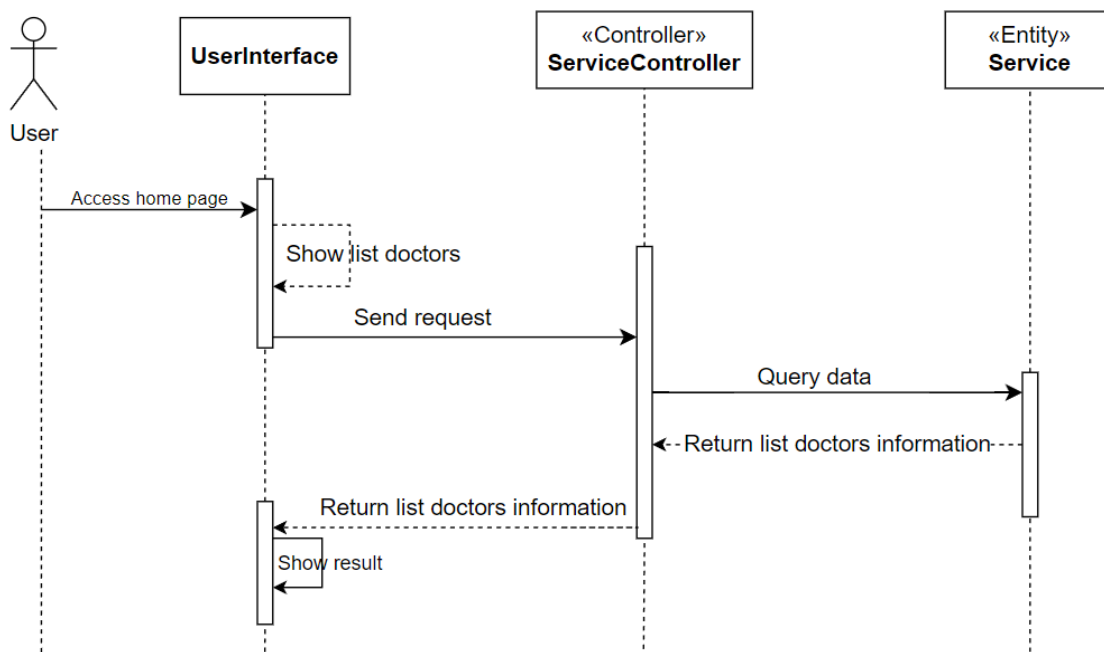


Figure 3.3.8.2.1 View doctor's information sequence diagram

3.3.9 View List of Doctors

3.3.9.1 Class Diagram

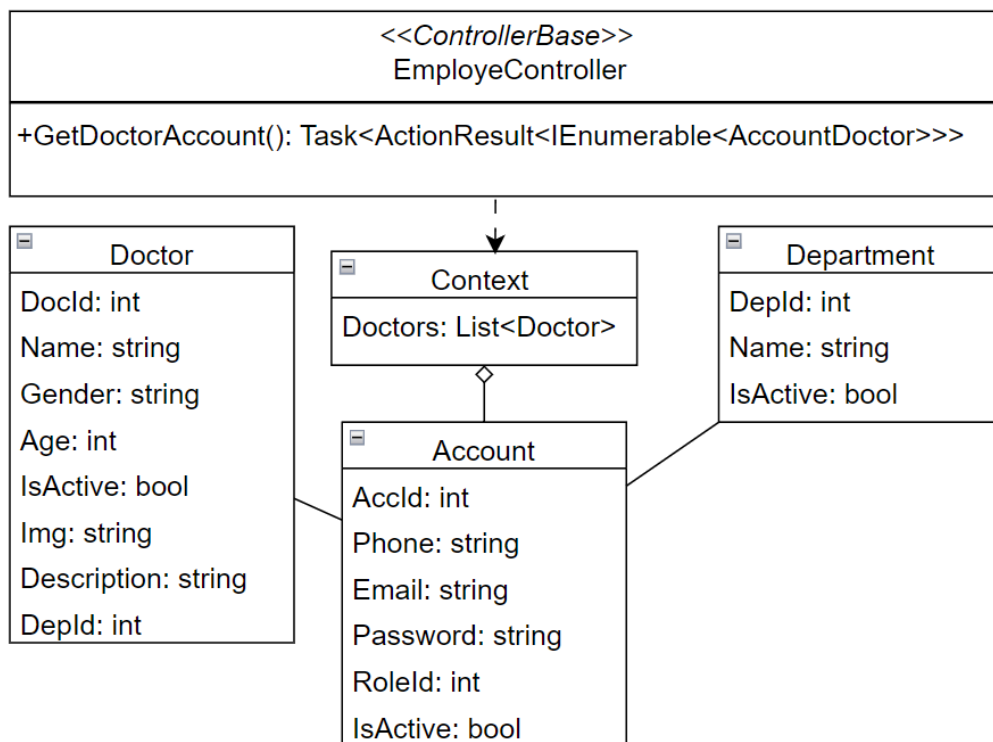


Figure 3.3.9.1.1 View list doctors class diagram

3.3.9.2 Sequence Diagram

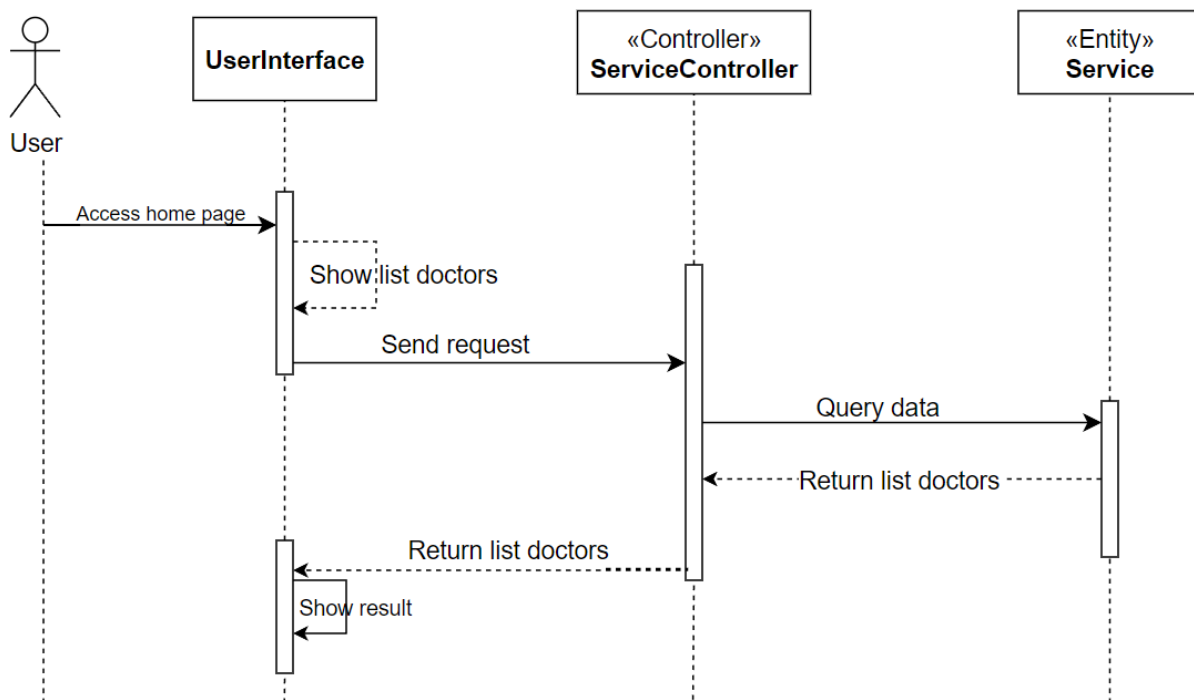


Figure 3.3.9.2.1 View list doctors sequence diagram

3.3.10 Update answer

3.3.10.1 Class Diagram

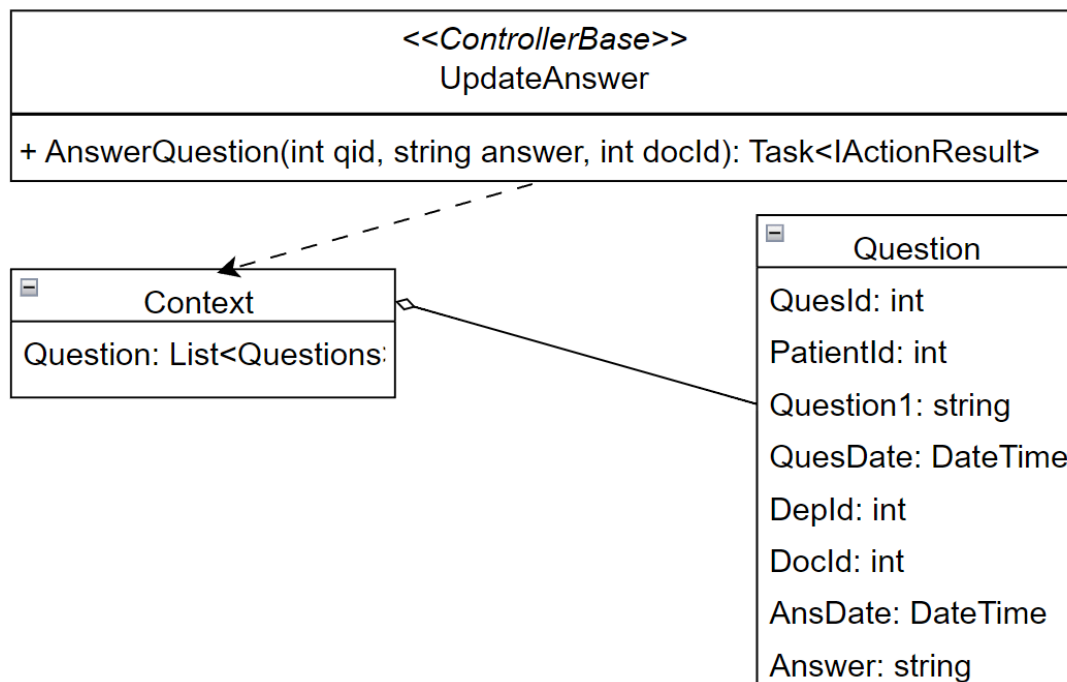


Figure 3.3.10.1.1 Update answer class diagram

3.3.10.2 Sequence Diagram

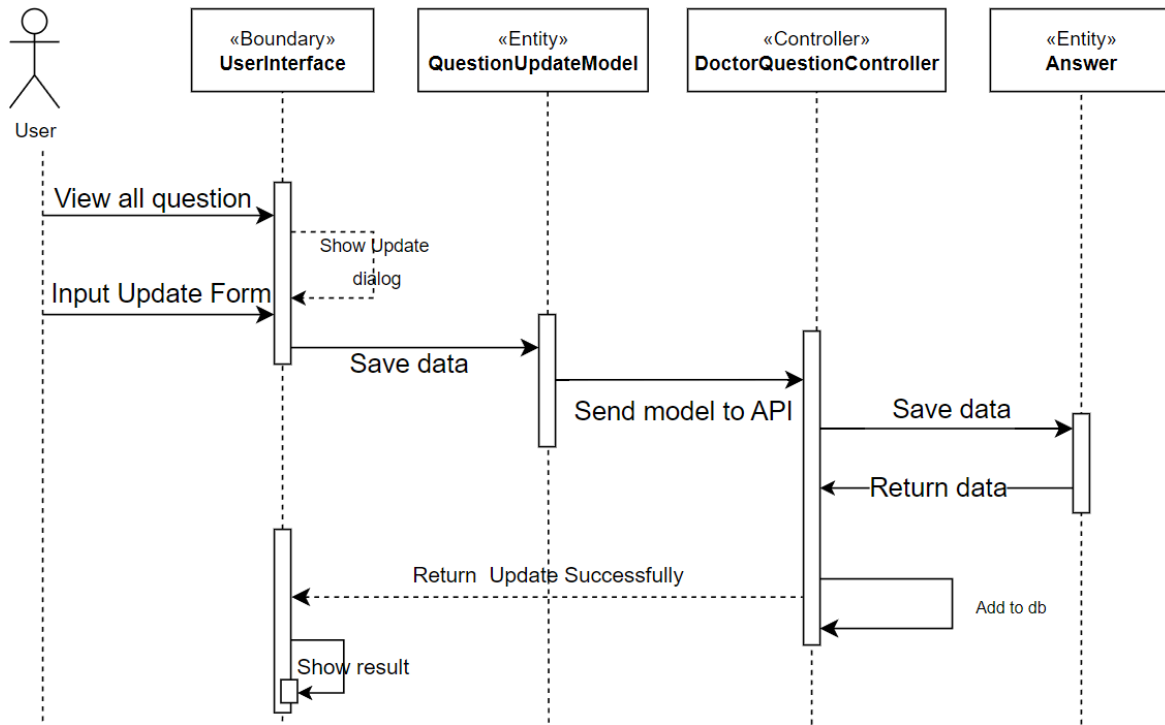


Figure 3.3.10.2.1 Update answer sequence diagram

3.3.11 Update question

3.3.11.1 Class Diagram

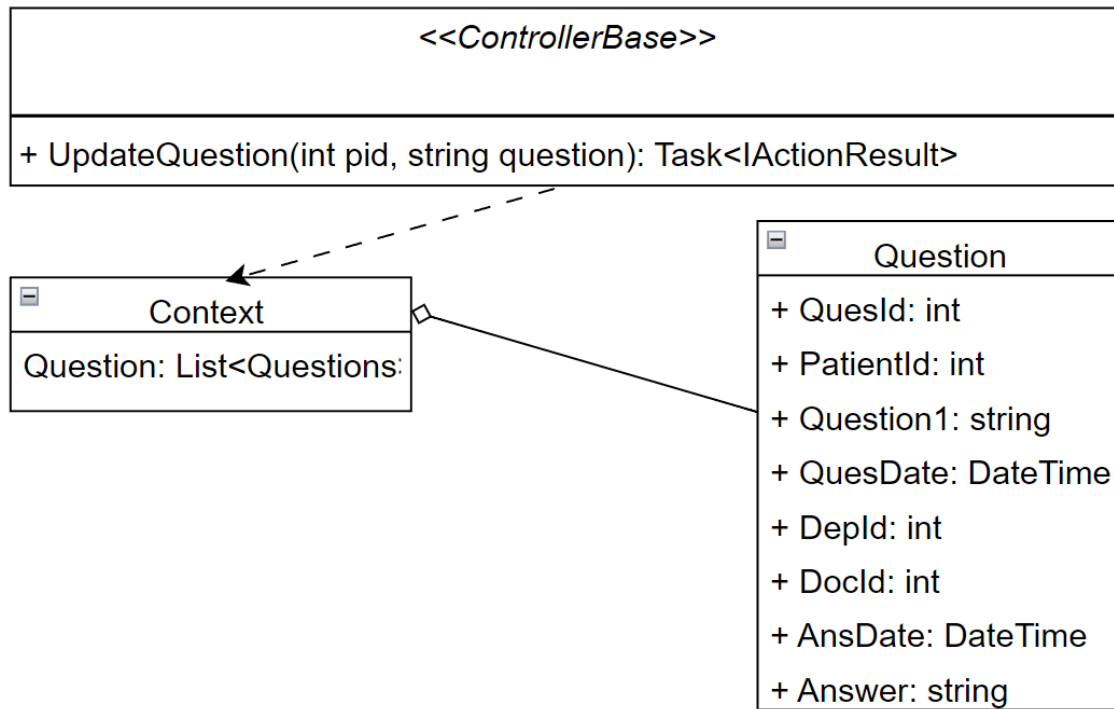


Figure 3.3.11.1.1 Update question class diagram

3.3.11.2 Sequence Diagram

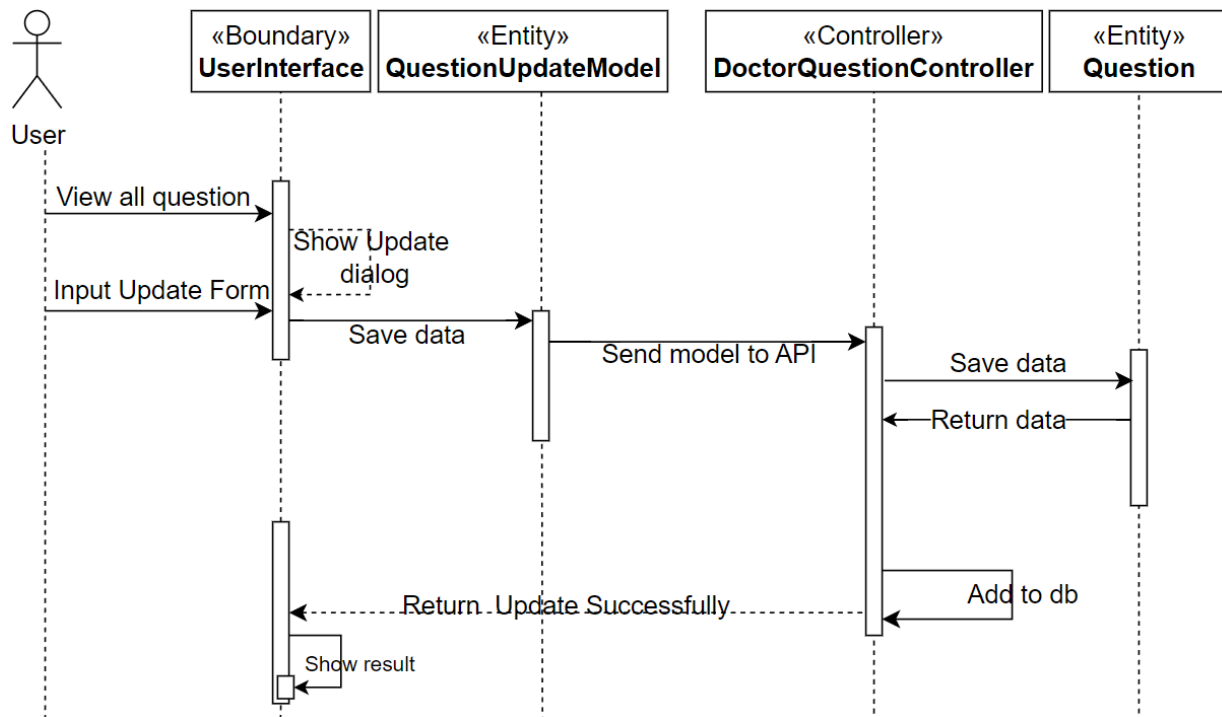


Figure 3.3.11.2.1 Update question sequence diagram

3.3.12 Create Question for doctor

3.3.12.1 Class Diagram

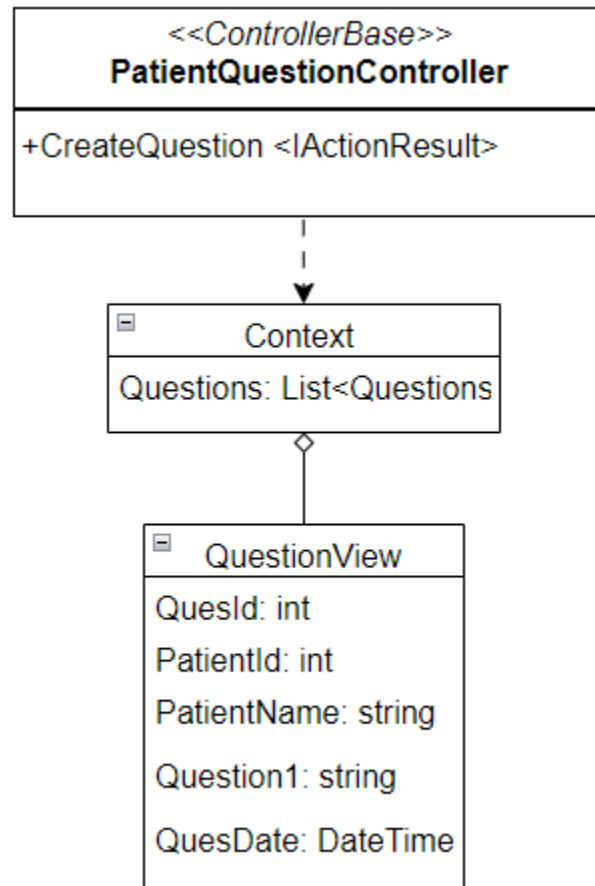


Figure 3.3.12.1.1 Create question for doctor class diagram

3.3.12.2 Sequence Diagram

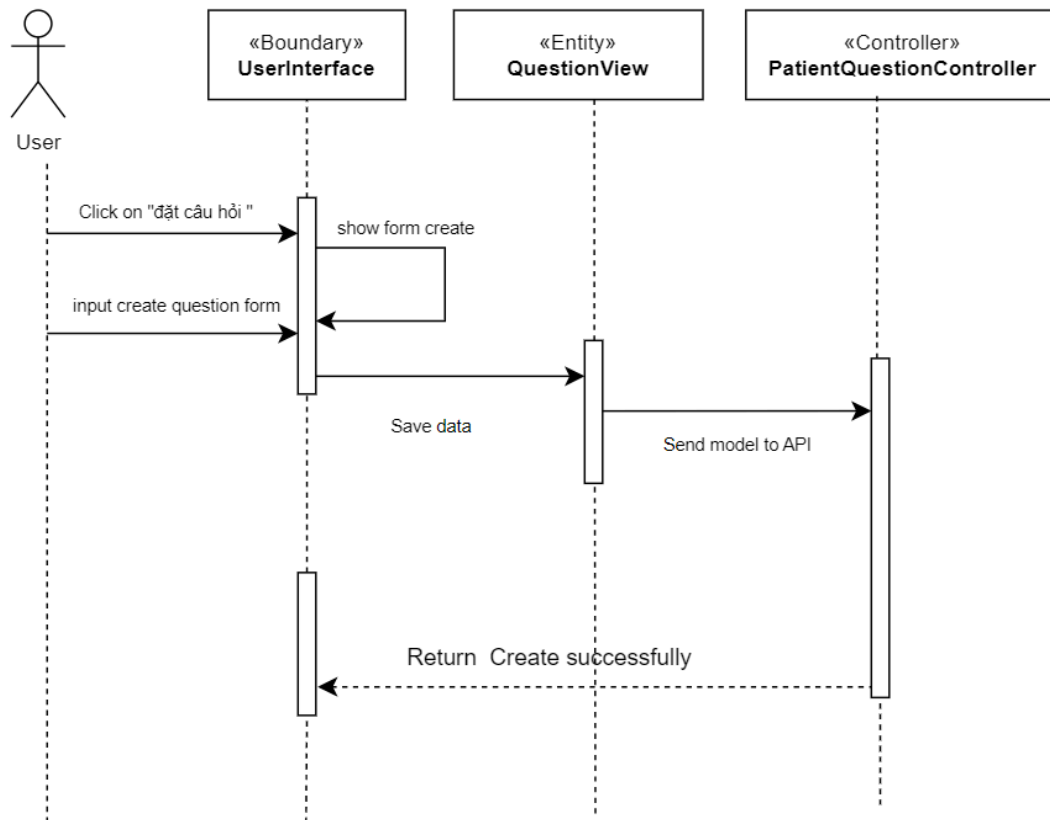


Figure 3.3.12.2.1 Create question for doctor sequence diagram

3.3.13 View customer feedback

3.3.13.1 Class Diagram

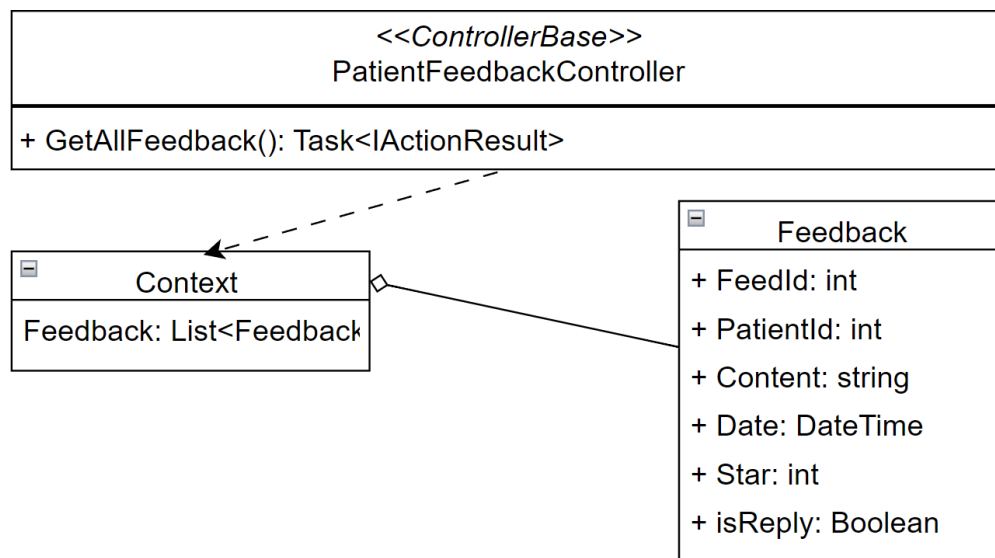


Figure 3.3.13.1.1 View customer feedback class diagram

3.3.13.2 Sequence Diagram

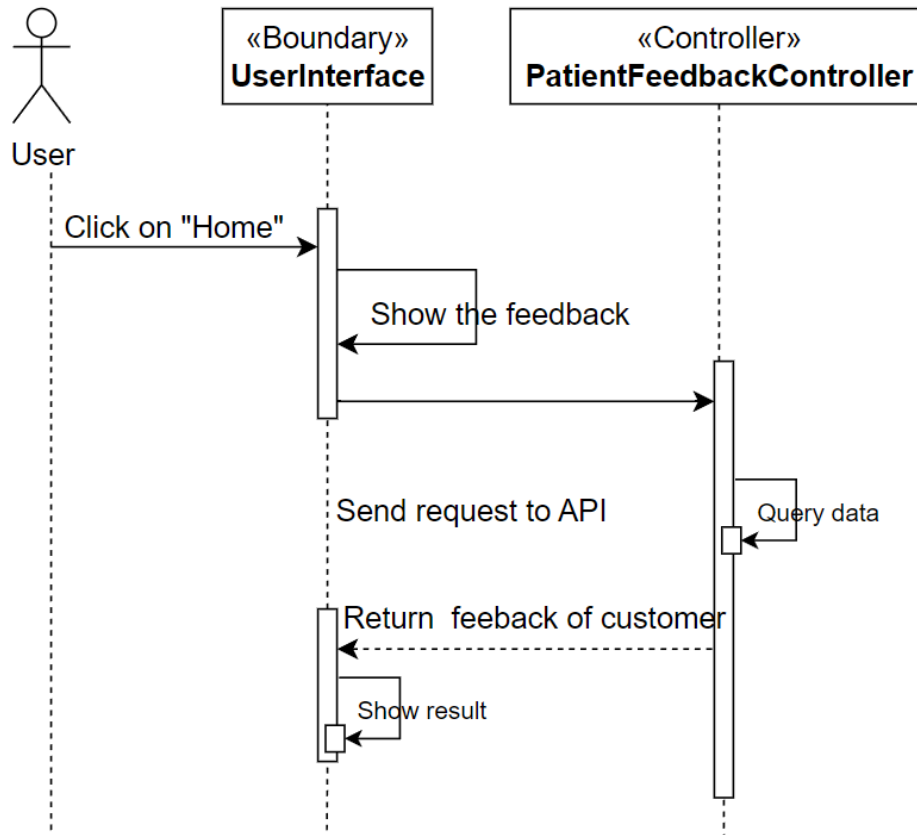


Figure 3.3.13.2.1 View customer feedback sequence diagram

3.4 Medical Notebooks

3.4.1 Doctor View Medical Notebooks

3.4.1.1 Class Diagram

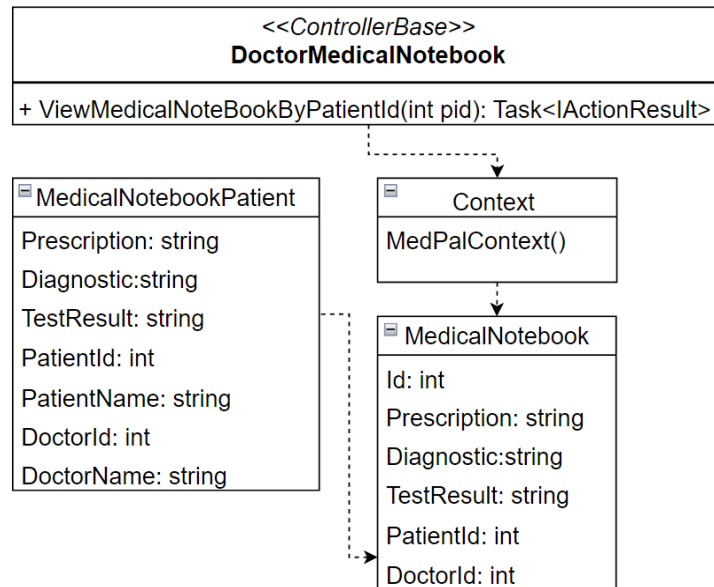


Figure 3.4.1.1: Doctor view medical notebook class diagram

3.4.1.2 Sequence Diagram

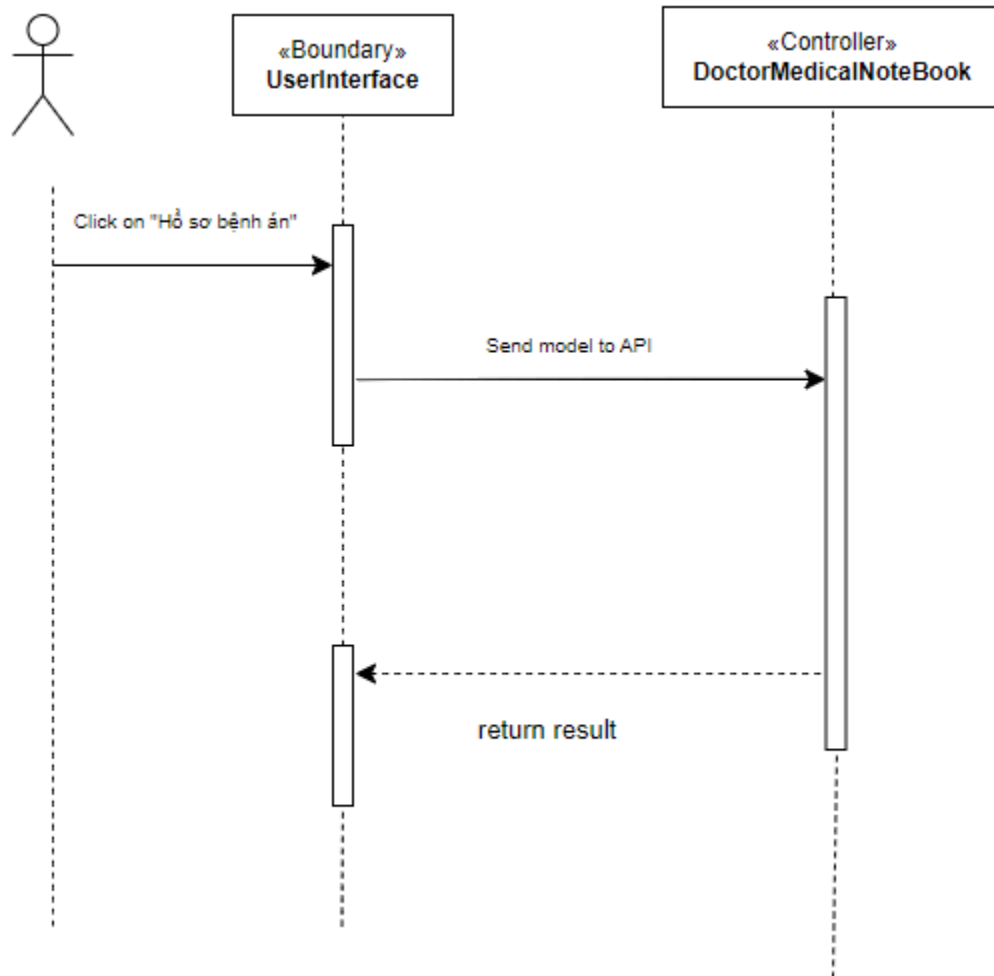


Figure 3.4.1.2.1 Doctor view medical notebook sequence diagram

3.4.2 Receptionist View Medical Notebooks

3.4.2.1 Class Diagram

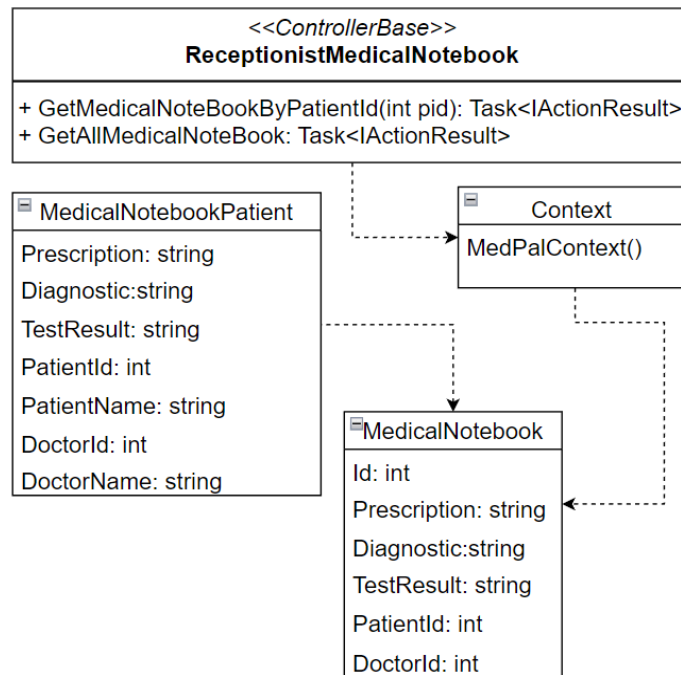


Figure 3.4.2.1.1 Receptionist view medical notebook class diagram

3.4.2.2 Sequence Diagram

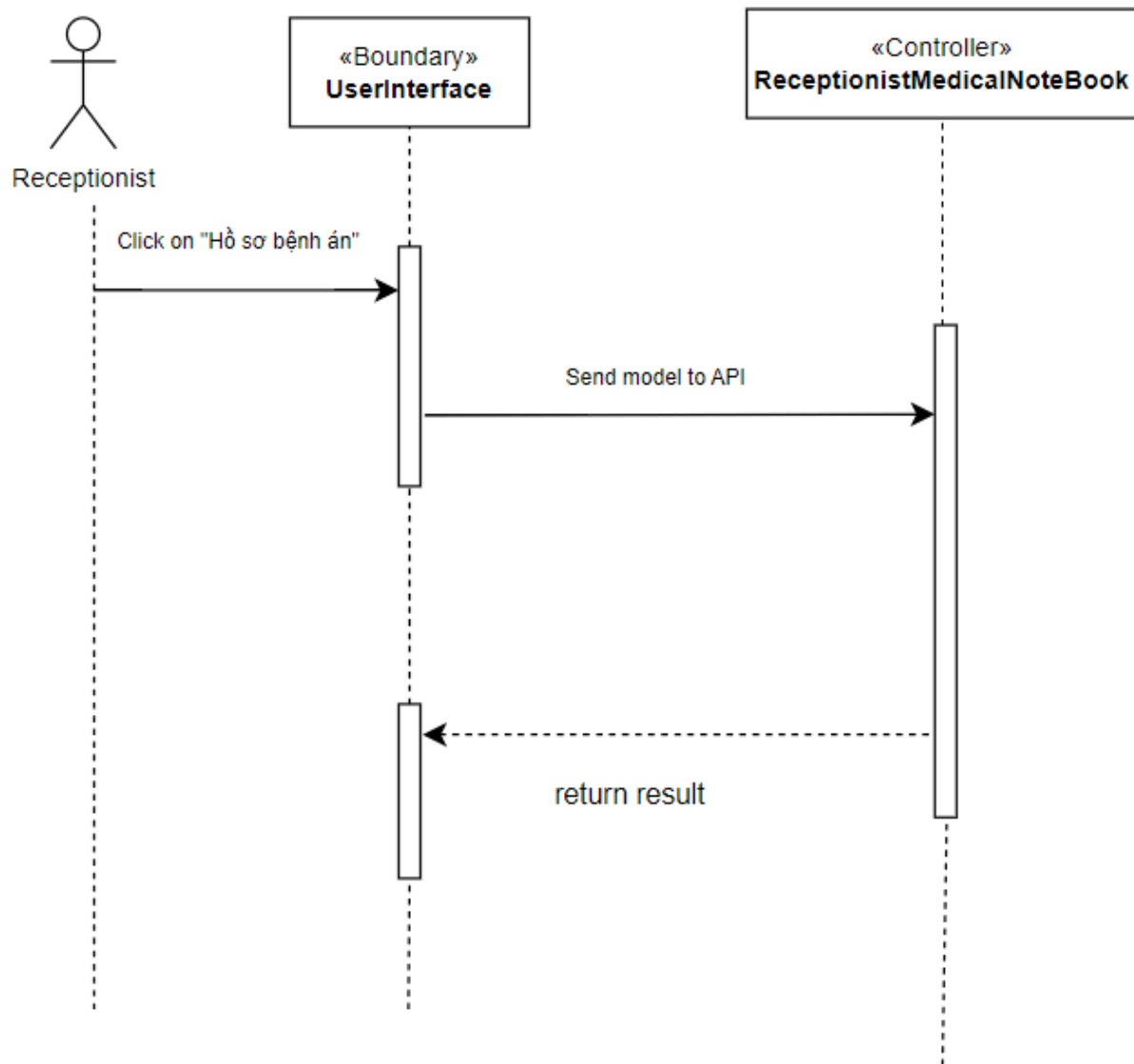


Figure 3.4.2.2.1 Receptionist view medical notebook sequence diagram

3.4.3 Doctor create medical Notebooks

3.4.3.1 Class Diagram

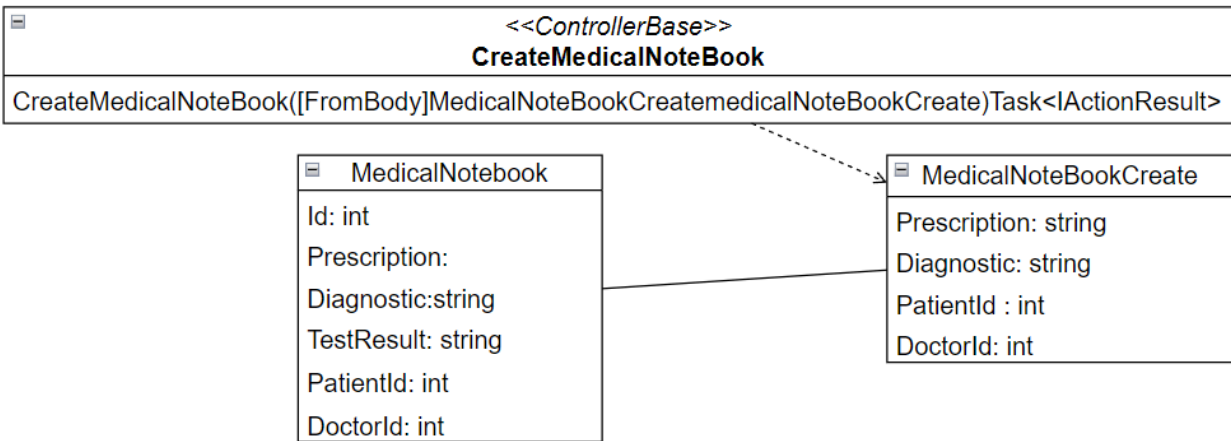


Figure 3.4.3.1.1 Doctor create medical notebook class diagram

3.4.3.2 Sequence Diagram

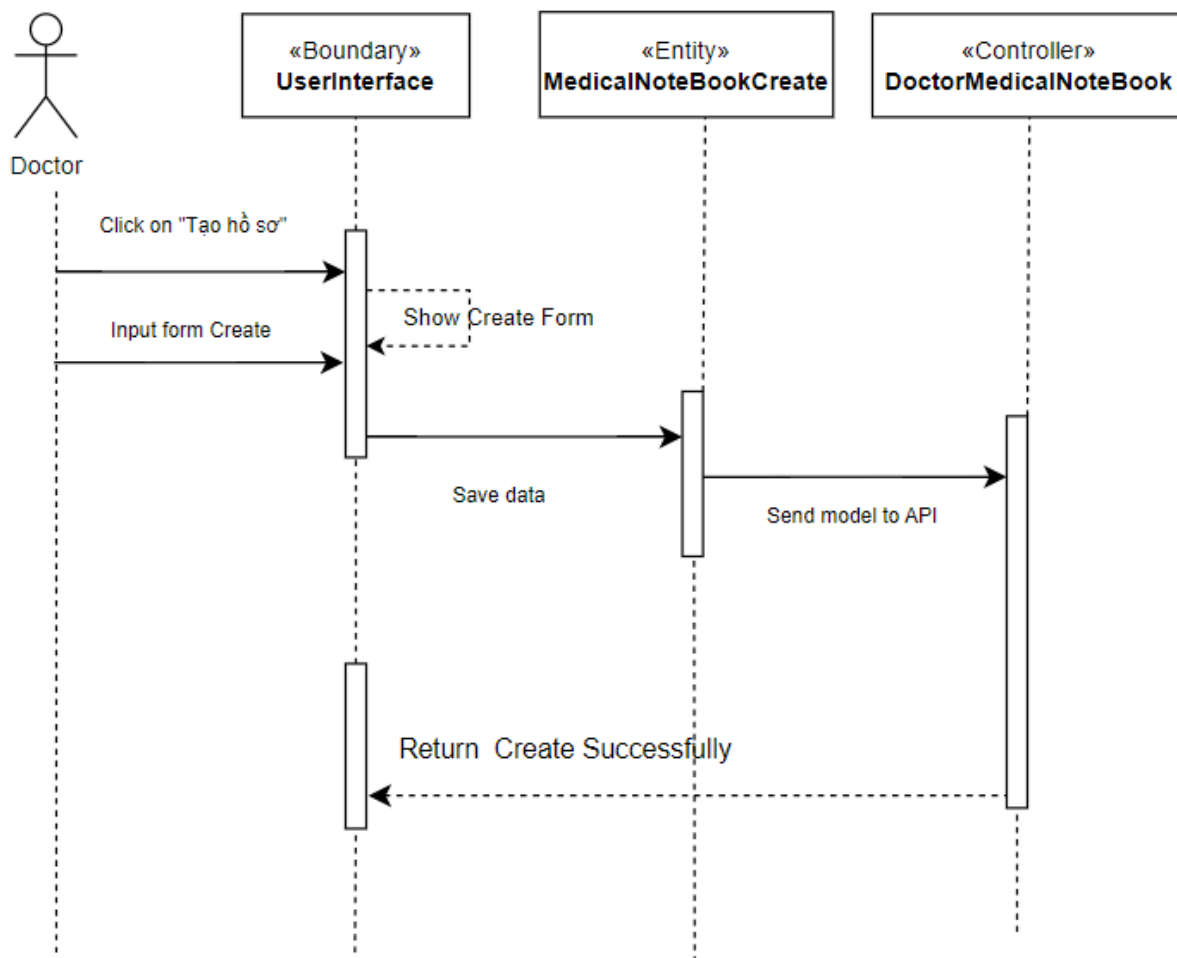


Figure 3.4.3.2.1 Doctor create medical notebook sequence diagram

3.4.4 Patient View Medical Notebooks

3.4.4.1 Class Diagram

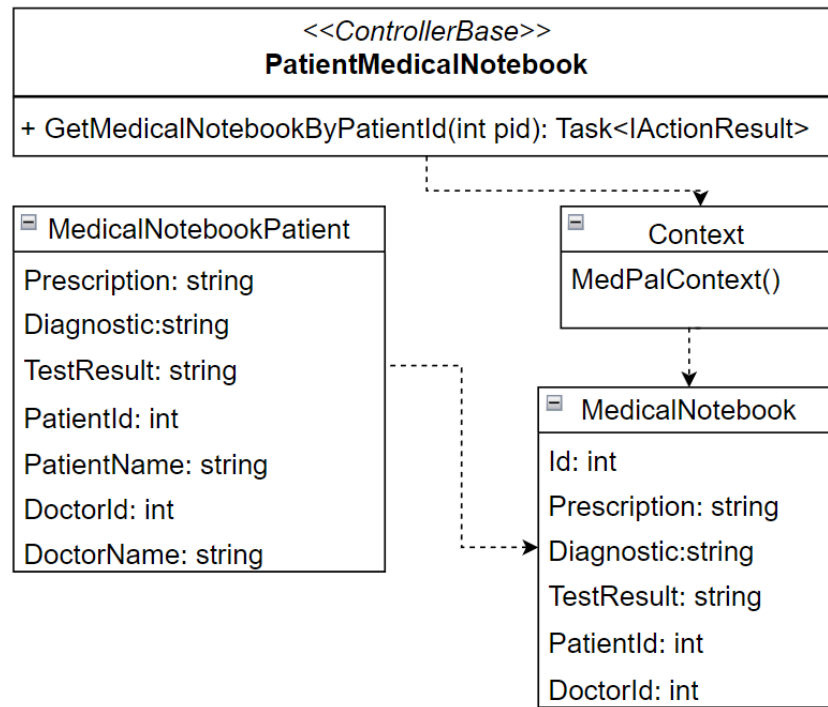


Figure 3.4.4.1.1 Patient view medical notebook class diagram

3.4.4.2 Sequence Diagram

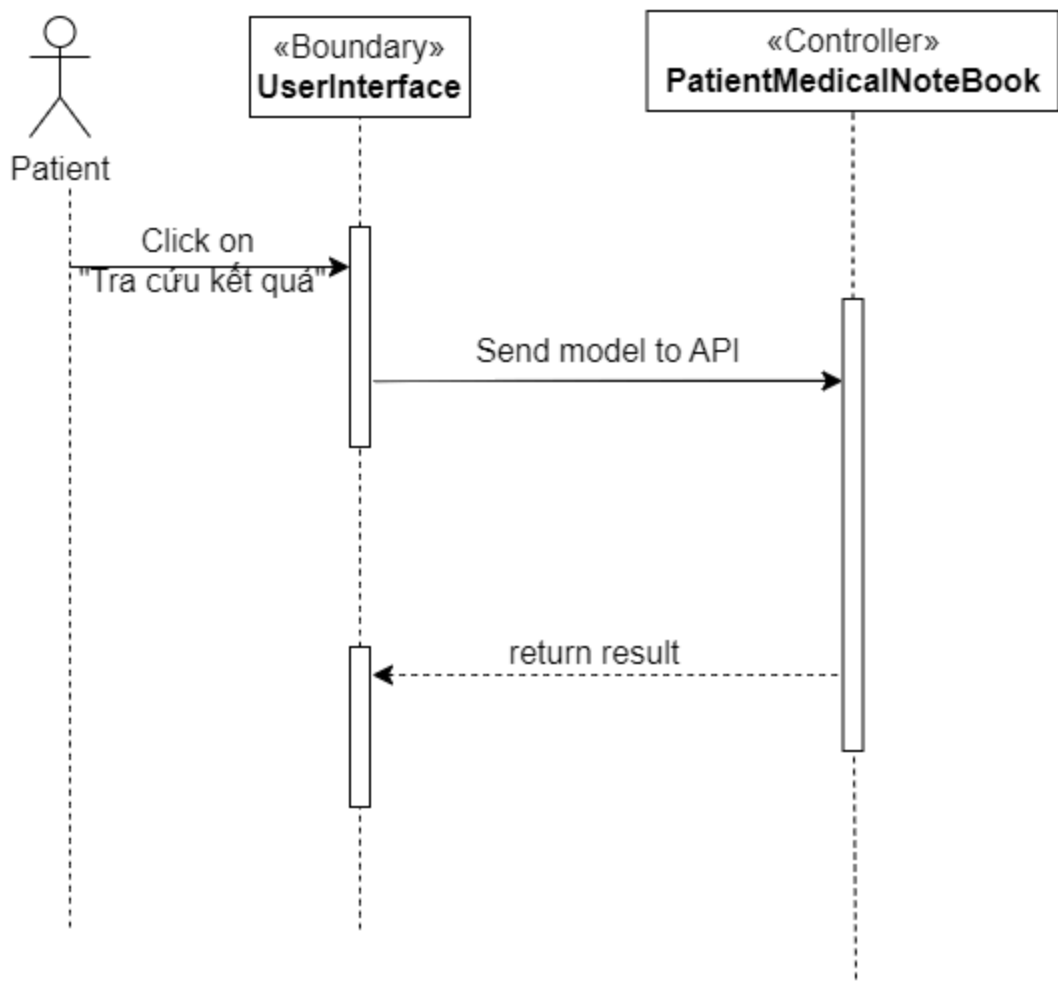


Figure 3.4.4.2.1 Patient view medical notebook sequence diagram

3.4.5 Active patient status

3.4.5.1 Class diagram

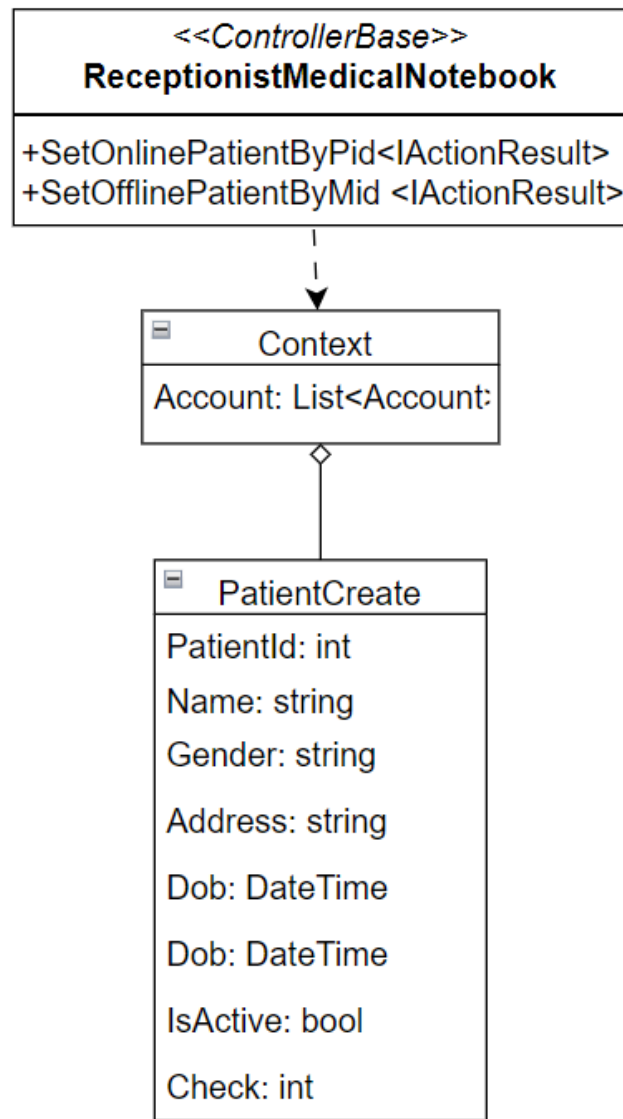


Figure 3.4.5.1.1 Active patient status Class diagram

3.4.5.2 Sequence diagram

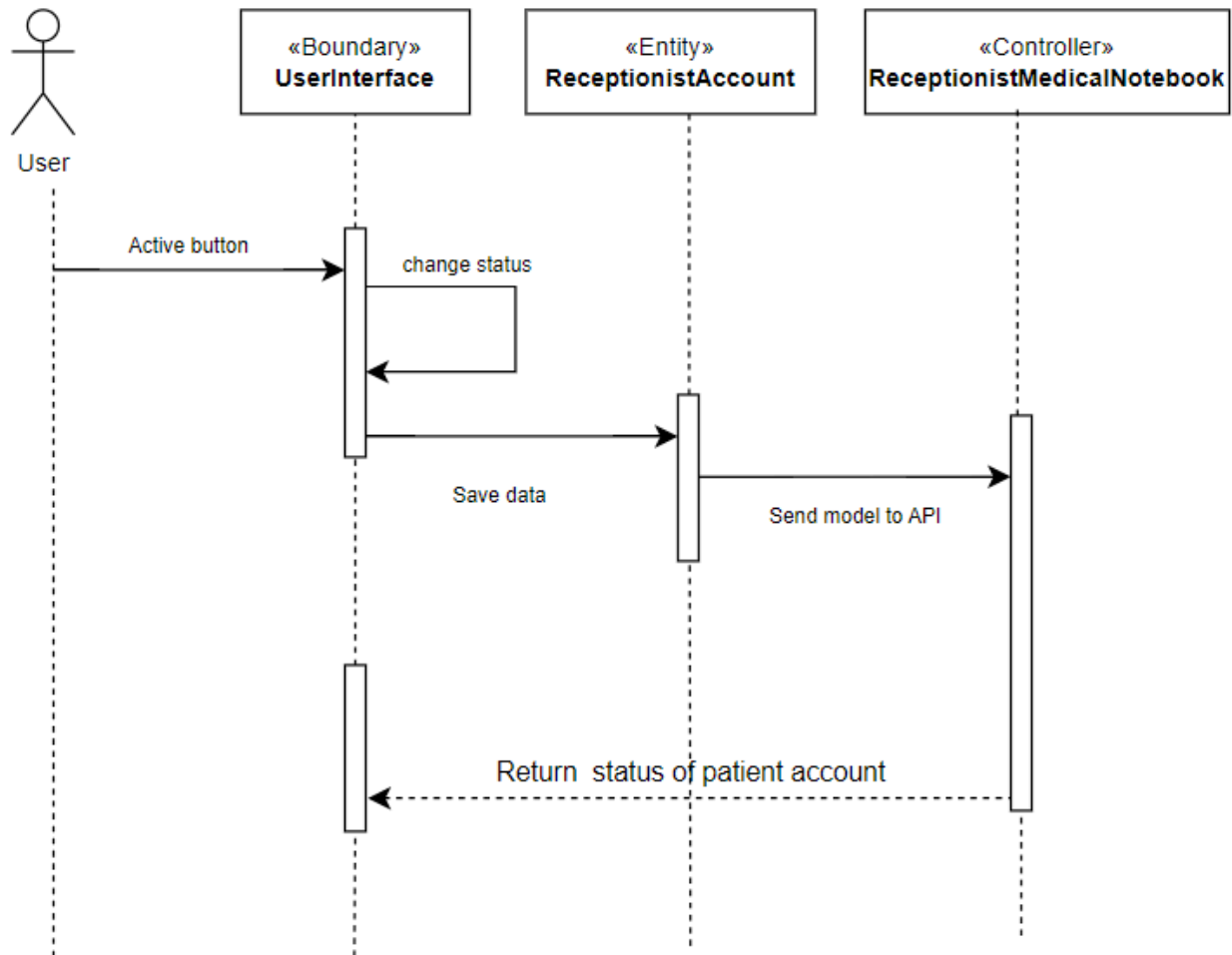


Figure 3.4.5.2.1 Active patient status Sequence diagram

3.4.6 Import result image

3.4.6.1 Class Diagram

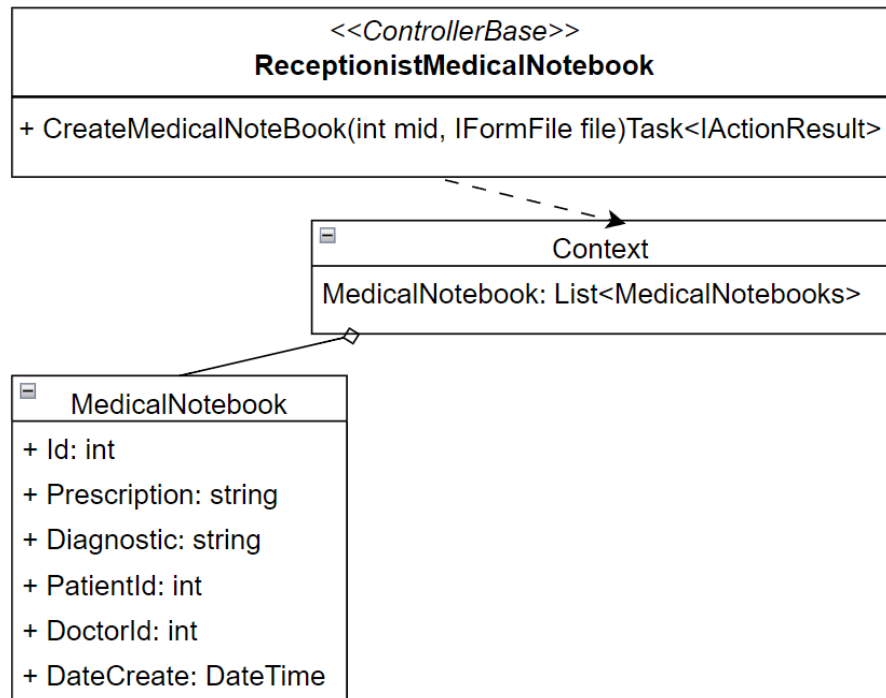


Figure 3.4.6.1.1 Import result image class diagram

3.4.6.2 Sequence Diagram

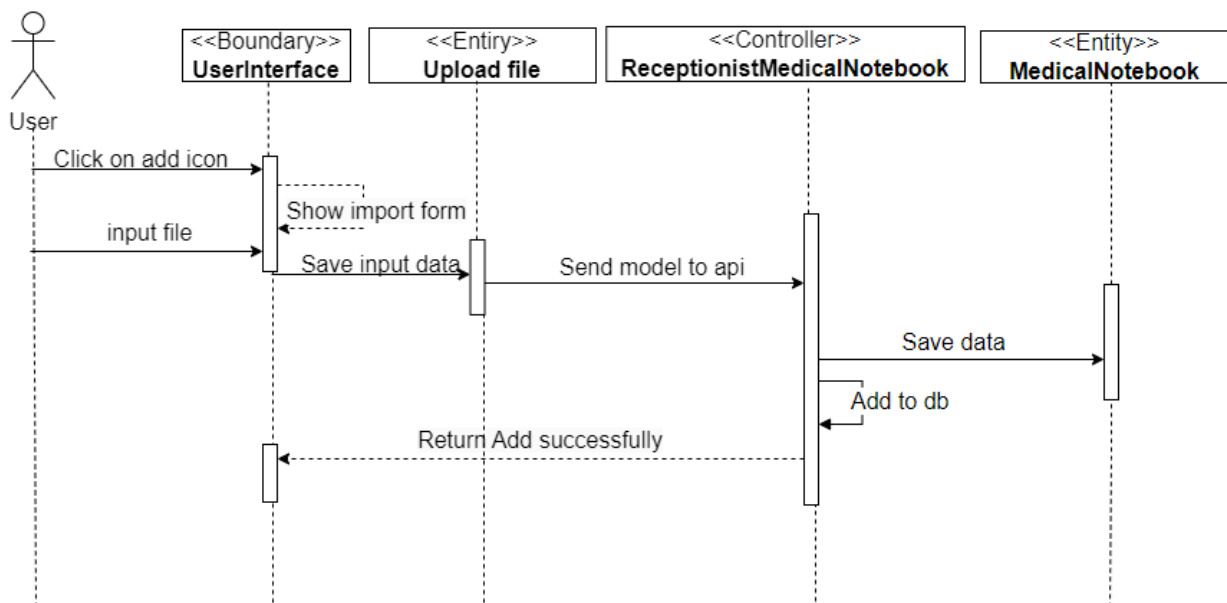


Figure 3.4.6.2.1 Import result image sequence diagram

3.4.7 Search Medical Notebook

3.4.7.1 Class Diagram

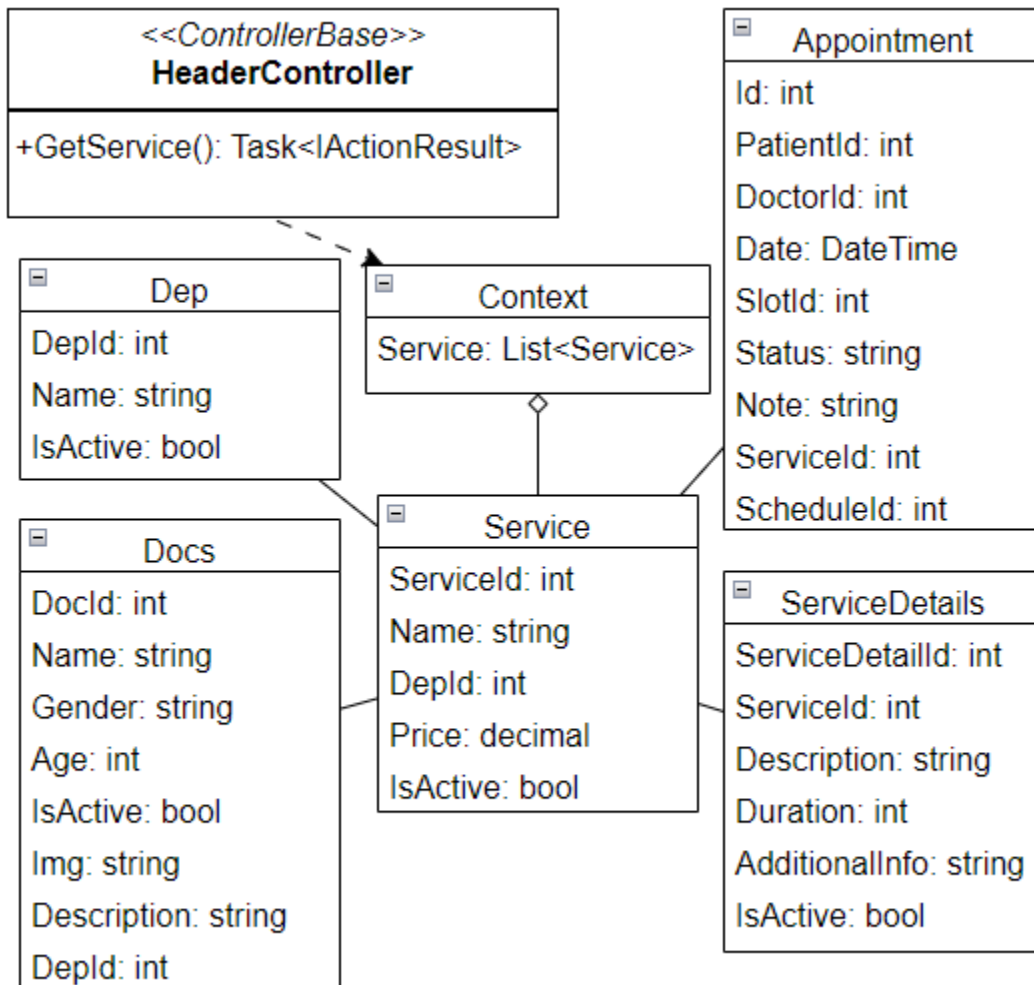


Figure 3.4.7.1.1 Search Medical Notebook Class diagram

3.4.7.2 Sequence Diagram

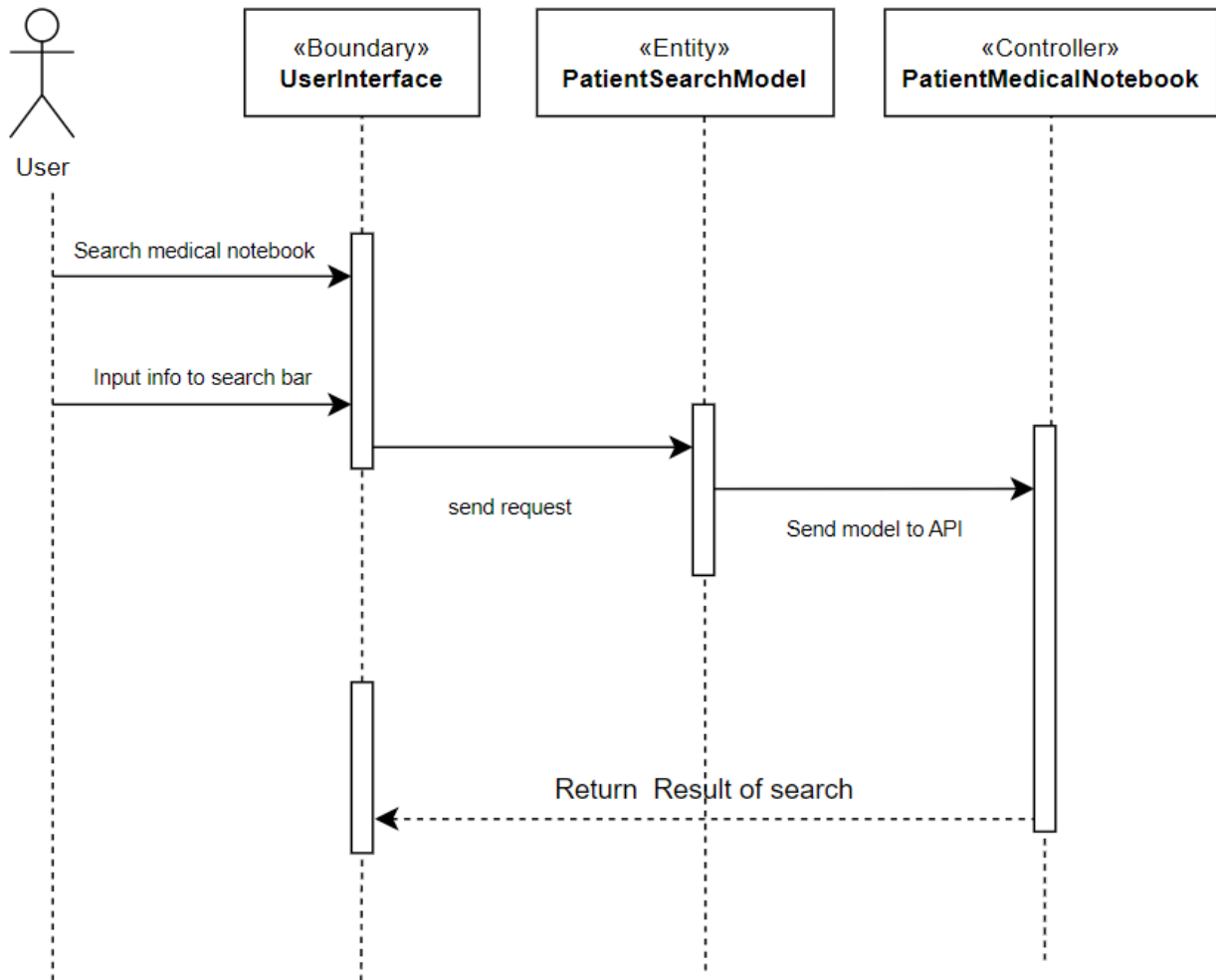


Figure 3.4.7.2.1 Search medical notebook Sequence diagram

3.5 Online Consulting

3.5.1 Reminder follow-up appointment(patient)

3.5.1.1 Class Diagram

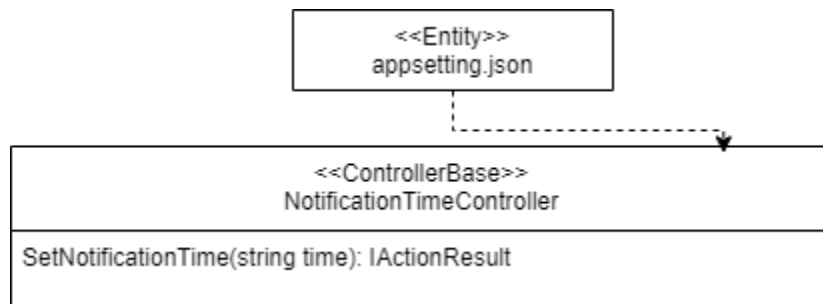


Figure 3.5.1.1.1 Reminder follow-up appointment(patient) class diagram

3.5.1.2 Sequence Diagram

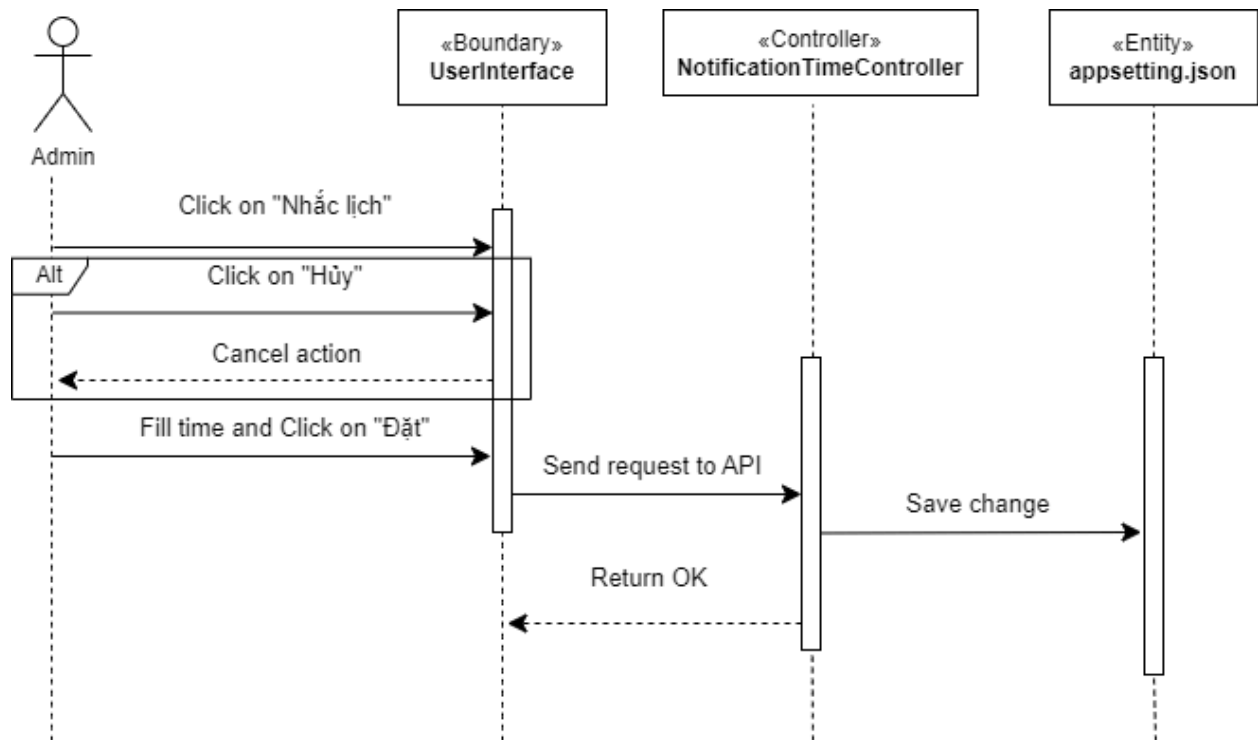


Figure 3.5.1.2.1 Reminder follow-up appointment(patient) sequence diagram

3.5.2 Reminder follow-up appointment(admin)

3.5.2.1 Class Diagram

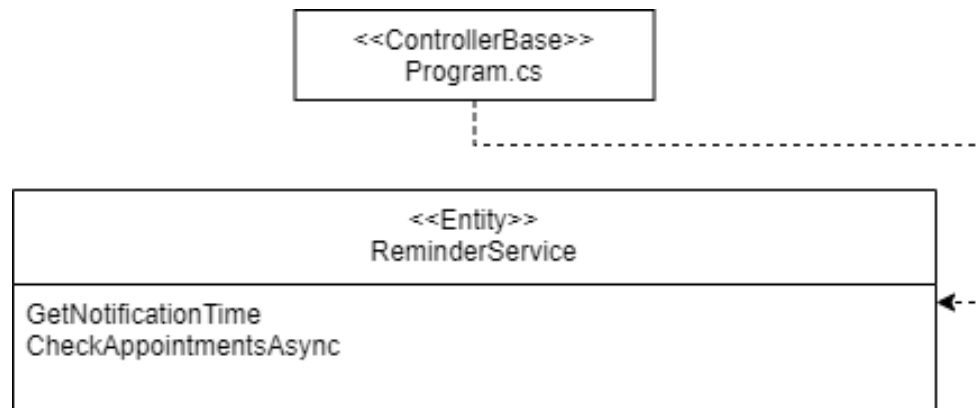


Figure 3.5.2.1.1 Reminder follow-up appointment(admin) class diagram

3.5.2.2 Sequence Diagram

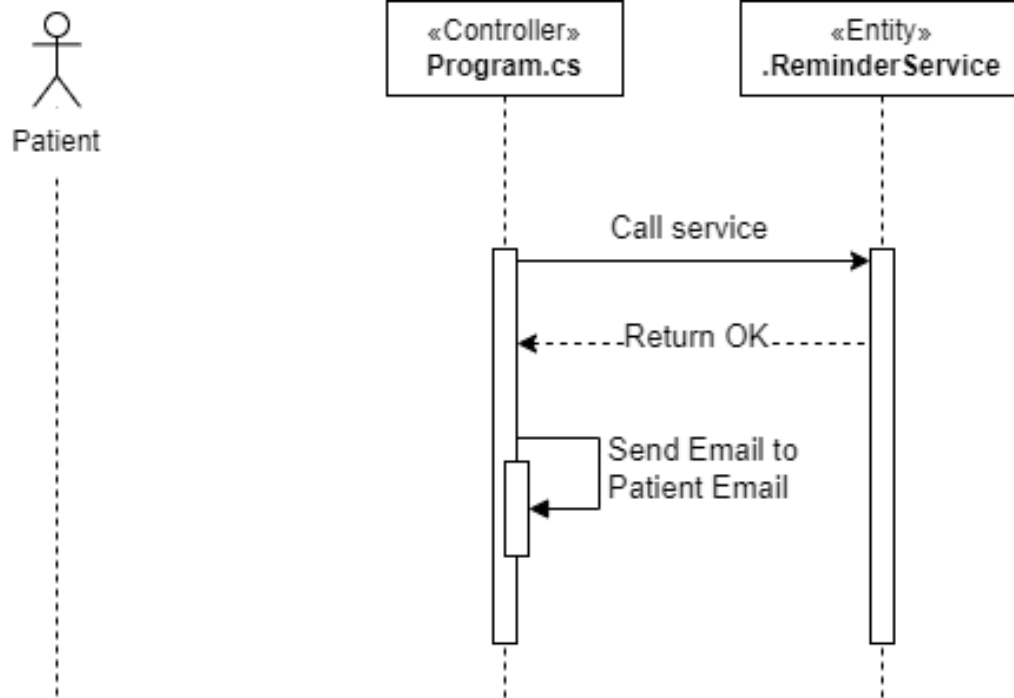


Figure 3.5.2.2.1 Reminder follow-up appointment(admin) sequence diagram

3.5.3 Post treatment follow-up(Doctor)

3.5.3.1 Class Diagram

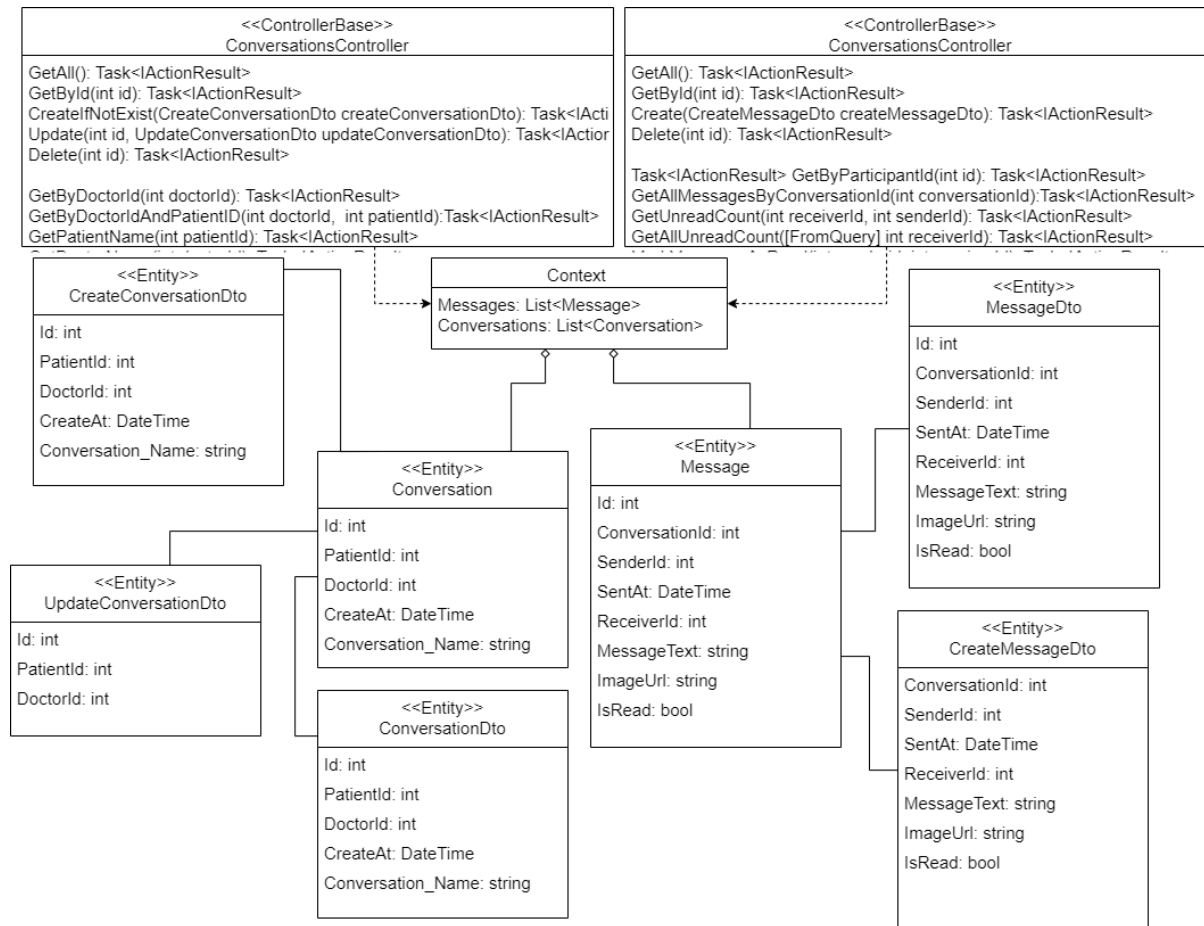


Figure 3.5.3.1.1 Post treatment follow-up(Doctor) class diagram

3.5.3.2 Sequence Diagram

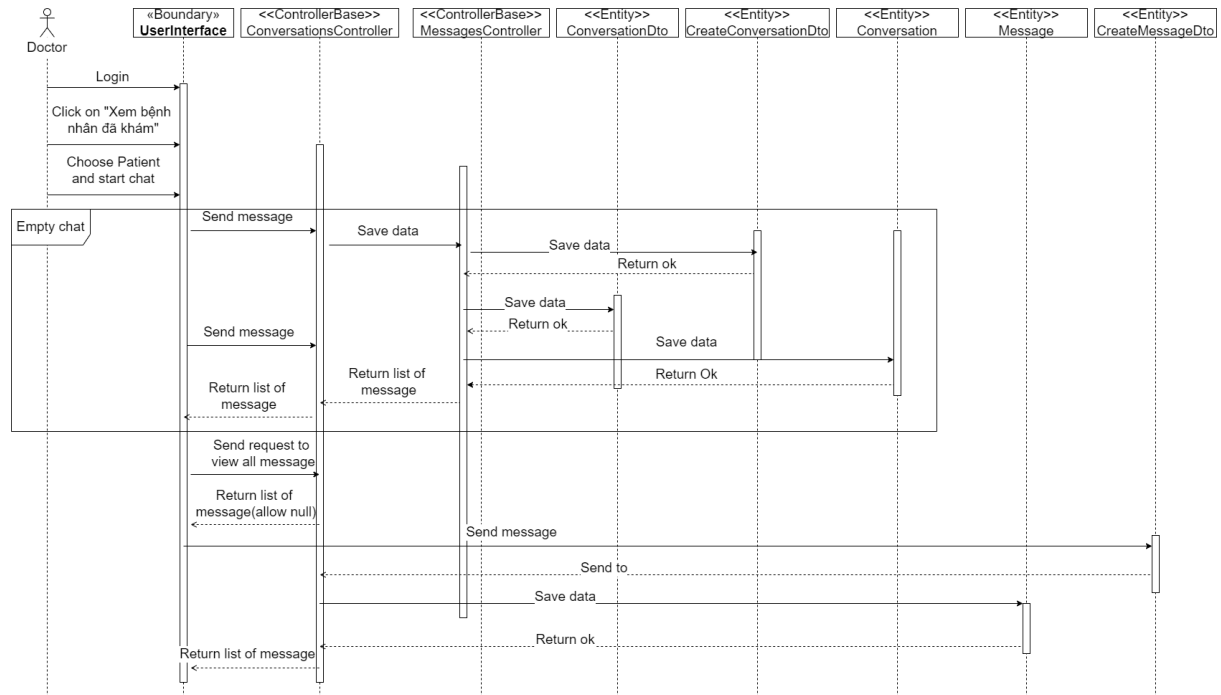


Figure 3.5.3.2.1 Post treatment follow-up(Doctor) sequence diagram

3.5.4 Online advising(Patient)

3.5.4.1 Class Diagram

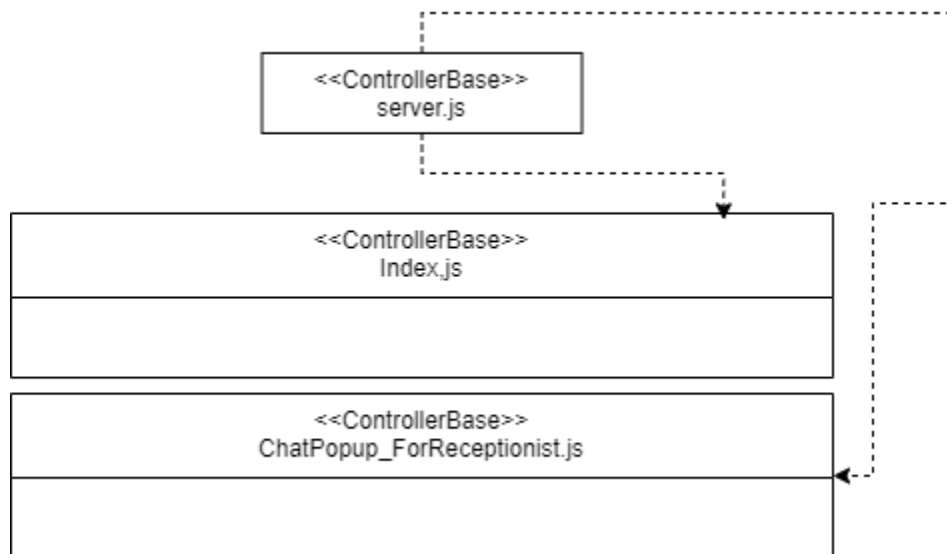


Figure 3.5.4.1.1 Online advising(Patient) class diagram

3.5.4.2 Sequence Diagram

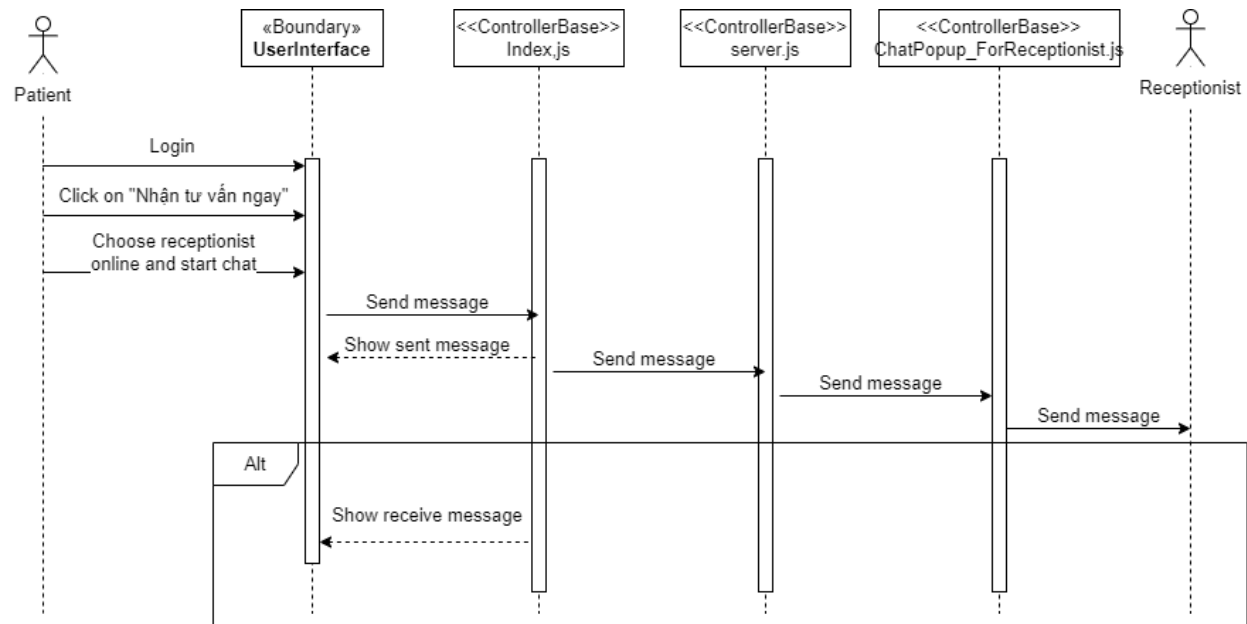


Figure 3.5.4.2.1 Online advising(Patient) sequence diagram

3.5.5 Online advising(Receptionist)

3.5.5.1 Class Diagram

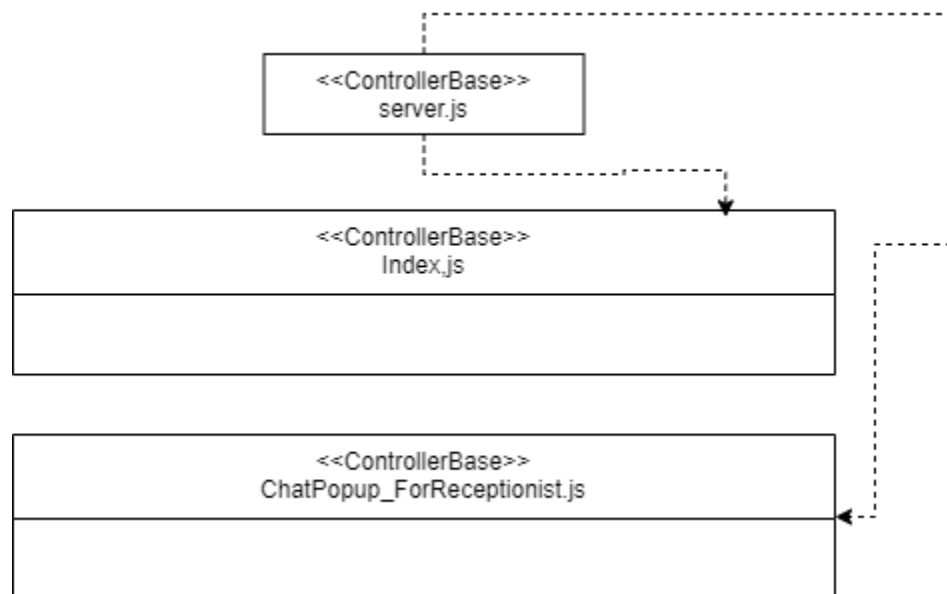


Figure 3.5.5.1.1 Online advising(Receptionist) class diagram

3.5.5.2 Sequence Diagram

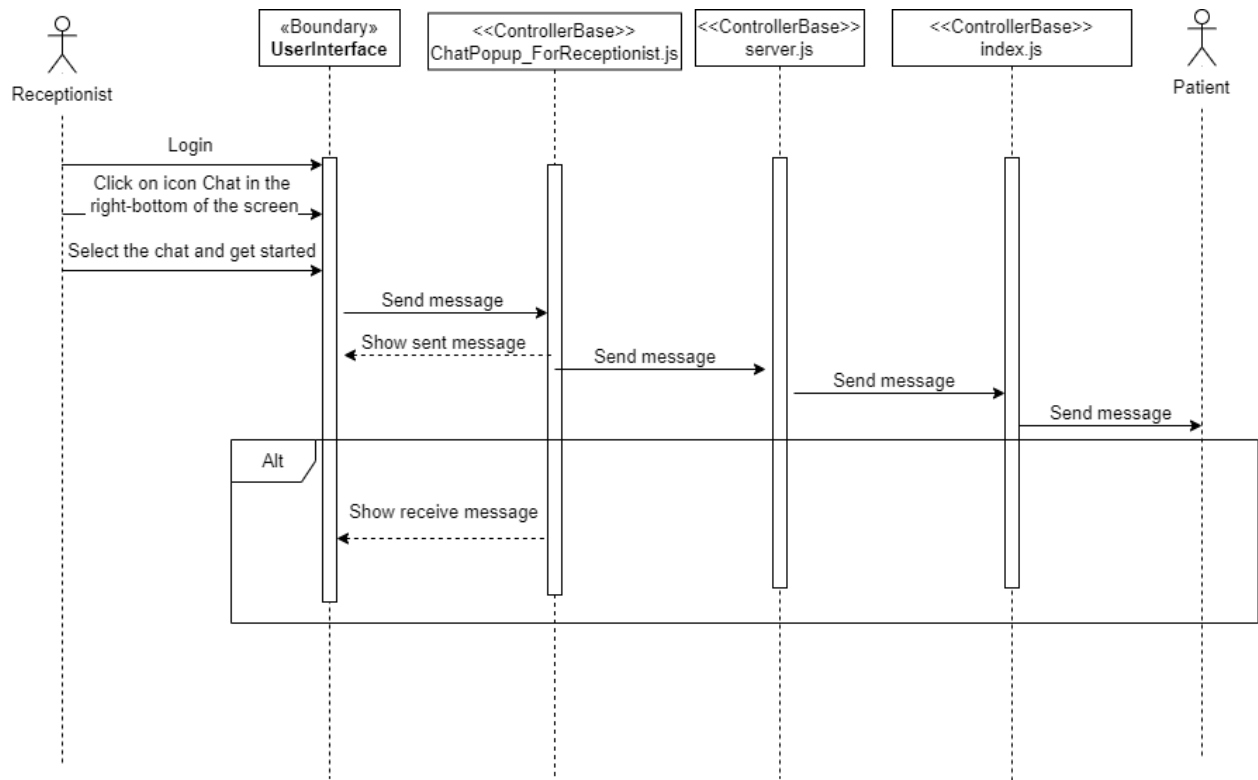


Figure 3.5.5.2.1 Online advising(Receptionist) sequence diagram